

Operating instructions

Die casting machine

Original operating instructions
700026276-1-en-US
2015-05-28

ECOLINE 84

精锐



Machine number: 700026276
Woo Su AMS
22222 Ulsan, South Korea



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1 Important information

1.1 Scope

This manual is valid for the following versions:

Designation	Type
Ecoline	84

1.2 Applicable documents

The following documents form an integral part of this manual:

Designation	Register
Dimensional drawings	2
Circuit diagram	3
Hydraulic diagram	4
Interface descriptions	5
Spare parts catalog	6
Certificates	7
Third-party documentation	8

1.3 Contact

- ▶ If you have any questions, please contact the nearest branch of Bühler Group. See www.buhlergroup.com.
- ▶ Keep the instructions ready at hand.
- ▶ Keep the machine number ready to hand.

1.3.1 Address

Bühler AG
CH-9240 Uzwil, Schweiz
+41 (71) 955 12 12
+41 (71) 955 25 88
die-casting.info@buhlergroup.com
www.buhlergroup.com/die-casting

1.3.2 Feedback regarding this manual

Requests and suggestions relevant to this manual
technical.documentation@buhlergroup.com.

1.4 Personnel qualifications

1.4.1 Users

Persons who are instructed in operation of the machine, who know and apply the safety regulations.

1.4.2 Machine setters

Persons who can execute the set-up tasks on the machine. Machine setters must be technically trained or have attended and passed the appropriate training course offered by the manufacturer.

1.4.3 Owner's service personnel

Individuals who operate the machine must be technically trained or have attended and passed the manufacturer's training.

1.4.4 Manufacturer's personnel

Manufacturer's personnel who execute installation and commissioning, as well as complex service and maintenance tasks.

1.4.5 Mechanics

Individuals who work on mechanical equipment must be technically trained or have attended and passed the manufacturer's training.

1.4.6 Electrician

Individuals who work on electrical equipment must be technically trained or have attended and passed the manufacturer's training.

1.4.7 Welders

Personnel who weld on the equipment must have specialized training.

1.5 Presentation conventions

1.5.1 Explanation of warnings

DANGER

Type and source of the hazard.
The hazard will result in death or severe injuries.
► Precautionary measures to mitigate the hazard.

WARNING

Type and source of the hazard.
The hazard may result in death or severe injury.
► Precautionary measures to mitigate the hazard.

 CAUTION

Type and source of the hazard.

The hazard may result in injuries.

- Precautionary measures to mitigate the hazard.

NOTICE

Type and source of the hazard.

The hazard may result in material damage.

- Precautionary measures to prevent material damage.

2 Safety

2.1 Intended use

The machine has been designed exclusively for die casting of the following materials:

- Aluminum alloys
- Magnesium alloys
- Zinc alloys
- Copper alloys

Any use beyond the specified use shall be considered non-intended use.

- ▶ Only use the machine as intended.
- ▶ Operate the machine only in accordance with the instructions in this operating manual.

2.2 Misuse

2.2.1 Wrong die

The machine must not be operated with dies that cannot withstand the set force.

- ▶ Verify that the dimensions of the die are within the specifications of the machine.
- ▶ Match the die to the metal pressure.
- ▶ Check the die for flashing and spitting.

2.3 Technical condition

If the machine is in poor technical condition, safety, proper operation and availability are compromised.

- ▶ Operate the machine only when it is in faultless technical condition.
- ▶ Comply with the maintenance schedule.
- ▶ Only use original spare parts listed in the spare parts catalog.
- ▶ If the operating behavior of the machine changes, check the machine for faults.
- ▶ Rectify faults immediately.
- ▶ The machine must not be modified or altered without approval from the manufacturer.

2.4 Personnel qualifications

Unqualified personnel cannot recognize risks and are thus exposed to greater danger.

- ▶ Allow only technically qualified personnel to perform the activities described in these operating instructions.
- ▶ Ensure that personnel comply with local laws and regulations for safe and hazard-conscious work.
- ▶ Define and communicate work responsibilities. Only give keys and passwords to appointed staff.

2.5 Personal protective equipment

During operation, machine surfaces are hot and touching them can result in burns due to liquid metal and hot surfaces.

- ▶ Wear protective clothing.
- ▶ Wear safety shoes.
- ▶ Wear hand protection.
- ▶ Wear head and eye protection.

2.6 Ear protection

Loud machine noises can cause damage to hearing.

- ▶ Always wear ear protection when staying at the machine.

2.7 Hazardous substances

- ▶ In addition to the operating manual, also comply with the instructions in the hazardous substance data sheets and the information provided by the manufacturer.

2.7.1 Nitrogen

Explosions can occur if using impure nitrogen.

- ▶ Only use nitrogen gas of purity class ≥ 4.5 , $N_2 = 99.99\%$ by volume.

2.7.2 Service water

Service water becomes contaminated through environmentally harmful and otherwise harmful substances, such as hydraulic fluids, spray agents, etc.

- ▶ Operate a closed service water circuit with water treatment.
- ▶ Only discharge service water through a wastewater cleaning system.

2.7.3 Hydraulic fluid

Hydraulic fluid that leaks and fouls the foundation.

- ▶ Collect hydraulic fluid in a collection trough and dispose of it properly.
- ▶ In addition to complying with the operating manual, also comply with the instructions in the hazardous substance data sheets and the information provided by the manufacturer.

2.7.4

Lubricant

Lubricant contaminates the foundation.

- ▶ Collect lubricant in a collection trough and dispose of it properly.
- ▶ In addition to complying with the operating manual comply with the instructions in the hazardous substance data sheets and the information provided by the manufacturer of the hazardous substances.

2.7.5

Fumes and smoke

Toxic fumes and smoke are health hazards for the employees.

- ▶ Exhaust fumes and smoke.
- ▶ In addition to complying with the operating manual, also comply with the instructions in the hazardous substance data sheets and the information provided by the manufacturer.

2.8

Securing the machine to prevent unexpected start-up

If the machine starts up unexpectedly, individuals working on the machine could be injured severely. The machine can be secured in the following manner:

- ▶ Stop the machine, press the EMERGENCY STOP button at the control column and carry the key for the EMERGENCY STOP button with you.
- ▶ Set the master switch at the control cabinet to "0", secure it with a padlock, and carry the key with you.

2.9

Protective devices

People are at risk when safety equipment is not functioning effectively.

- ▶ Before operating the machine, ensure that all protective devices are functioning effectively.

2.10

Safety signs

If safety signs are not noticeable, people are at risk.

- ▶ Replace safety signs which are not noticeable.
- ▶ Do not remove or cover safety signs.

2.11

Safe working environment

2.11.1

Slipping and tripping hazards

Slippery surfaces and tripping hazards may result in serious accidents.

- ▶ Keep aisles, handles, steps, ladders, platforms and handrails free of grease, oil and other types of dirt.
- ▶ Do not use the machine as a climbing aid or storage area. Use only the steps and platforms provided.
- ▶ Wear anti-slip safety shoes.

2.12

One-person operation

If several people are working on the machine at the same time, life-threatening injuries can occur.

- ▶ Only have one person operate the machine.
- ▶ In set-up mode, ensure that a second person is not in the danger zone of the machine.
- ▶ If a second person is necessary in set-up mode, maintain eye contact.

2.13

Metal fires

Metal fires are intensified if fought with the wrong extinguishing agent.

- ▶ Use dry sand or category D fire extinguishers to extinguish metal fires.

2.14

Danger of explosion

Danger of explosion and severe bodily injury if water comes into contact with liquid metal.

- ▶ Do not immerse wet ladles or other wet objects into the liquid metal.
- ▶ Keep the danger zone of the machine dry and clean.

2.15

Control cabinet

A cooling device that is not working correctly can cause a defect.

- ▶ Always keep the doors closed.
- ▶ For the installation, ensure that the cooling device is not leaking.
- ▶ Ensure that water drains.
- ▶ Set temperature to 35°C.

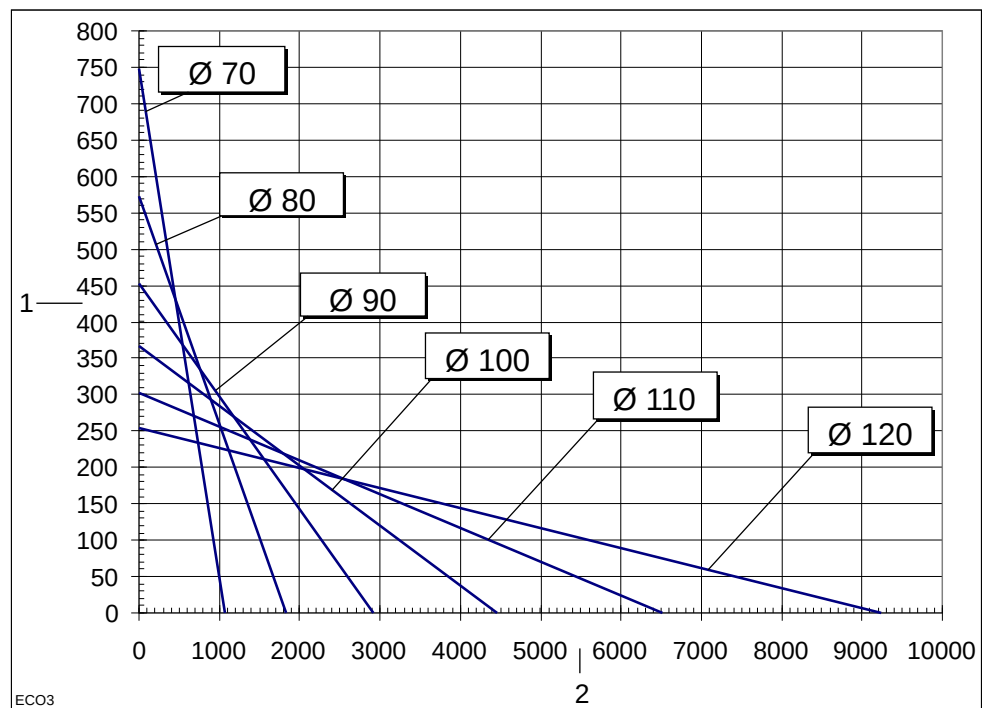
3 Technical data

3.1 Ambient conditions

Description	Value	Unit
Temperature range	+5 to +50	°C
Relative humidity, max.	95	%

3.2 Performance characteristics

3.2.1 p/Q² diagram



(1) Specific metal pressure in bar

(2) Metal volume flow in (dm³/s)²

Description	Value	Unit
Specific metal pressure, minimum	255	bar
Specific metal pressure, maximum	749	bar
Metal volume flow, minimum	1069	(dm³/s)²
Metal volume flow, maximum	9232	(dm³/s)²

Plunger diameter 70 mm

Description	Value	Unit
Shot volume, maximum	1796	cm ³
Shot weight, maximum	4.49	kg
Shot pressure, maximum	2004	bar
Projected area, maximum	419	cm ²

Plunger diameter 80 mm

Description	Value	Unit
Shot volume, maximum	2346	cm ³
Shot weight, maximum	5.86	kg
Shot pressure, maximum	1535	bar
Projected area, maximum	547	cm ²

Plunger diameter 90 mm

Description	Value	Unit
Shot volume, maximum	2969	cm ³
Shot weight, maximum	7.42	kg
Shot pressure, maximum	1213	bar
Projected area, maximum	693	cm ²

Plunger diameter 100 mm

Description	Value	Unit
Shot volume, maximum	3665	cm ³
Shot weight, maximum	9.12	kg
Shot pressure, maximum	982	bar
Projected area, maximum	855	cm ²

Plunger diameter 110 mm

Description	Value	Unit
Shot volume, maximum	4435	cm ³
Shot weight, maximum	11.09	kg
Shot pressure, maximum	812	bar
Projected area, maximum	1035	cm ²

Plunger diameter 120 mm

Description	Value	Unit
Shot volume, maximum	5278	cm ³
Shot weight, maximum	13.19	kg
Shot pressure, maximum	682	bar
Projected area, maximum	1232	cm ²

Shot weight, maximum "DIN 24480" =

$\frac{2}{3} \times \text{plunger diameter} \times \text{surface} \times \text{density (Al, 2.5 kg/dm}^3\text{)}$

3.3

Operating supplies

3.3.1

Hydraulic fluids

The following purity class requirements are met with the installed filter system, provided the maintenance schedule has been complied with.

Purity classes

Description	Value	Unit
Purity class to ISO 4406: 1999	17/15/12	
Purity class to NAS 1638	6	

Viscosity range

Description	Value	Unit
Permissible viscosity range	20 to 250	cSt

Hydraulic fluid ISO VG 46 - HFC

Description	Value	Unit
Specification in accordance with DIN 51502	HFC, highly-flammable	
Viscosity class	ISO VG 46	
Density at 15°C in accordance with DIN 51717	< 1.1	kg/dm ³
Water proportion	35 ... 50	%
pH value	9 ... 10	
Optimal operating temperature	43	°C
Permissible operating temperature	2 ... 55	°C

3.3.2

Central lubricating oil CGLP

Requirements:

- Ambient conditions of the machine
- Data sheets of the operating materials suppliers

Oil ISO VG 150 - CGLP

Description	Value	Unit
Specification in accordance with DIN 51502	CGLP	
Viscosity class	ISO VG 150	
Operating temperature range	+18 ... +55	°C

Oil ISO VG 220 - CGLP

Description	Value	Unit
Specification in accordance with DIN 51502	CGLP	
Viscosity class	ISO VG 220	
Operating temperature range	+25 ... +63	°C

3.3.3

CLP gear oil

Requirements:

- Ambient conditions of the machine
- Data sheets of the operating materials suppliers

Oil ISO VG 680 - CLP

Description	Value	Unit
Specification in accordance with DIN 51517	CLP	
Viscosity grade	ISO VG 680	
Application temperature range	-25 ... +120	°C

3.3.4

Grease for standard requirements

Application area:

- Fixing surfaces
- Die casting position adjustment

Grease

Description	Value	Unit
Application temperature range	-25 ... +125	°C
Welding force in accordance with DIN 51350	~ 4 200 ... 5 200	N
Penetration in accordance with DIN 51804	~ 250	mm/10
Product recommendation	Dow Corning Molycote DX	

3.3.5 Grease for higher requirements

Application area:

- Tie bar bearings of the fixed plate
- Tie bar nuts
- Split nut

Grease

Description	Value	Unit
Application temperature range	-40 ... +200	°C
Dry lubricating capacity	> 500	°C
Welding force in accordance with DIN 51350	~ 4 200 ... 5 200	N
Penetration in accordance with DIN 51804	~ 300	mm/10
Product recommendation	Dow Corning Molycote P74	

3.3.6 NLGI 2 grease

Application area:

- Bearings of the drive motor

NLGI 2 grease

Description	Value	Unit
Food grade approval	-	
Consistency class	NLGI 2	
Application temperature range	-20 ... +130	°C

3.3.7 Lubricant

Application area:

- Plunger

Description	Value	Unit
Viscosity range	20 ... 11 000	cSt

3.3.8 Cooling water

Description	Value	Unit
pH value	7.5 ... 9.0	
Water hardness	max. 6	°dH
Chloride content	max. 70	mg/l
Nitrate content	max. 70	mg/l
Sulfate content	max. 70	mg/l

3.3.9**Nitrogen N₂**

Description	Value	Unit
Purity class NAS	≥ 4.5	
Nitrogen N ₂	> 99.99	% by volume
Oxygen O ₂	50	vpm
Hydrogen H ₂	~ 30	vpm

3.3.10**Operating materials for maintenance tasks**

Cleaning the heat exchanger:

- 5% phosphoric acid or 5% oxalic acid

Cleaning the tie bars:

- Gas-oil mixture

Cleaning the sieve inserts and filter elements:

- Compressed air

3.4**Consumption data****3.4.1****Cooling - die and injection unit**

Description	Value	Unit
Operating pressure	3 ... 10	bar
Temperature range cooling water	5 ... 30	°C
Flow rate per cooling circuit	0 ... 30	l/min
Flow rate, flow regulator at operating temperature of 5 bar	300	l/min

3.4.2**Cooling, hydraulic fluid**

Description	Value	Unit
Operating pressure	3 ... 10	bar
Temperature range, cooling water	5 ... 30	°C
Consumption at a water temperature of 30°C	2.0	m³/h

3.5 Airborne noise emission

3.5.1 Measured values

The measured values do not take ambient effects into account.

Description	Value	Unit
Sound power level of the machine L_w	88	dB (A)
Equivalent workplace emission level L_p	68	dB (A)
Accuracy class	in accordance with ISO EN 3744	

3.6 Weights

3.6.1 Machine in ready-to-operate state

Description	Value	Unit
Weight	33 000	kg

3.7 Dimensions

3.7.1 Machine dimensions

Detailed dimensions of the machine and the components are specified in the dimension drawings.

3.8 Electrical specifications

3.8.1 Control system

Description	Value	Unit
Connected load	42	kW
Voltage ($\pm 10\%$)	380	V
Frequency	60	Hz
Degree of protection (DIN 40050)	IP54	

3.8.2 Drive

Description	Value	Unit
Connected load	37	kW

3.9 Locking unit

Locking force

Description	Value	Unit
Locking force, maximum	8 400	kN

Die

Description	Value	Unit
Die height	400 ... 1 000	mm
Die opening stroke	800	mm
Surface pressure, maximum	100	N/mm ²
Clamping dimensions, minimum	720 × 720	mm

Ejector

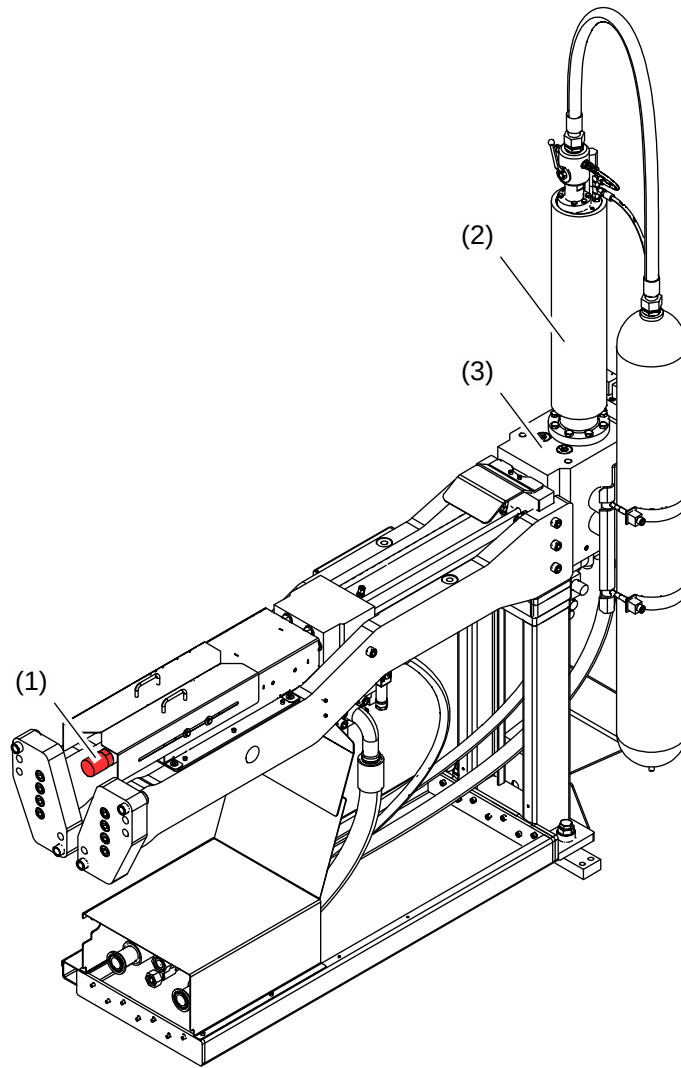
Description	Value	Unit
Ejector stroke	175	mm
Ejector force	225	kN

Tie bars

Description	Value	Unit
Clear dimensions of the tie bars	900 × 900	mm
Tie bar diameter	170	mm
Tie bar extension from the fixing surface of the fixed plate.	7 100	mm

Central lubrication system

Description	Value	Unit
Filling quantity	2.7	dm ³



- (1) Plunger
- (2) Nitrogen system, piston accumulator
- (3) Die casting position adjustment

Plunger

Description	Value	Unit
Dynamic injection force	288	kN
Injection force intensification pressure	771	kN
Plunger stroke	700	mm
Plunger diameter	70 ... 120	mm
Plunger speed, maximum	8.5	m/s

Plunger lubrication system

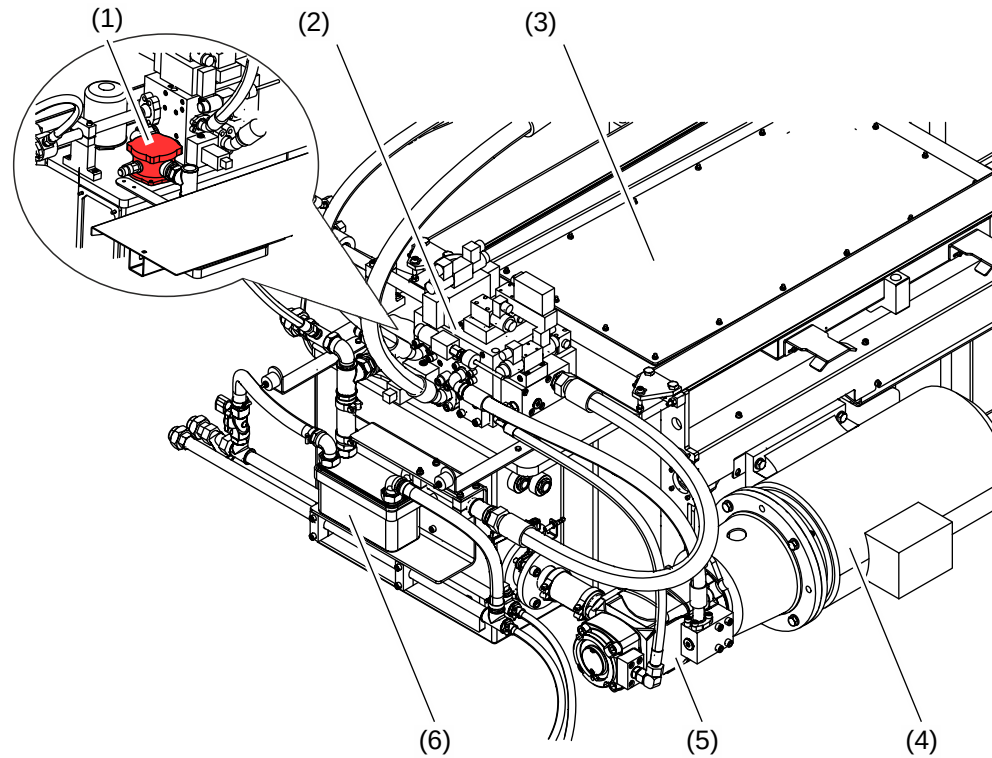
Description	Value	Unit
Filling quantity	10	dm ³

Nitrogen system, piston accumulator

Description	Value	Unit
Filling quantity	115	dm ³
Filling pressure, maximum	155	bar
Pressure with charged piston accumulator, maximum	210	bar

3.11

Drive unit



- | | |
|--------------------|--------------------------|
| (1) Return filter | (4) Drive motor |
| (2) Pump block | (5) Pumps |
| (3) Hydraulic tank | (6) Plate heat exchanger |

Drive motor

Description	Value	Unit
Drive power	37	kW
Speed	1 450	rpm

Pumps

Description	Value	Unit
Delivery rate - fixed displacement pump 1	145	l/min
Delivery rate - fixed displacement pump 2	72	l/min

Partial-flow filter

Description	Value	Unit
Grade of filtration	10	µm

Plate heat exchanger

Description	Value	Unit
Operating pressure, cooling water	3 ... 10	bar
Cooling water temperature at the inlet	5 ... 30	°C

Pump block

Description	Value	Unit
Operating pressure	0 ... 210	bar
Operating temperature, optimal	43	°C

3.12

Machine base

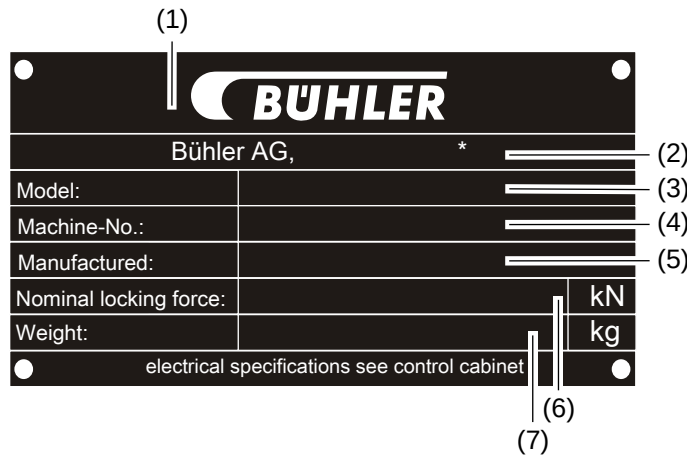
Hydraulic tank

Description	Value	Unit
Filling quantity	1 180	dm³
Filling quantity if the “early warning signal - level too low” is indicated when the machine has stopped	235	dm³
Filling quantity if the “early warning signal - level too low” is indicated during operation	205	dm³

4 Description

4.1 Identification

4.1.1 Type plate

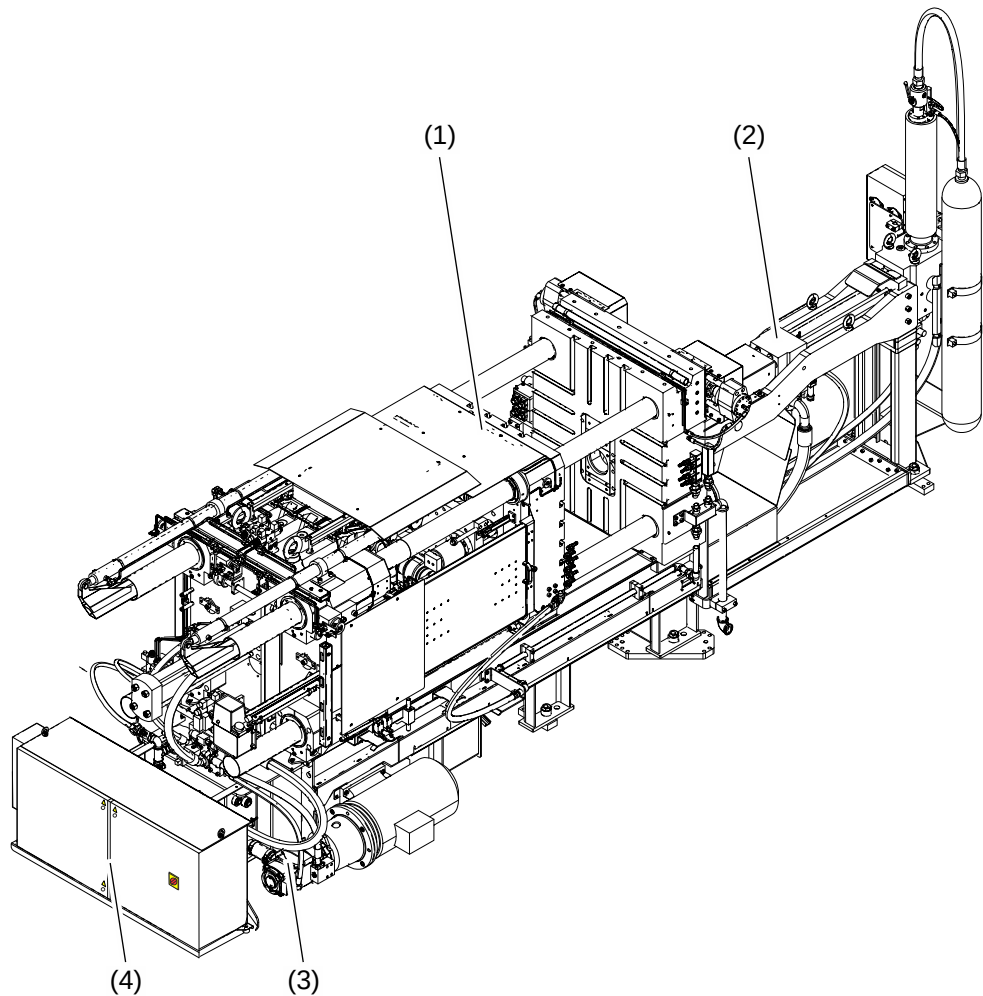


- (1) Manufacturer
- (2) Place of manufacture
- (3) Machine type
- (4) Machine number
- (5) Year of manufacture
- (6) Die locking force
- (7) Weight

4.1.2 Type code

Series	Ecoline
Overall size	84

The machine is a cold-chamber die-casting machine with horizontal plunger and injects alloys of aluminum, magnesium, zinc, and copper. The machine consists of four main subassembly groups: the locking unit, the injection unit, the drive unit, and the switch cabinet.



(1) Locking unit

(2) Injection unit

(3) Drive unit

(4) Control cabinet

Casting occurs in cycles. At the beginning of a machine cycle the closing cylinder of the die closes. A knee lever system keeps the die closed with high die locking force. A metal ladling device fills the liquid metal into the shot sleeve. The injection unit presses the liquid metal into the die. In the intensification pressure phase the plunger presses the metal into the die at high pressure to compress the cast part and compensate for the rigidity deficit. When the cast part is rigid, the die opens. The ejector pushes the cast part out of the die half of the moving plate. The parts extractor removes the cast part. A spray device cools, lubricates, and cleans the die. The machine cycle is concluded.

The metal ladling device, the parts extractor, and the spray device are peripheral devices that automate the cycle.

4.3.1

Set-up mode

In set-up mode, all movements are executed with reduced speed. To execute movements, a key must be depressed without interruption, or a switch must be activated.

The movements are executed with the protective doors open. The casting process is disabled.

4.3.2

Manual mode

In manual mode all movements of the machine are executed at the programmed speed. To execute movements, a key must be depressed without interruption, or a switch must be activated.

When the protective doors are closed, the die closes. The casting process is ready.

4.3.3

Semi-automatic mode

In semi-automatic mode, all movements of the machine are executed in a defined sequence. Each machine cycle is started manually. At the end of the machine cycle the machine remains in home position.

If the machine is not equipped with peripheral devices, or is partially equipped with peripheral devices, individual production steps are executed manually. The end of a manually executed production step is confirmed via a key on the control column. The machine cycle will be continued.

If the machine is completely equipped with peripheral devices a machine cycle is automatically executed.

4.3.4

Automatic mode

In automatic mode, all movements of the machine are executed in a defined sequence. The first machine cycle is started manually.

If the machine is not equipped with peripheral devices, or is partially equipped with peripheral devices, individual production steps are executed manually. The end of a manually executed production step is confirmed via a key on the control column. The machine cycle will be continued.

If the machine is completely equipped with peripheral devices the machine cycle is automatically executed. After the end of a machine cycle a new machine cycle is started automatically.

4.4

EMERGENCY STOP

In case of danger, the danger can be averted or reduced by pressing the EMERGENCY STOP.

With the triggering of an EMERGENCY STOP the drive motor is shut down and the hydraulic system is depressurized. The drive of the peripheral devices is stopped, if the devices are integrated in the EMERGENCY STOP circuit.

An additional EMERGENCY STOP button must be placed in the danger zone, if one of the following conditions is present:

- A manually-activated protective door without step-behind safeguard is installed.
- The automatic protective door remains open during the machine cycle.

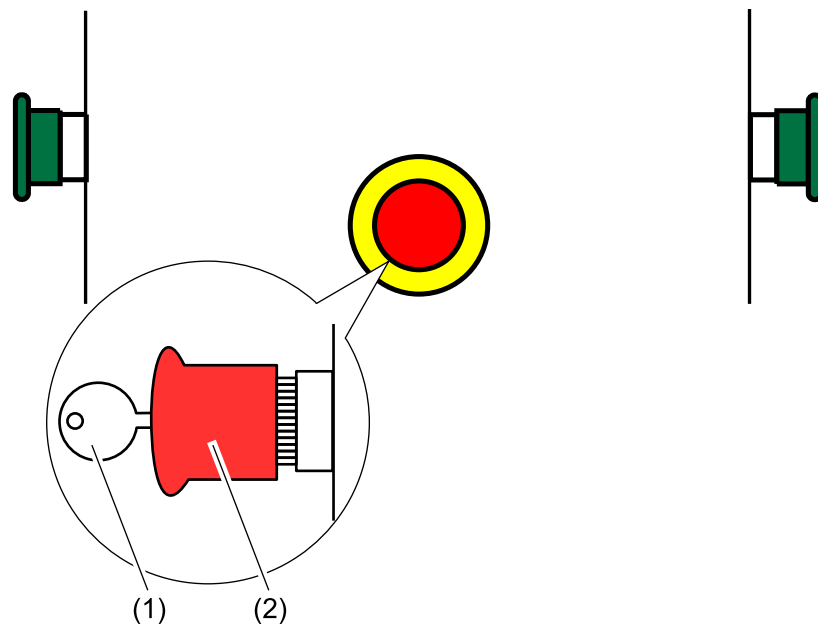
4.4.1

EMERGENCY STOP button

With the EMERGENCY STOP button, the machine can be shut down if there is danger.

There is an EMERGENCY STOP button on the control column.

The EMERGENCY STOP button on the control column is pressed to protect against unexpected start-up. In order to restart the machine, the EMERGENCY STOP button is unlocked with a key.



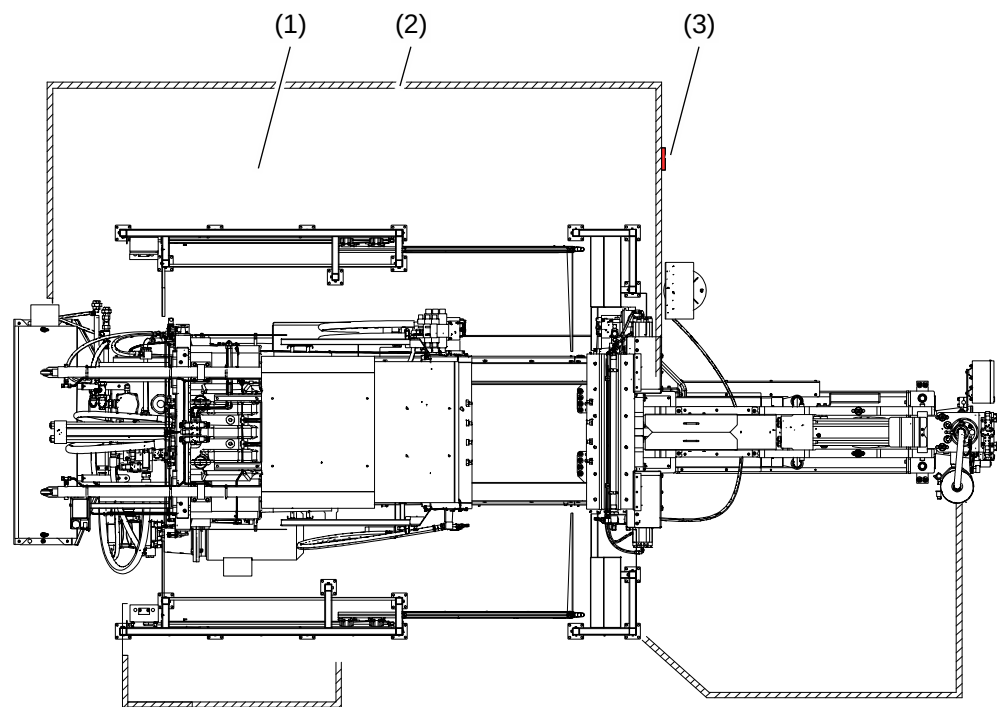
- (1) Key
(2) EMERGENCY STOP button

Additional EMERGENCY STOP buttons are located at the interlock switches of the protective fence

4.4.2

Emergency stop door switches

The emergency stop door switches bring the machine to a standstill if a door is opened after the machine is started.



- (1) Peripheral devices
- (2) Protective fence
- (3) Emergency stop door switches

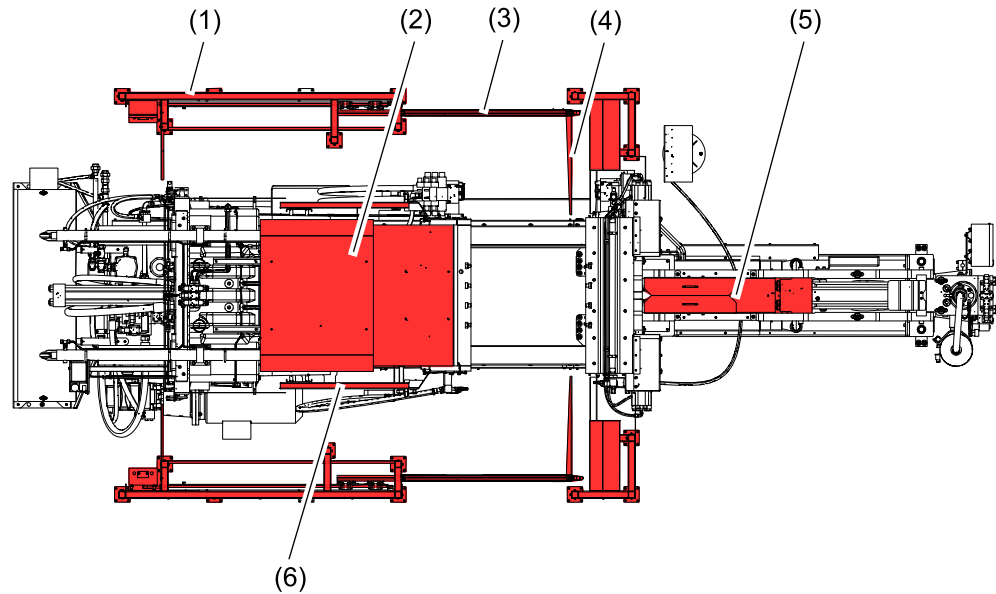
4.5

Protective devices

4.5.1

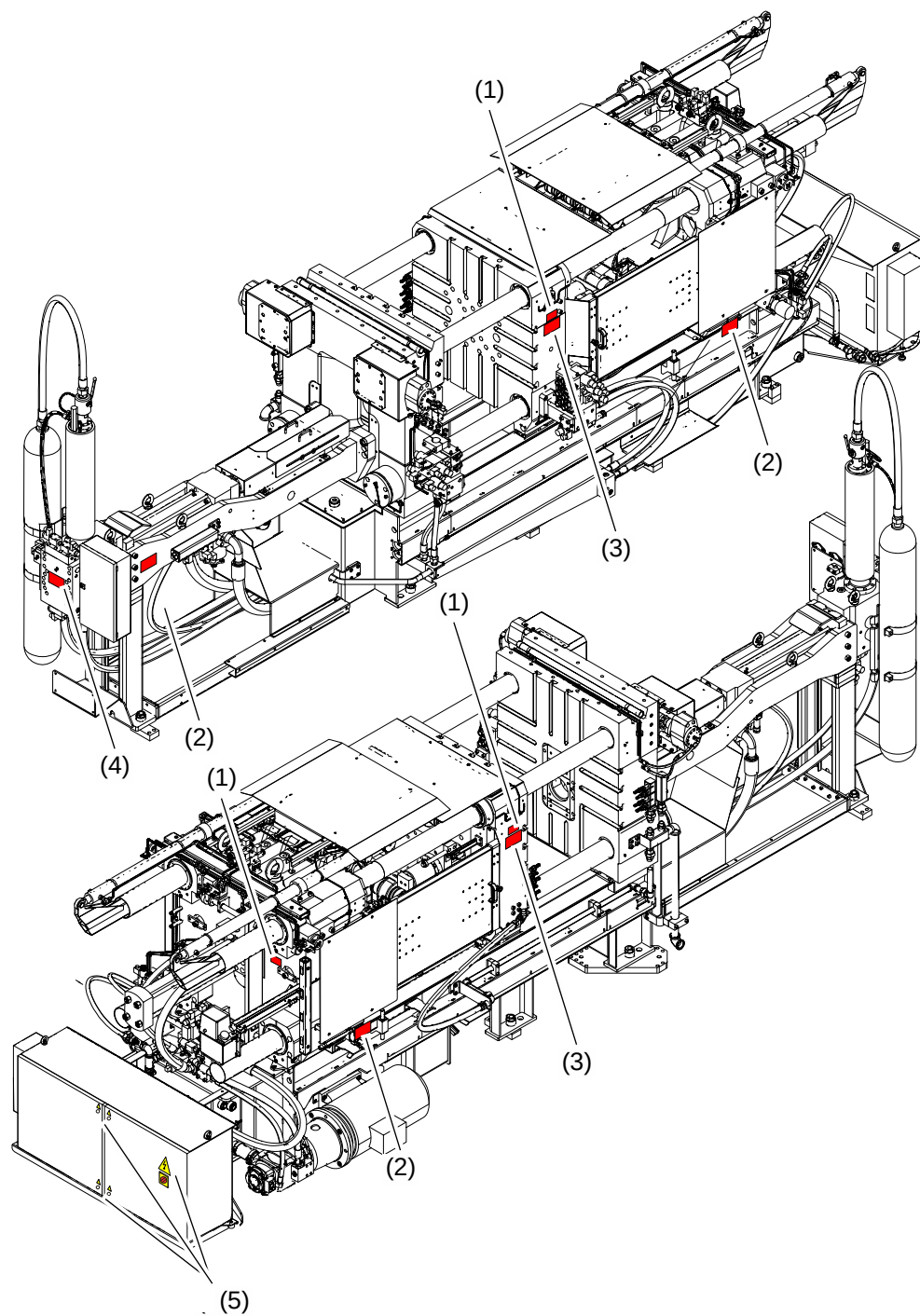
Separating safety devices

The safety devices protect against danger associated with operating the machine.



- | | |
|--------------------------------|--------------------------------------|
| (1) Protective enclosure | (4) Anti-trap device |
| (2) Guard panel, top | (5) Protective hood, plunger |
| (3) Automatic protective doors | (6) Protective fence, linkage system |

The safety symbols warn of dangers.

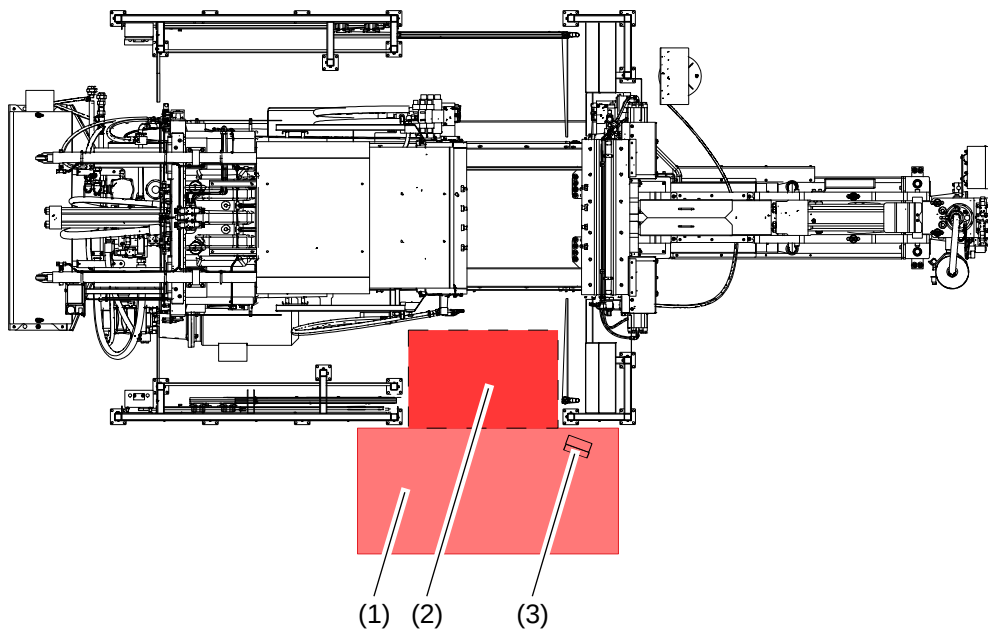


Item	Safety symbol	Meaning
(1)		For tie bars with manual tie bar pull: Only for manual tie bar pulling
(2)	CAUTION! Prior to displacing or transporting the machine discharge the nitrogen systems and drain the reservoirs of the hydraulic fluid. Always lift the machine according to the hoisting instructions.	
(3)	CAUTION! During repair, maintenance and set-up tasks in the danger zone of moving machine parts, SWITCH OFF DRIVE MOTOR	
(4)	CAUTION, Pressure accumulator system! The system must be depressurized on the oil side before starting any repair task.	
(5)		Electric shock warning

4.7

Workstations

The machine is operated by one person.

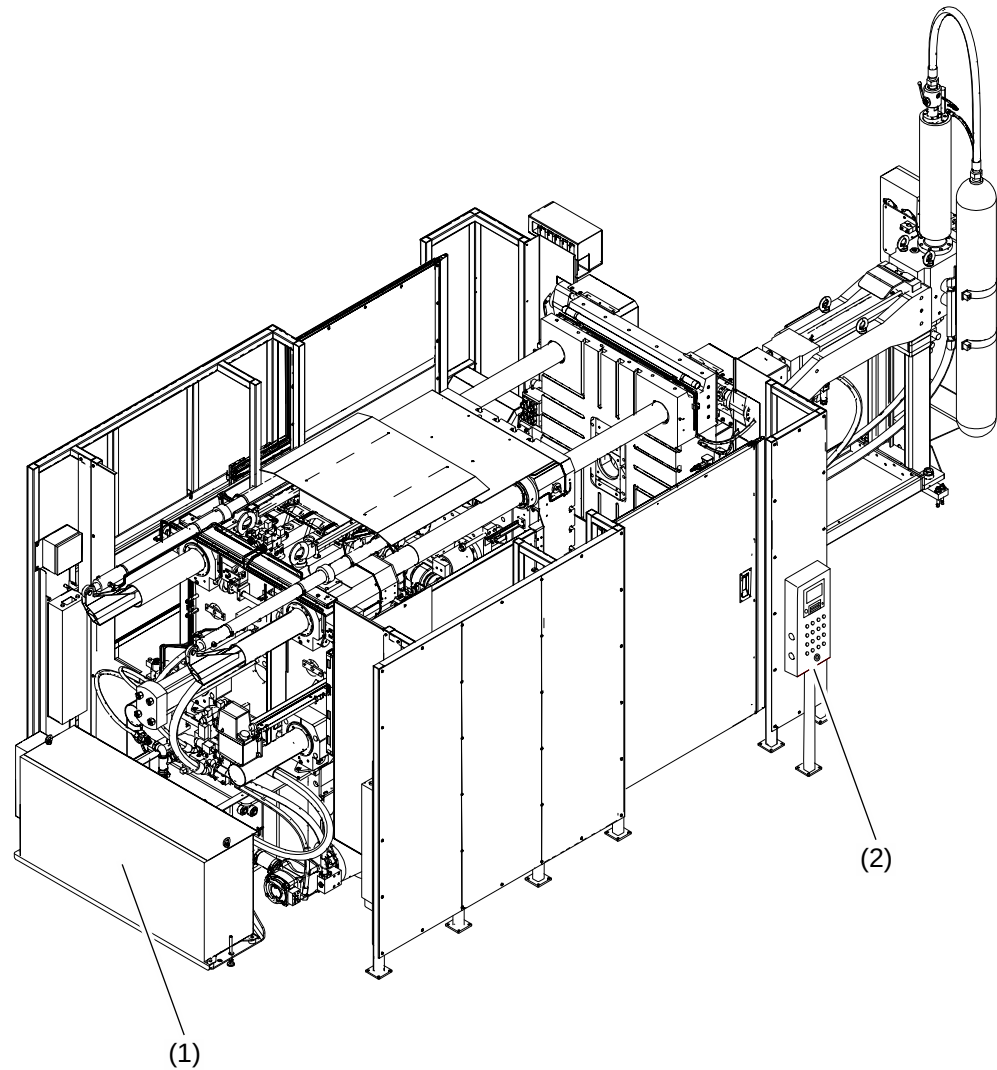


- (1) Operator workstation
- (2) Additional operator workstation: For manual operations with the protective door open
- (3) Control column

4.8

Operating and display elements

The operating elements and display elements control the machine.

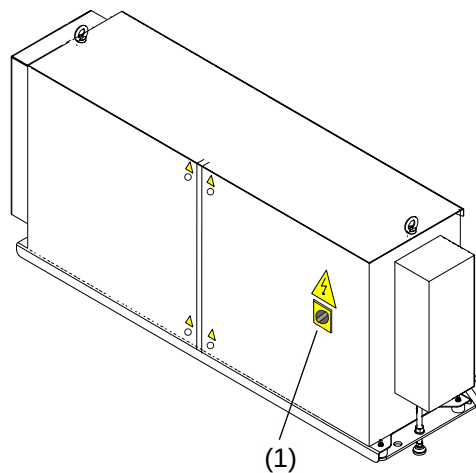


- (1) Control cabinet
- (2) Control column

4.8.1

Master switch

The master switch with which the machine is switched on and off is on the control cabinet.

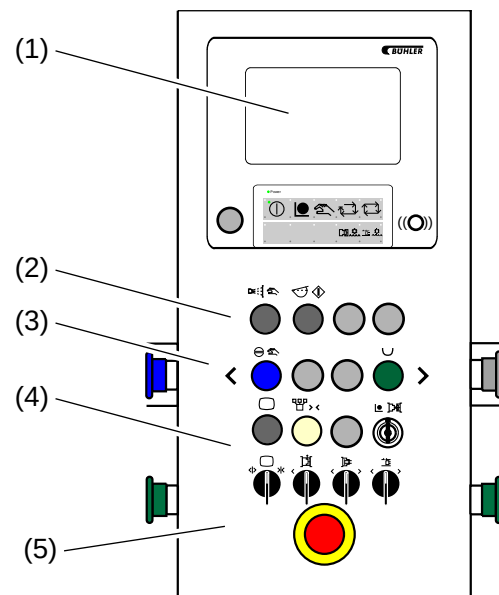


(1) Master switch

4.8.2

Control column

The control column controls the machine and the peripheral devices.



(1) Operating panel

(2) Spraying and dosing

(3) Manual operation

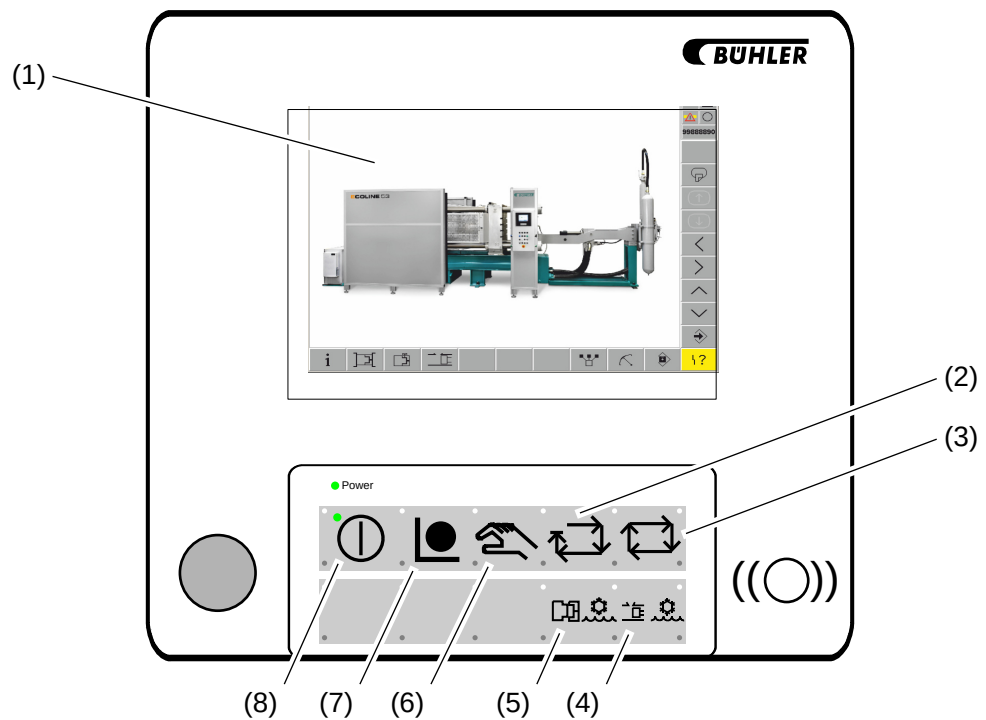
(4) Movement switch for set-up mode and manual mode

(5) Cycle start, emergency stop

With the control column the machine is started and stopped.

Operating panel

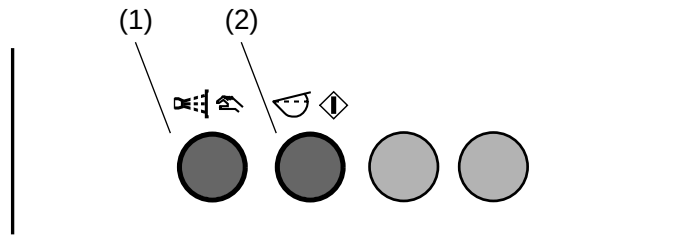
The machine is programmed and adjusted via the operating panel. Use the keys to activate a mode for the machine.



Item	Name	Function
(1)	Touch screen	Enables entries in the machine controller.
(2)	Semi-automatic mode (ES52/EH8)	Activates semi-automatic mode.
(3)	Automatic mode (ES53/EH9)	Activates automatic mode.
(4)	Cool injection set (ES190/EH190)	Activates and deactivates the cooling system for the casting outfit.
(5)	Cool die (ES191/EH191)	Activates and deactivates the cooling system for the die.
(6)	Manual mode (ES51/EH7)	Activates manual mode.
(7)	Set-up mode (ES50/EH6)	Activates set-up mode.
(8)	Drive motor On/Off (ES1/EH10)	Switches the drive motor on and off. When the drive motor is switched on, the LED is illuminated.

Spraying and dosing

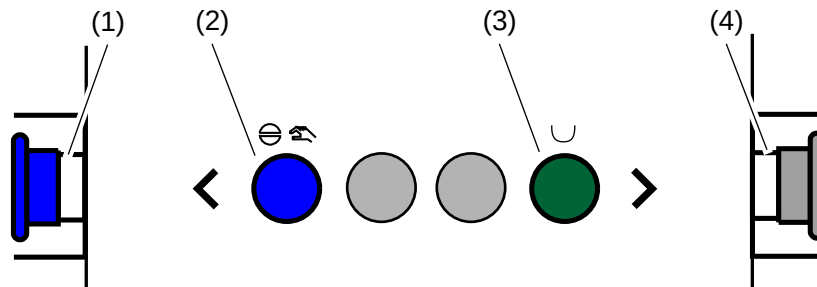
The spray device can be operated with the buttons.



Item	Name	Function
(1)	Manual spraying (ES13)	Sets manual spraying as a step in the machine cycle.
(2)	Start metal ladling (ES88)	Starts metal ladling in manual mode.

Manual operation

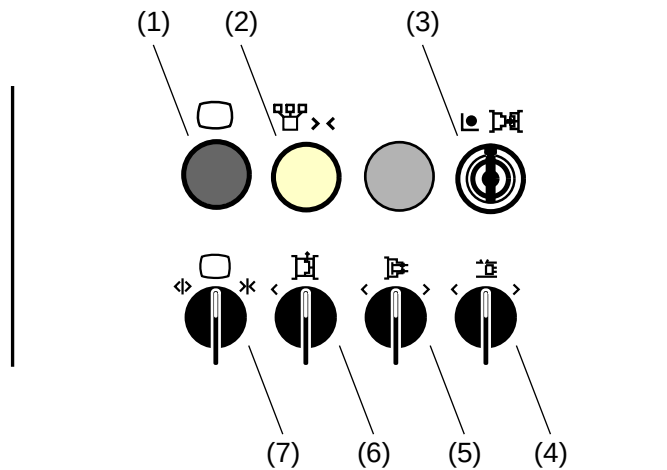
The buttons are used to request, acknowledge and execute manual operations. The illuminated pushbuttons signal alarms.



Item	Name	Function
(1)	Acknowledge manual operation (ES63)	Acknowledges a manual operation. Thereafter the machine cycle will be continued.
(2)	Manual operation (EH12)	Flashes, when a manual operation is required.
(3)	Ready for casting (EH3)	Is illuminated when the machine is ready for the casting process.
(4)	Move plunger forward (ES12)	Moves the plunger forward. The mushroom button is only active in the "ready for casting" status.

Movements in set-up mode and manual mode

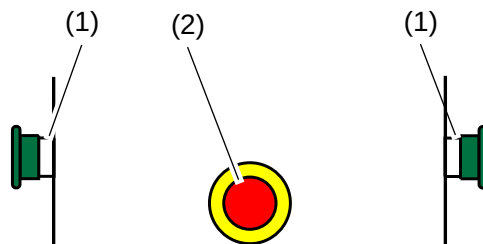
Use the buttons and switches to move machine elements in set-up mode and manual mode.



Item	Name	Function
(1)	Preselect set-up movement (ES90)	Displays the page "Die height, tie bars, core movement". Movements can be pre-selected on the page.
(2)	System home position (ES60)	Moves the system into home position. The key is illuminated until home position is reached. (In set-up mode only the machine and the die are moved into home position.)
(3)	Close die in set-up mode (ES94)	Disables or activates closing of the die in set-up mode.
(4)	Move plunger (ES11)	Moves the plunger.
(5)	Move ejectors (ES10)	Moves the ejectors.
(6)	Move die, core pulls (ES6)	Opens the die and extends the core pulls.
(7)	Run the preselected movement (ES91)	Runs the movement that was selected on the touch screen.

Cycle start, emergency stop

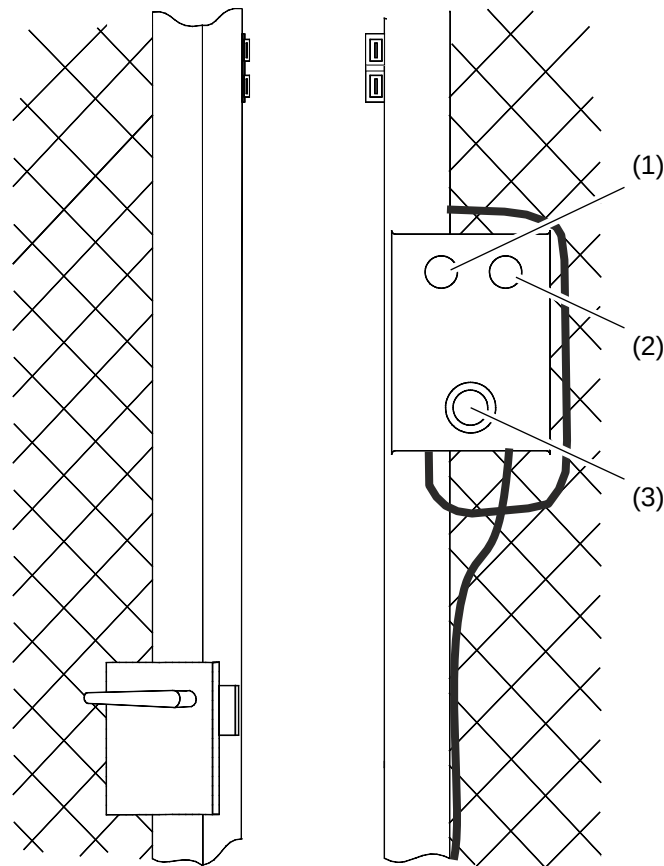
The buttons start the machine cycle. The emergency stop is used as protection in dangerous situations during operation.



Item	Name	Function
(1)	Two-hand start buttons (ES4/ES5)	Start the machine cycle or movements of the machine.
(2)	Emergency stop (ES2)	Stops the machine movement without delay.

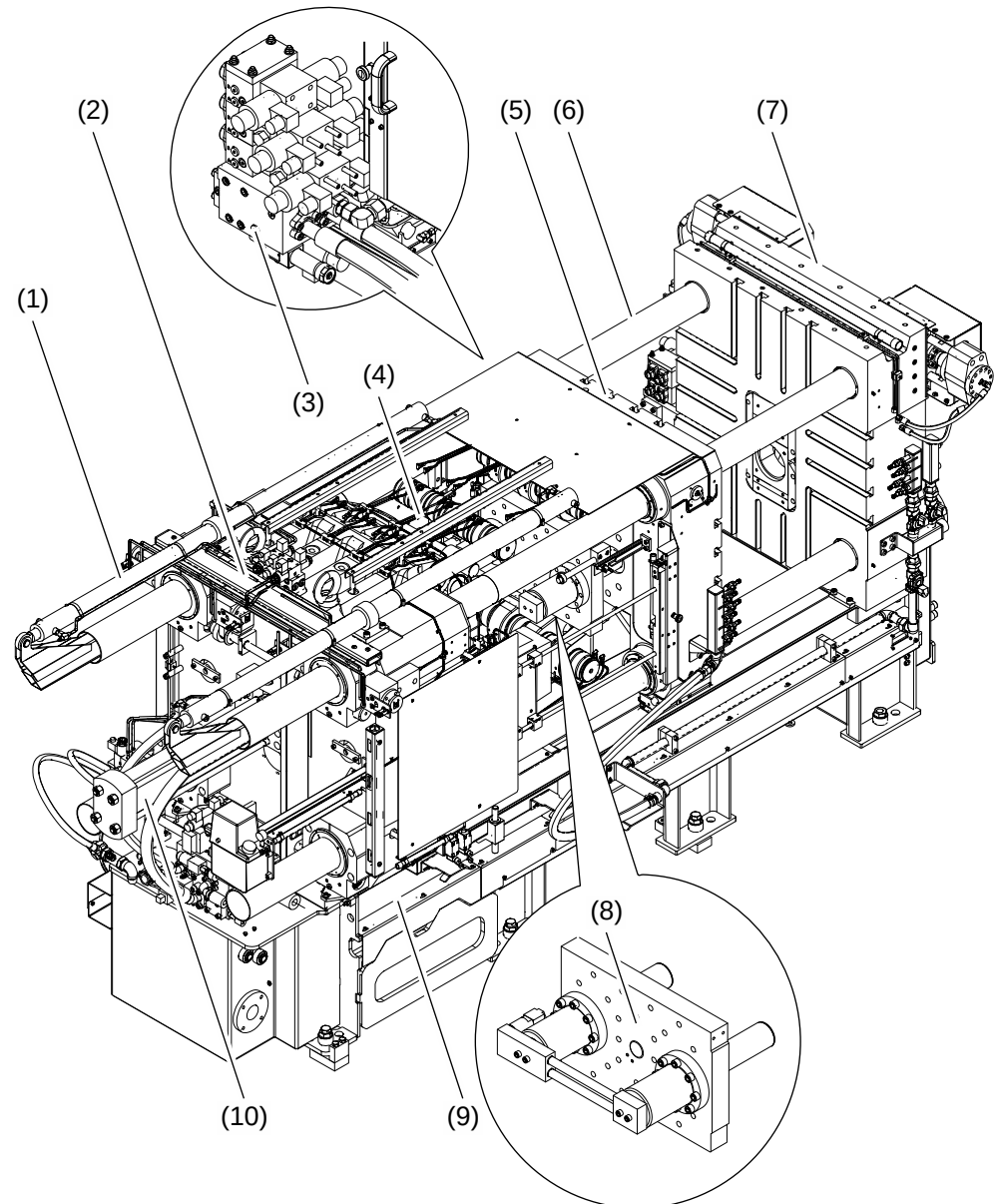
4.8.3 Safety interlock of the protective fence

The doors in the protective fence are locked and unlocked with the buttons of the operator panel. A spare tripping device is present for a de-energized state.



Item	Name	Function
(1)	Open door	Unlocks the door.
(2)	Lock door (blue)	When the button is illuminated: The door locks.
(3)	EMERGENCY STOP	Stops the machine movement without delay.

The locking unit closes and opens the die.



- | | |
|---------------------|-----------------------|
| (1) Tie bar pull | (6) Tie bar |
| (2) Cylinder plate | (7) Fixed plate |
| (3) Core pull units | (8) Ejector |
| (4) Linkage system | (9) Machine base |
| (5) Moving plate | (10) Closing cylinder |

The locking unit is a hydraulic 3-plate construction with a cylinder plate (2), a moving plate (5) and a fixed plate (7). The die is mounted on the fixing plates.

The locking unit closes the die in two phases:

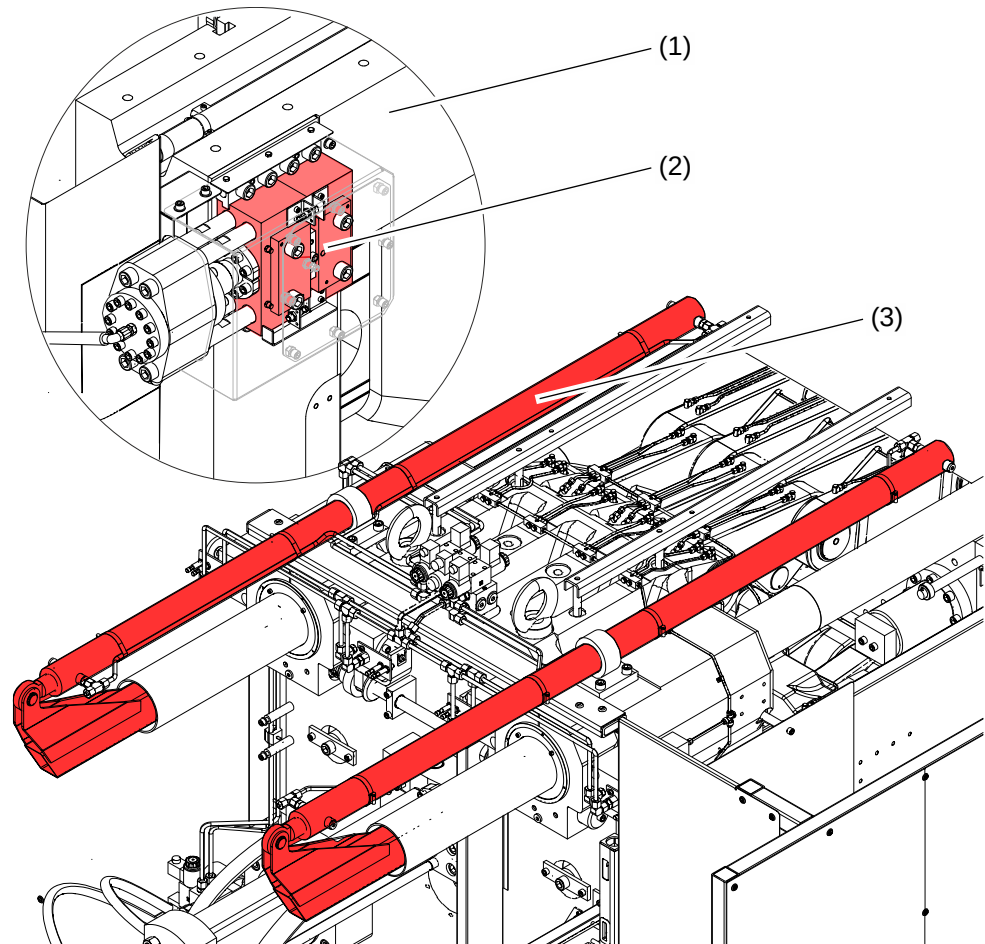
1. The closing cylinder (10) moves the moving plate (5) that is guided by the tie bars (6) and closes the die.
2. The linkage system (4) generates the die locking force with which the die is kept closed.

The locking unit opens the die in the reverse sequence. When the die is open, the ejector (8) pushes the cast part out of the die half of the moving plate (5).

4.9.1

Tie bar pull

To facilitate the change of the die, the upper tie bars are retracted and extended automatically with the tie bar pull.



- (1) Stationary fixing plate
- (2) Split nut
- (3) Tie bar pull cylinders

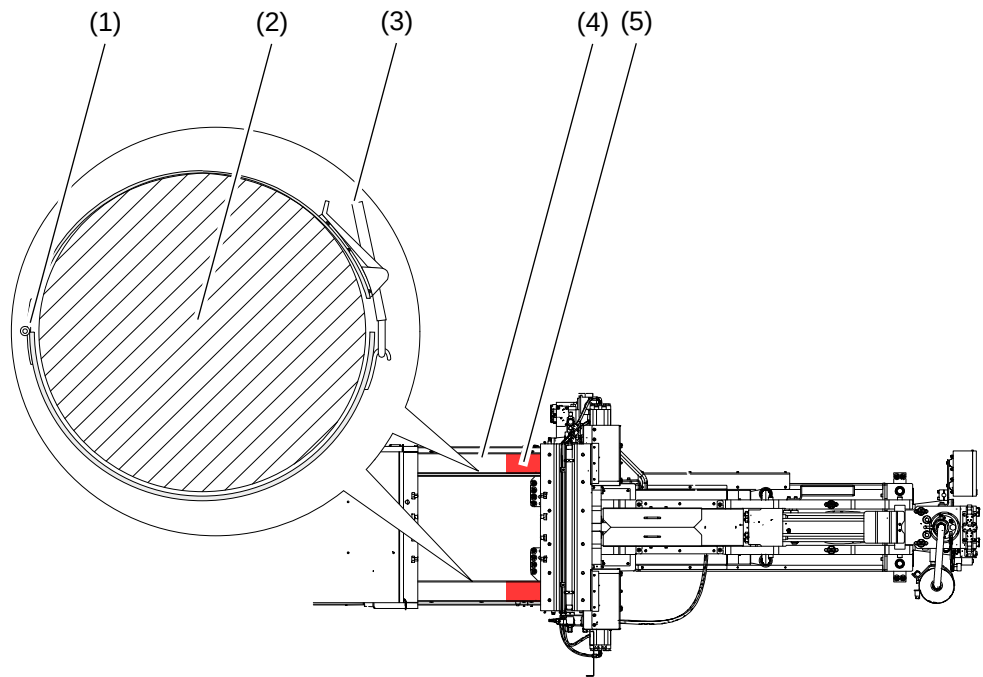
For automatic tie bar pulling, the split nuts (2) on the fixed plate (1) are opened and thus the tie bars are detached. The tie bar pull cylinders (3), pull the tie bars completely out of the die area. All movements are locked electrically. Proximity switches monitor the end position.

4.9.2

Tie bar protection

The tie bar protection has the following function:

- Protection against metal splashes in the area of the parting surfaces of the die
- Protection against damage
- Protection against excessive heat when the die is heating

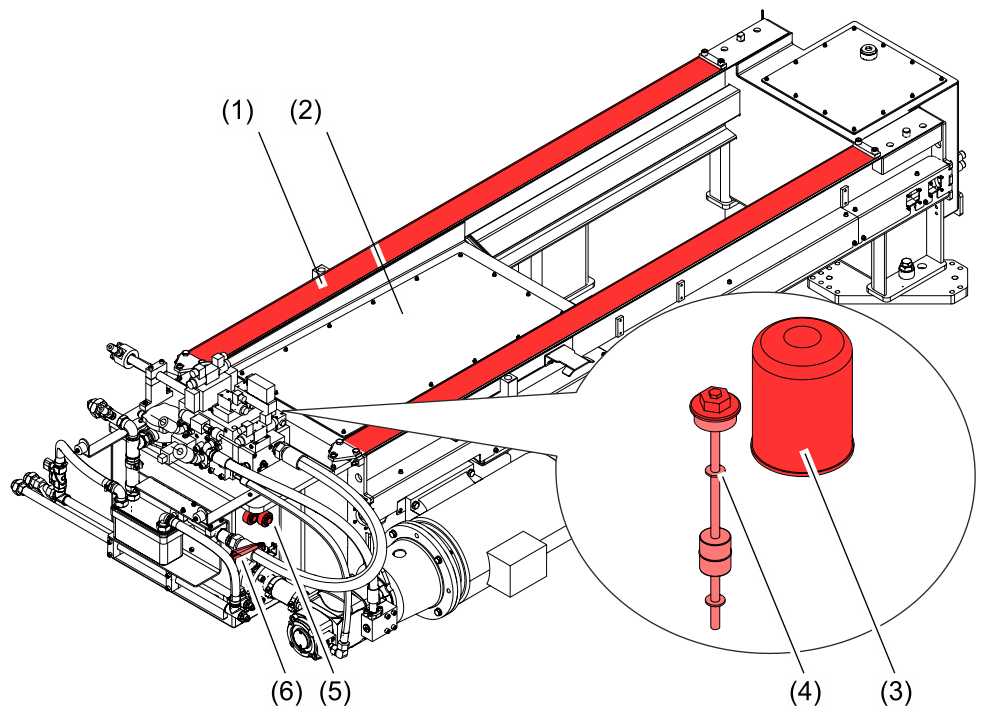


- | | |
|-------------|--------------------|
| (1) Hinge | (4) Tie bar |
| (2) Tie bar | (5) Protected area |
| (3) Closure | |

4.9.3

Machine base

The machine base distributes the static load of the locking unit. The hydraulic tank is integrated in the machine base.



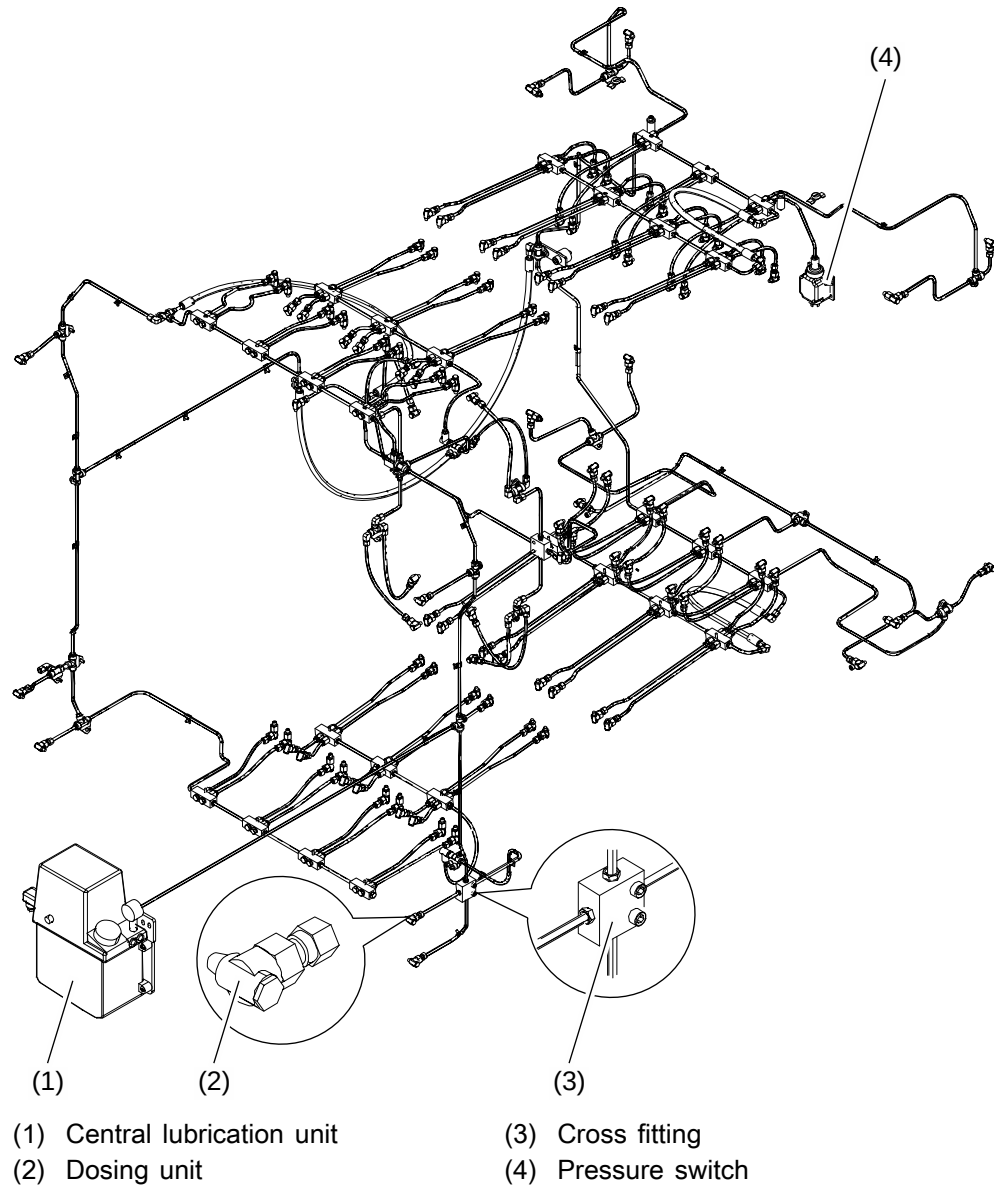
- | | |
|--------------------|--|
| (1) Slide rails | (4) Temperature and filling level sensor |
| (2) Hydraulic tank | (5) Oil sight gauges |
| (3) Tank aeration | (6) Ball valve |

The fixed plate is bolted to the machine base. The moving plate rests on the wear bands (1). The temperature and the filling level of the hydraulic fluid in the hydraulic tank (2) are monitored. The filling level can also be seen on the oil sight gauge (5). Air breather (3) ensures pressure equalization. The hydraulic tank is connected to the pumps via various hydraulic lines. The hydraulic tank is isolated from the system via the ball valves (6). Proximity switches are used to monitor the ball valves (6).

4.9.4

Central lubrication system

The central lubrication system supplies all slide surfaces and bearing bushings with lubricant at regular intervals.



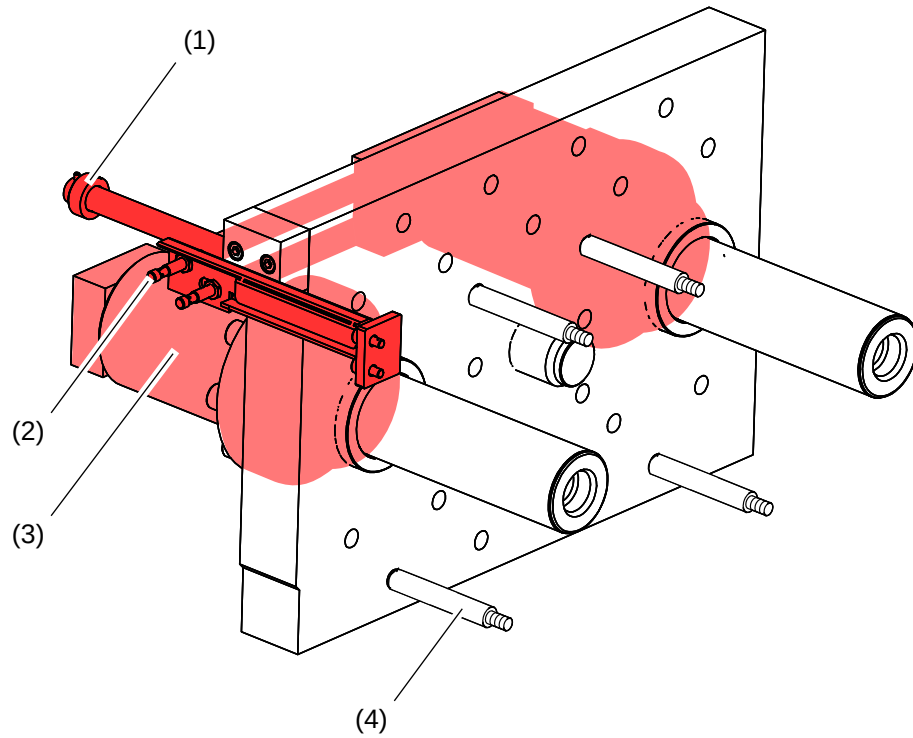
The central lubrication system is a consumption-based lubrication system. The lubricating oil is pumped to the dosing units (2) by the central lubrication unit (1) via a line system with cross fittings (3) and distribution strips. Dosing units (2) dose the lubricant directly at the lubrication points.

A pressure switch (4) monitors the function of the central lubrication unit (1).

4.9.5

Ejector

The ejector pushes the casting out of the open die half of the moving plate.



(1) Cam

(2) Proximity switch

(3) Hydraulic cylinder

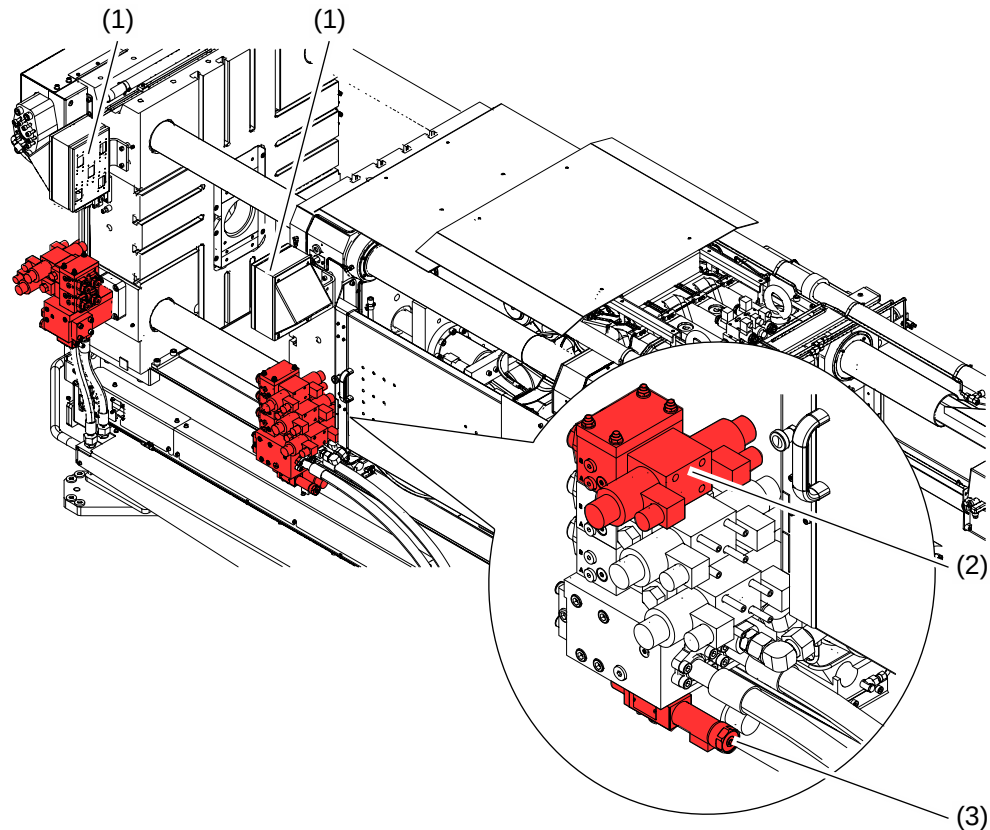
(4) Ejector rod

Two hydraulic cylinders (3) generate the force of the ejector. The ejector rods (4) transmit the motion to the ejector pins of the die. Limit switches (2) monitor the end position of the ejector.

4.9.6

Core pull units

The core pulls of the die are moved via the core pull units.



- (1) Power outlet box
- (2) Core pull units
- (3) Central pressure regulator

The core pulls can work with a pressure that is lower than the operating pressure of the machine. A central pressure regulator (3) regulates the pressure for all core pull units (2) on a die-mounting platen.

Limit switches that are connected on the power outlet box (1) are provided for monitoring of the core pulls.

Options for core pull units

- Pressure regulator: For operation with reduced operating pressure.
- Check valve: To keep the core pull in position.
- Core pull pressure relief: To discharge the pressure in the hydraulic lines.

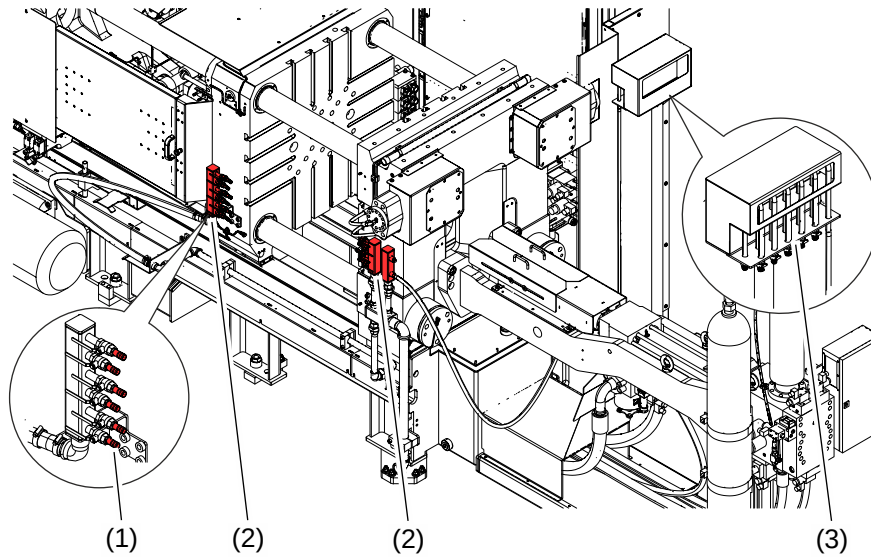
4.9.7

Cooling - die and injection set

The cooling system cools the die and injection set with water.

The cooling function is divided into three independent cooling systems, each is equipped with a solenoid valve.

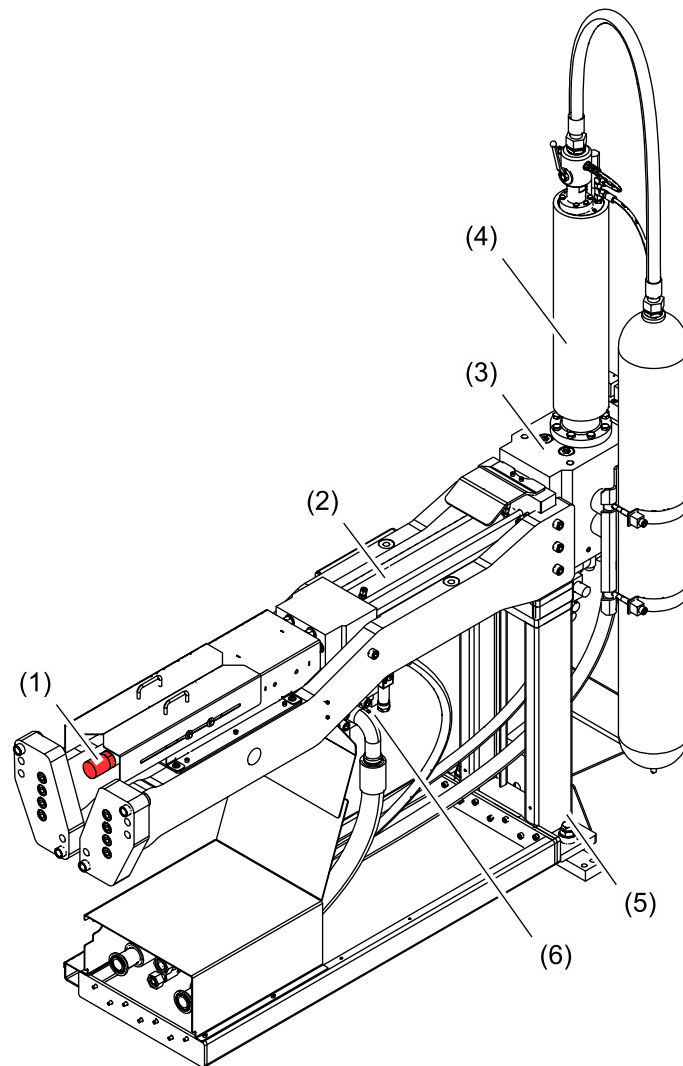
- Cooling - die, moving plate
- Cooling - die, fixed plate
- Cooling - injection set



- (1) Quick-release coupling
- (2) Cooling water connection
- (3) Water collector

Each cooling system consists of multiple cooling circuits. The cooling water connections (2) on the die-mounting platens are fitted with quick-release couplings (1). The cooling capacity of the cooling circuits is adjusted on the ball valves of the cooling water connections (2). The water that flows through the cooling circuit is collected at the water collector (3) and routed to the drain.

The injection unit presses the liquid metal out of the shot sleeve and into the die with the plunger.



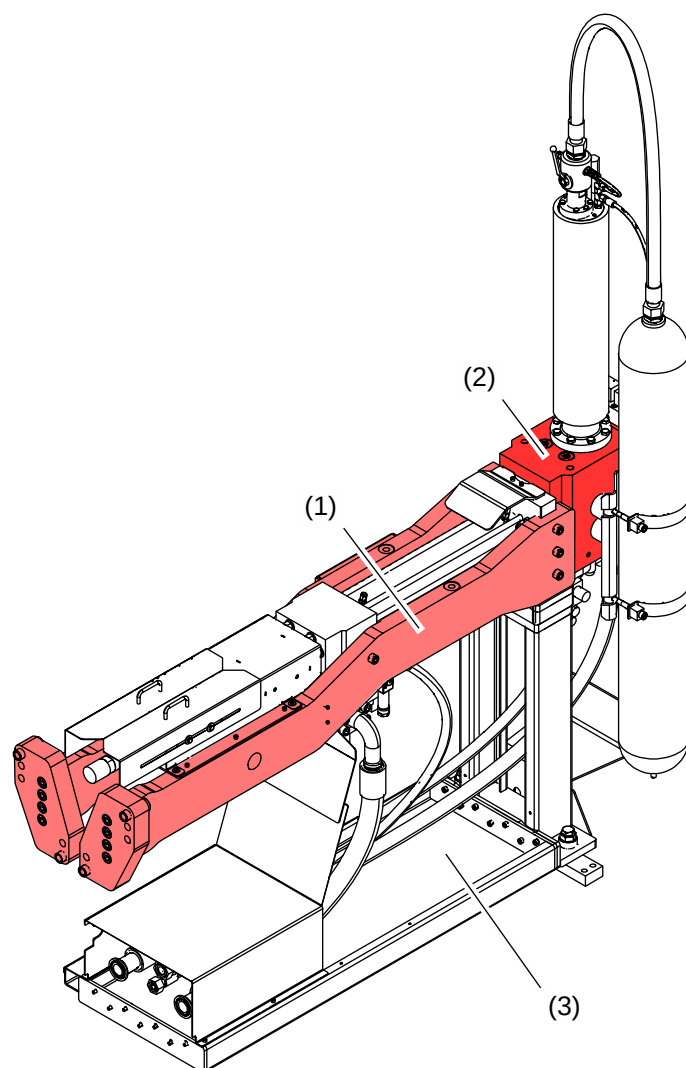
- | | |
|------------------------|------------------------|
| (1) Plunger | (4) Piston accumulator |
| (2) Injection cylinder | (5) Collection trough |
| (3) Control block | (6) Valve block |

The stored energy in the piston accumulator (4) moves the plunger (1) via the injection cylinder (2). The valves of the valve block (6) control the metal pressure and the injection velocity. In the filling phase the piston (1) injects the liquid metal into the die.

4.10.1

Connecting elements

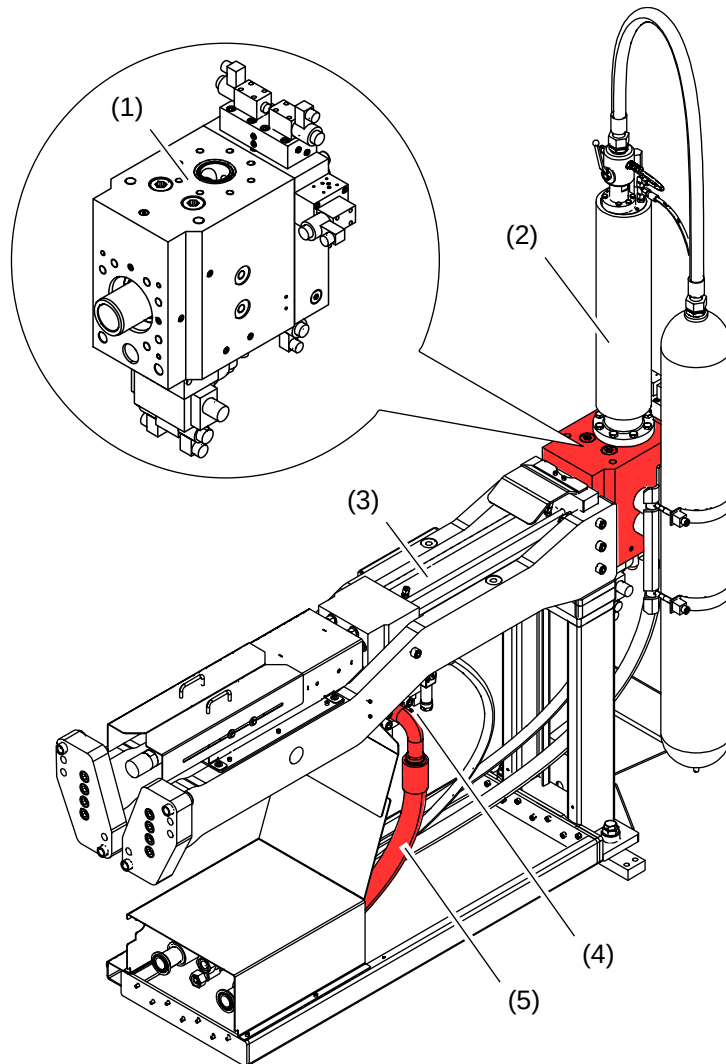
The connecting elements connect the injection unit with the locking unit.



- (1) Connecting element
- (2) Control block
- (3) Collection trough

The control block (2) absorbs the injection force via the connecting elements (1).

The injection cylinder generates the force with which the liquid metal is injected into the die. The valves of the valve block regulate the speed of the plunger in the filling phase and the metal pressure during the pressure intensification phase.



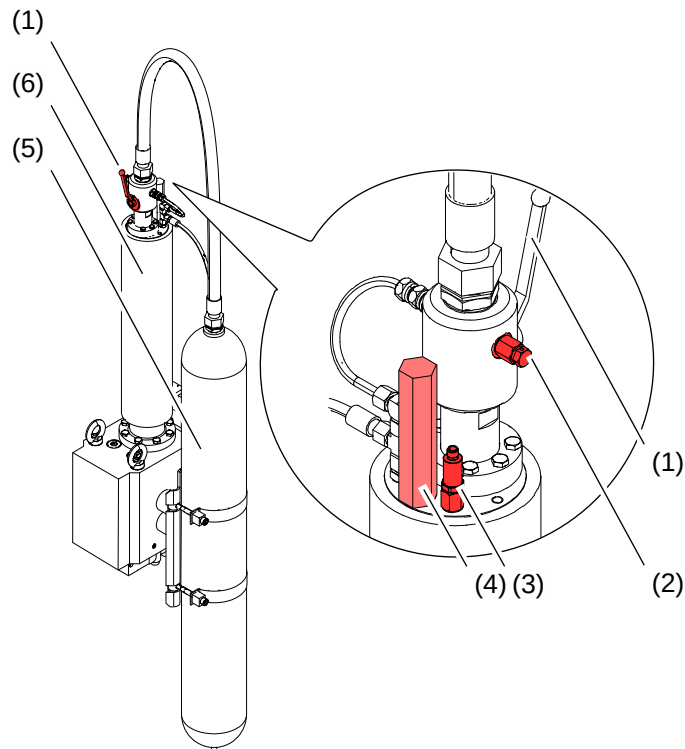
- | | |
|------------------------|--------------------|
| (1) Control block | (4) Valve block |
| (2) Piston accumulator | (5) Hydraulic hose |
| (3) Injection cylinder | |

The valves of the control block (1) control the injection unit. The hydraulic fluid is routed from the piston accumulator (2) to the primary side of the injection cylinder (3) via the control block (1). The valves of the valve block (4) meter the flow out of the secondary side of the injection cylinder. The hydraulic fluid flows back to the hydraulic tank via the hydraulic hose (5). Thus the speed of the plunger and the intensification pressure are controlled.

4.10.3

Piston accumulator

The piston accumulator stores pressure. The stored pressure is given off to the injection cylinder during the filling phase.

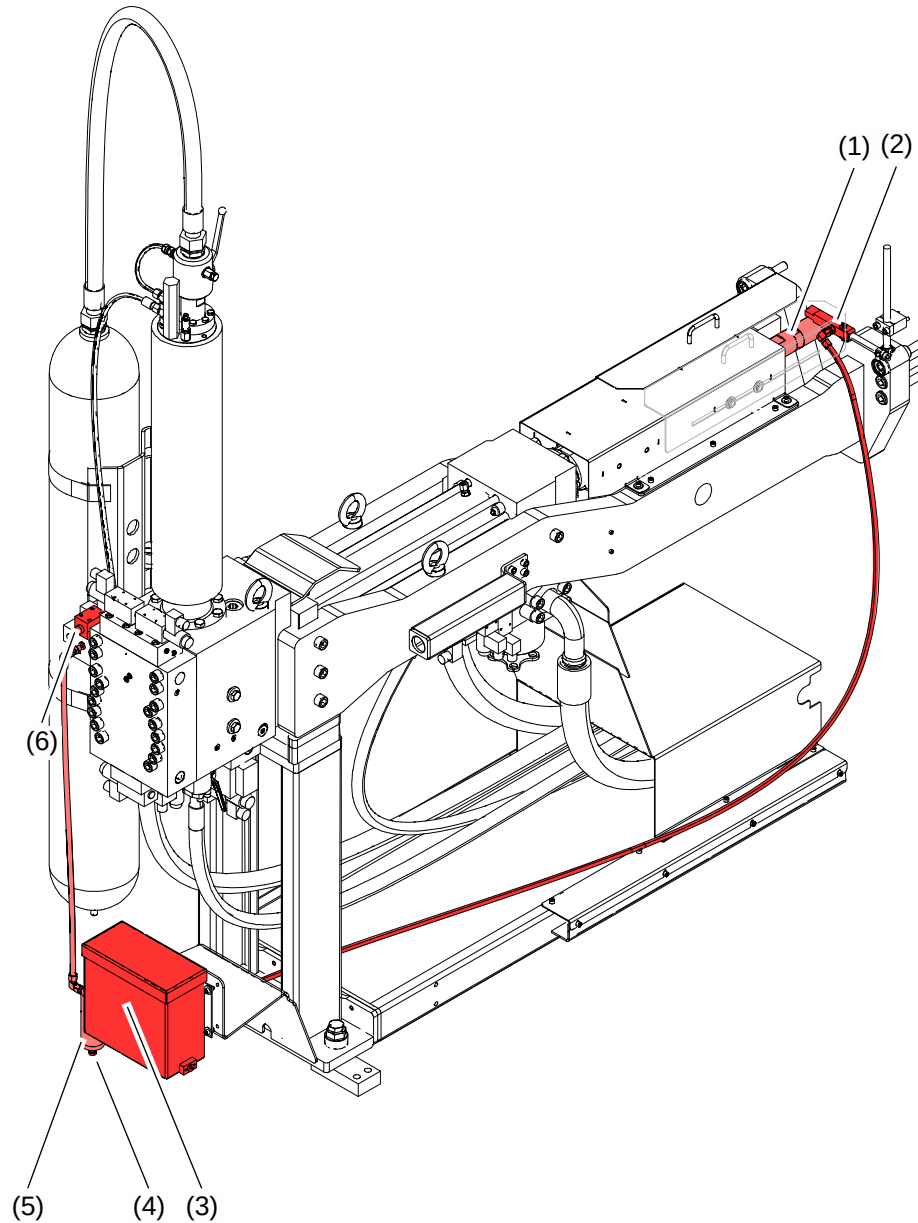


- | | |
|---------------------|----------------------------|
| (1) Ball valve | (4) Limit position control |
| (2) Rupture disk | (5) Nitrogen bottle. |
| (3) Pressure sensor | (6) Piston accumulator |

A piston accumulator (6), in which nitrogen is stored, is used as pressure accumulator. When the piston accumulator is charged the limit position control (4) is activated. A pressure sensor (3) measures the nitrogen pressure that is displayed on the operator panel.

The piston accumulator is connected to the nitrogen bottles (5) via a ball valve (6). A rupture disk (2) prevents excessive pressure in the nitrogen system.

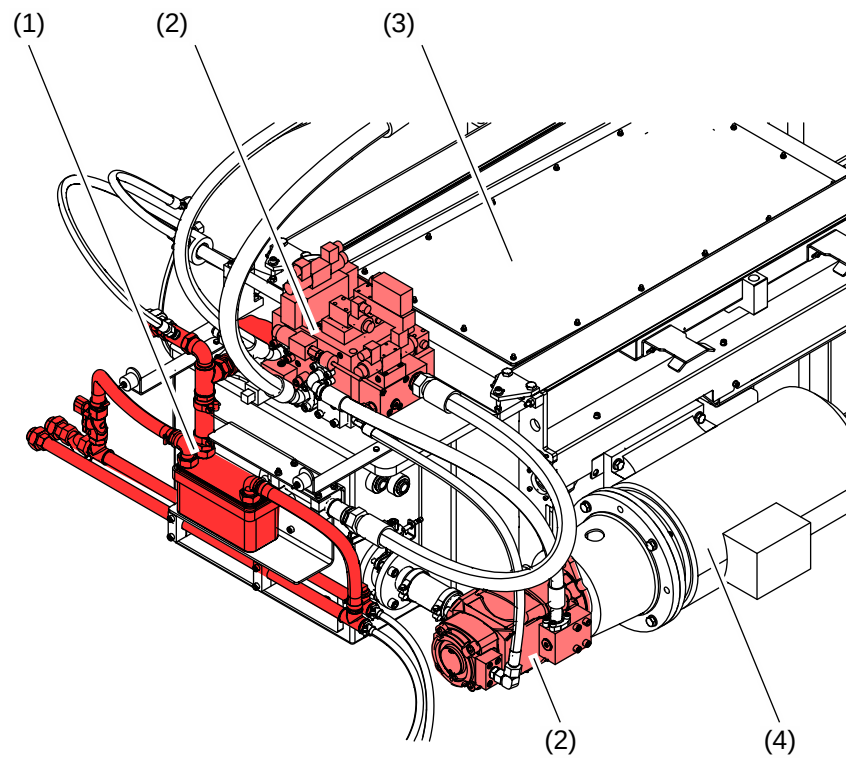
The plunger lubrication system provides the plunger with lubricant.



- | | |
|---------------------------|----------------------|
| (1) Plunger | (4) Adjusting screw |
| (2) Lubricant distributor | (5) Lubrication pump |
| (3) Lubricant tank | (6) Solenoid valve |

The plunger lubrication system is an electrohydraulically controlled once-through lubrication system. The solenoid valve (6) controls the lubrication pump (5). The lubrication pump (5) pumps lubricant out of the lubricant tank (3) to the lubricant distributor (2). The lubricant distributor (2) applies a protective film of lubricant on to the plunger (1). The number of lubrication pump strokes is specified on the operating panel. The adjusting screw (4) is used for dosing the lubricant.

The drive group supplies the hydraulic system of the machine with the required hydraulic fluid.



(1) Cooling and filter circuit of the hydraulic system

(2) Pump system

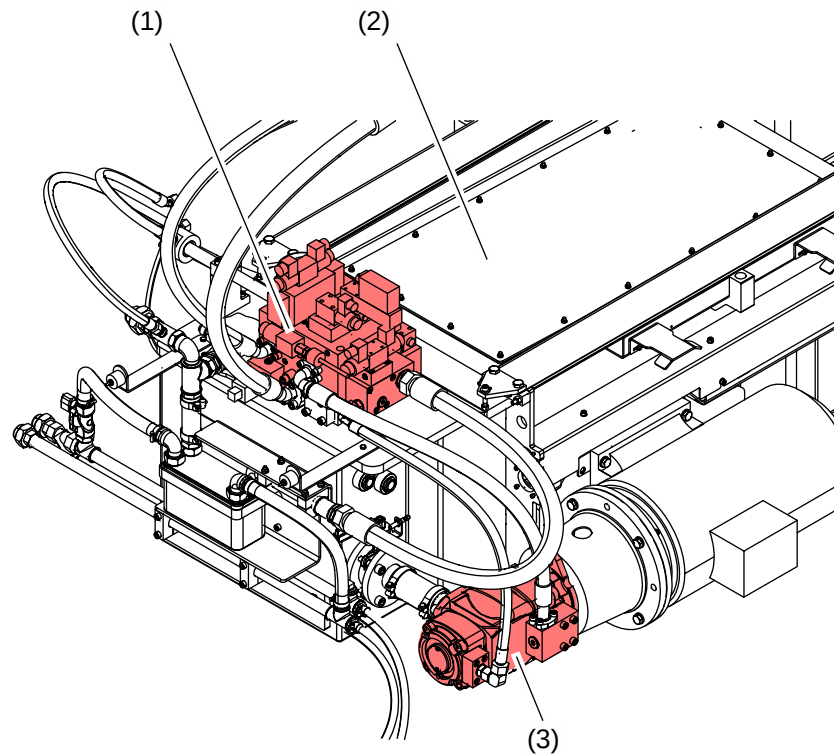
(3) Hydraulic tank

(4) Drive motor

4.11.1

Pump system

The fixed displacement pumps generate the pressure in the hydraulic system of the machine. The pump block limits the maximum operating pressure of the fixed displacement pumps and controls the hydraulics of the machine.



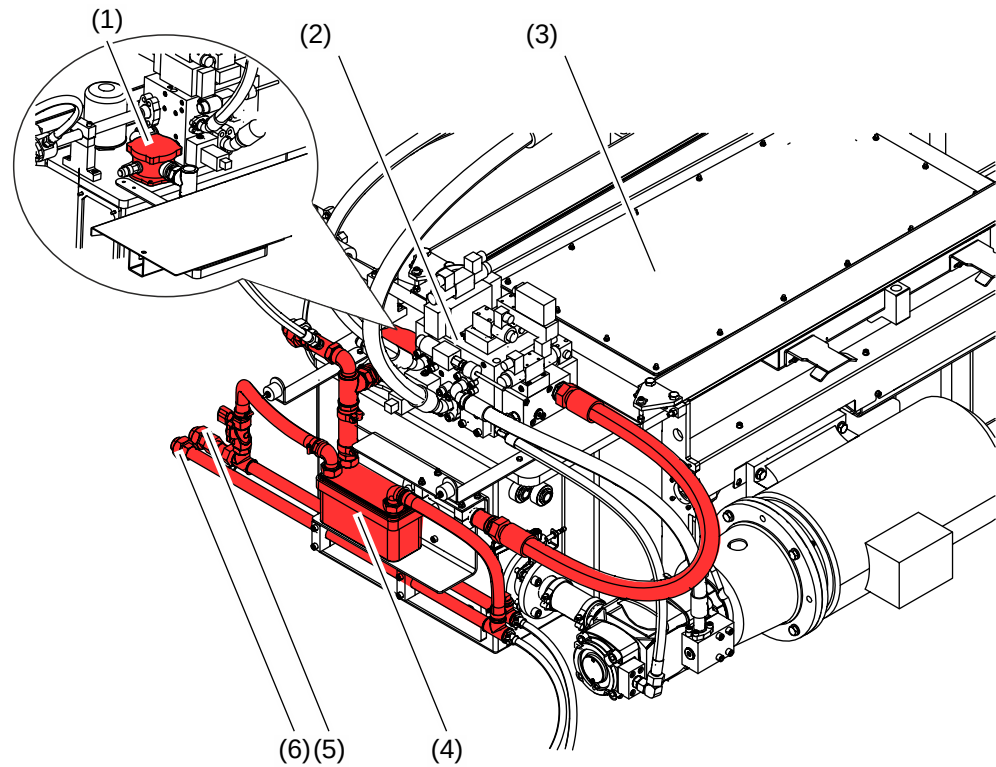
- (1) Pump block
- (2) Hydraulic tank
- (3) Fixed displacement pumps

The fixed displacement pumps (3), are powered by a drive motor and pump hydraulic fluid out of the hydraulic tank (2) to the pump block (1). The pump block (1) limits and regulates the operating pressure of the machine.

4.11.2

Cooling and filter circuit of the hydraulic system

Hydraulic fluid is filtered and cooled in the cooling and filter circuit.



- | | |
|--------------------|--------------------------|
| (1) Return filter | (4) Plate heat exchanger |
| (2) Pump block | (5) Water inlet |
| (3) Hydraulic tank | (6) Water return |

In the plate heat exchanger (4) the hydraulic fluid coming back out of the hydraulic system is cooled with water. The hydraulic fluid is cleaned by the return filter (1) and is routed back into the hydraulic tank (3).

5 Transport

5.1 Safety

5.1.1 Round slings

Load-bearing force reduction and tearing of the round slings.

- ▶ Select round slings and type of attachment based on the specified transport weight.
- ▶ Use round slings in accordance with EN 1492-2.
- ▶ Comply with the maximum angle of inclination of the round slings.
- ▶ Use edge protection on sharp corners.
- ▶ Ensure uniform load distribution.

5.1.2 Collision protection

Transport accidents due to insufficient stability of the vehicle superstructure.

- ▶ Up to a transport weight of 12.5 t, use transport vehicles in accordance with EN 12642 Code L.
- ▶ Up to a transport weight of 25.5 t, use transport vehicles in accordance with EN 12642 Code XL.
- ▶ For transport weight greater than 25.5 t, use special transport vehicles.
- ▶ Ensure positive fit for the load.

5.1.3 Lashing material

Transport accidents due to improper handling of the lashing material.

- ▶ Select lashing material based on the specified transport weight.
- ▶ Use lashing straps in accordance with EN 12195-2.
- ▶ Use lashing chains in accordance with EN 12195-3.
- ▶ Secure the load in accordance with VDI 2700 "Securing of loads on road vehicles".
- ▶ Uniformly tension chain tensioners.
- ▶ Check the tension of the lashing chains, prior to starting transport, and en route.

5.1.4 Shackles

Transport accidents due to improper handling of shackles.

- ▶ Select shackles based on the specified transport weight.

5.1.5

Transport crates

Transport accidents due to improper handling of transport crates.

- ▶ Note packaging symbols.
- ▶ Note the total weight specified on the transport crate.
- ▶ Only attach hoisting devices on the marked points.
- ▶ If using a forklift: introduce the forks parallel to the floor skids.

5.1.6

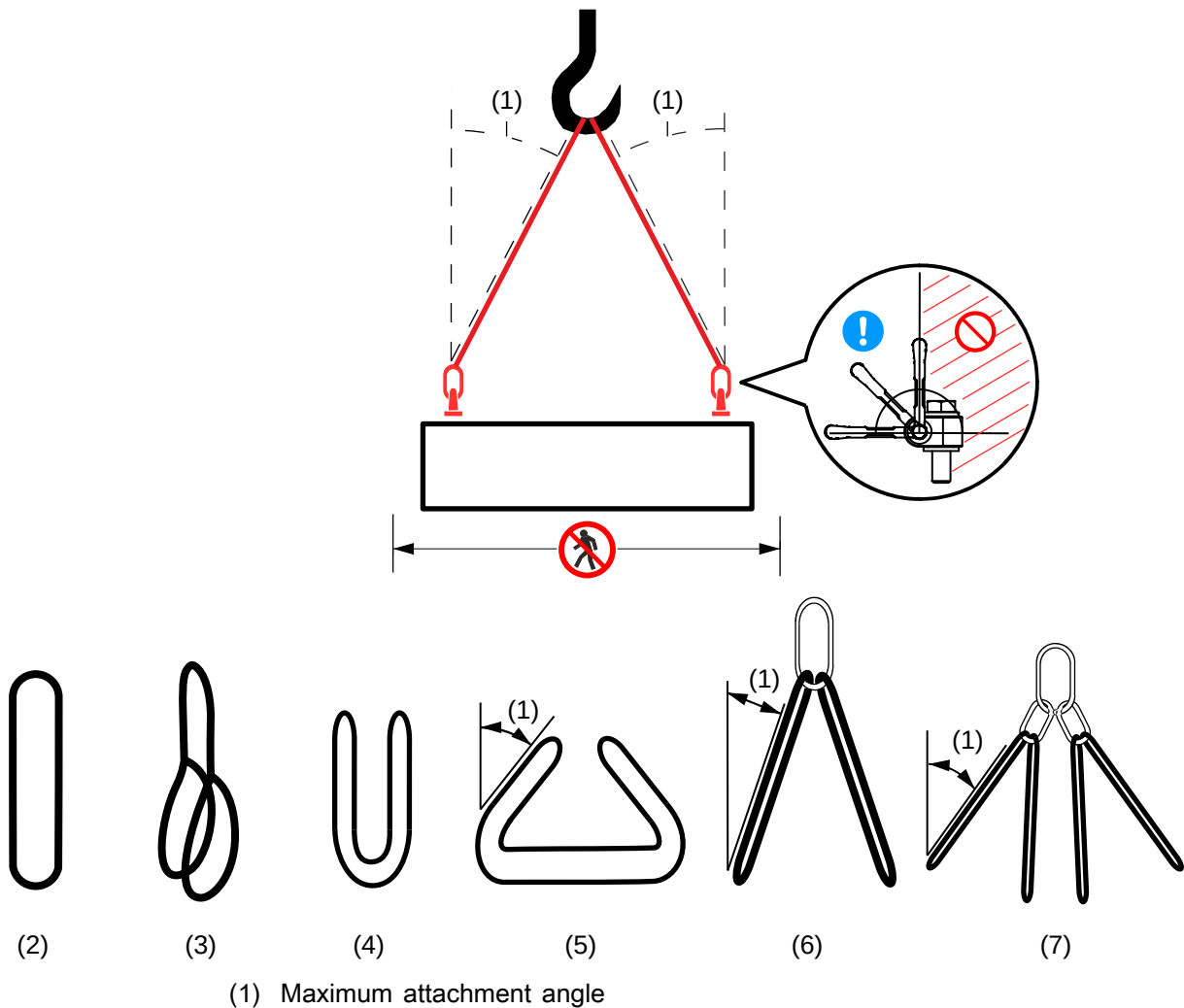
Transport by ship

Increased risk of damage with ship transport.

- ▶ Pack transport units in plastic foil.
- ▶ If possible, transport transport units in HPE standard transport crates.
- ▶ If this is not possible, transport transport unit on HPE standard transport bases.

5.2

Attachment factors of the round slings



(1) Maximum attachment angle

Item	Attachment type	Attachment factor for symmetric load
(2)	Direct	1
(3)	Securing the load	0.8
(4)	Wrapped around parallel	2

Item	Attachment type	Attachment factor for symmetric load
(5)	Wrapped around	Attachment angle 0° to 45°: 1.4 Attachment angle 45° to 60°: 1
(6)	Two-strand round sling	Attachment angle 0° to 45°: 1.4 Attachment angle 45° to 60°: 1
(7)	Three-strand and four-strand round sling	Attachment angle 0° to 45°: 2.1 Attachment angle 45° to 60°: 1.5

5.3 Checking delivery

- Check for completeness of delivery according to the delivery note.
- Report any missing parts or transport damage. See page 8, chapter "Contact".

5.4 Packaging symbols



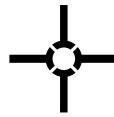
(1)

(1) Top



(2)

(2) Protect against moisture



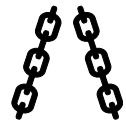
(3)

(3) Center of gravity



(4)

(4) Fragile



(5)

(5) Attach here

- Note packaging symbols.

5.5 Temporary storage

- Note packaging symbols.
- Leave the machine and machine parts in the original packaging until starting assembly work.
- Do not store the machine outdoors.
- Protect the machine from the effects of weather.
- Prevent temperature fluctuations.

5.6 Transport material (manufacturer)

For transports from the manufacturer's locations to the customer, the following transport materials are made available by the manufacturer:

- Bolts (eye bolts, load stands, transport bolts, etc.)
- Round slings (only for loading)
- Transport devices and frames (**as loan items**: They must be sent back, otherwise the customer will be billed for these items.)

5.7

Transport material (freight carrier)

The following transport materials must be provided by the freight carrier:

- Transport vehicle
 - Sufficient load-bearing capacity of the lashing points
 - Collision protection (requirements / use in accordance with weight and acceleration)
 - Loading area swept clean and undamaged
- Lashing material (requirements / use in accordance with weight and acceleration)
- Clean and undamaged non-slip mats (requirements / use in accordance with weight and acceleration)
- Clean and undamaged heavy-duty mats (requirements / use in accordance with weight and acceleration)
- Covering tarps
 - Adequately dimensioned (see transport dimensions)
 - Completely intact and watertight

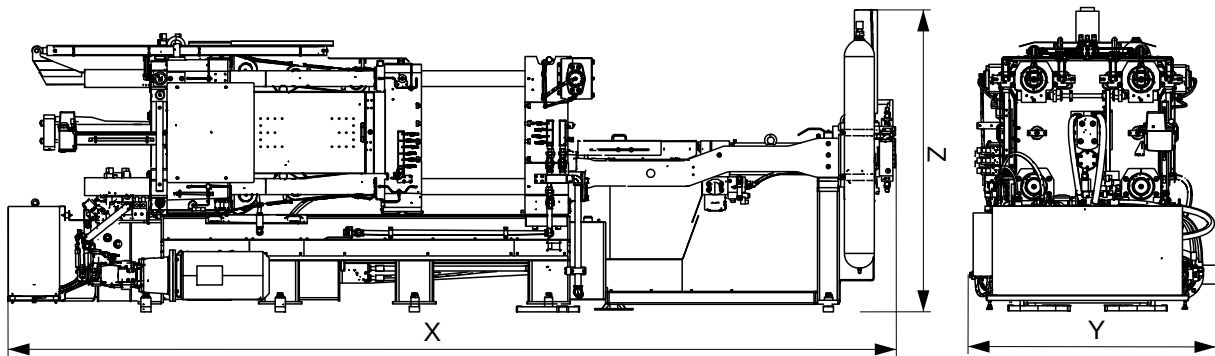
Important

Missing correct transport material must be purchased by the freight carrier on-site. Transport without correct transport material is not permitted.

- ▶ Provide and bring along correct transport material before starting the transport.
- ▶ Comply with the information on the transport units and the transport materials in this document.
- ▶ If purchasing transport materials on site: Ensure payment.

5.8

Transportmasse

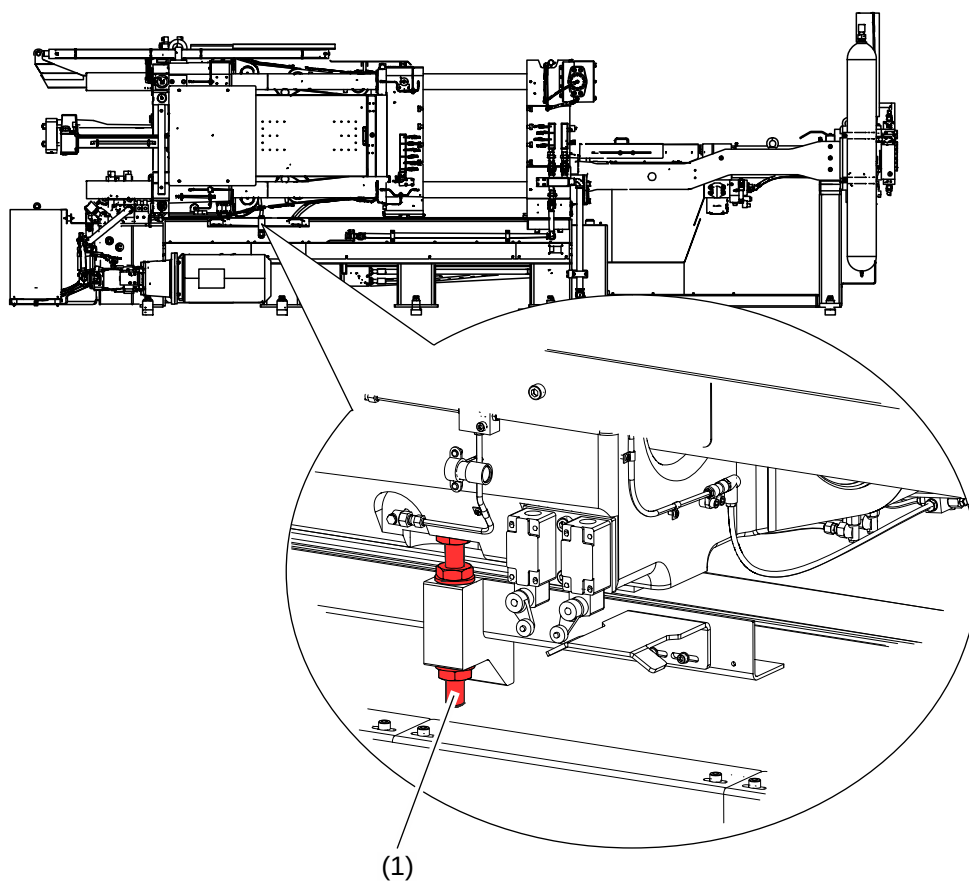


Description	Value	Unit
X	7 950	mm
Y	2 050	mm
Z	3 000	mm
Gewicht	32 000	kg

5.9

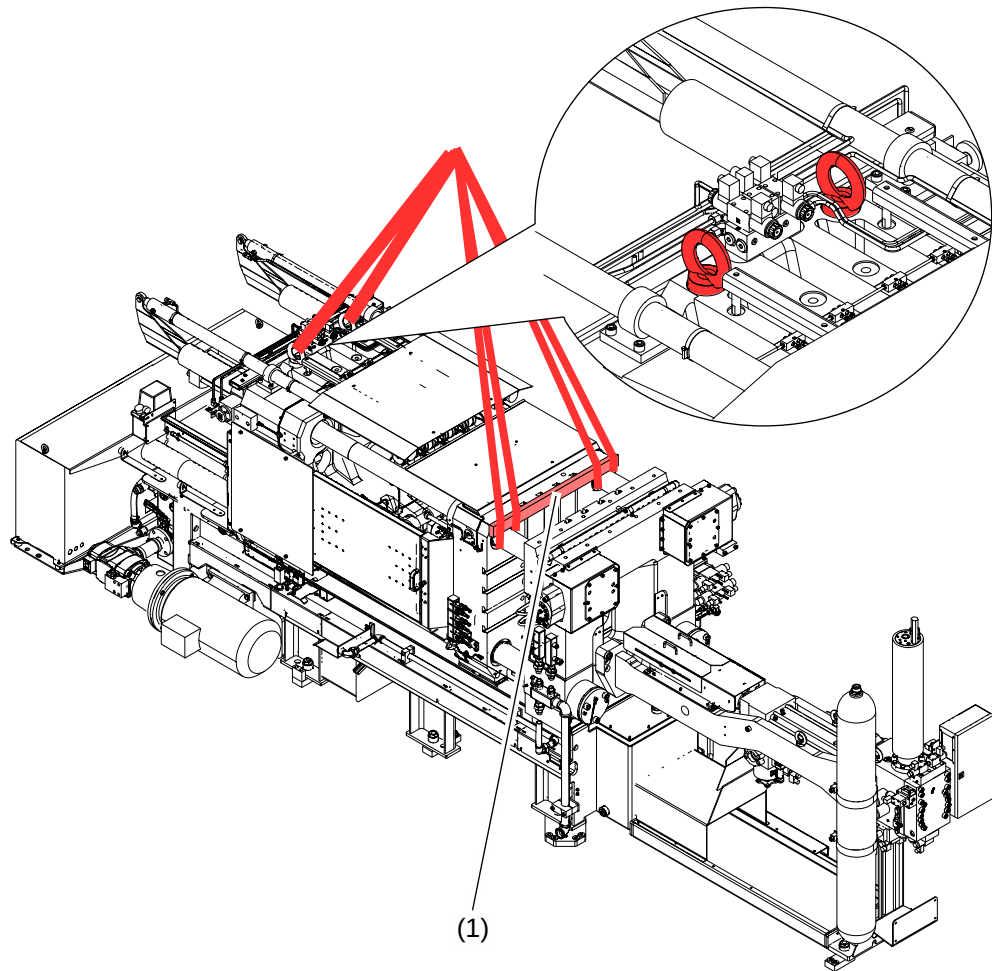
Lifting gear

Description	Value	Unit
Maximum attachment angle	30	°
2 x eyebolt, M56	7 800	kg



(1) Transport locking device

1. Empty hydraulic tanks.
2. Discharge the nitrogen in the pressure cylinders.
3. Install and tighten the transport safeguards (1).



(1) Edge protection

WARNING



Suspended load.

Serious injuries or death.

- ▶ Use only undamaged hoisting devices with a sufficient rating.
- ▶ Do not step or stand under suspended loads.
- ▶ Ensure that there is no one in the danger zone.

NOTICE

Lifting at an angle.

Damage to components.

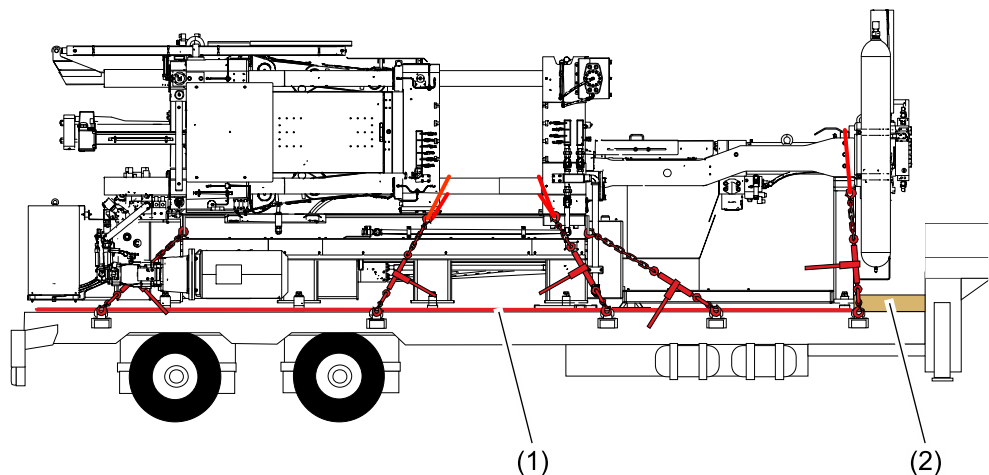
- ▶ Adjust hoisting device.
- ▶ Lift machine horizontally.

1. Attach hoisting devices on the specified attachment points.
2. Attach the edge protection (1).
3. Slowly lift the locking unit.

Freight carrier

Description	Value	Unit
9 x lashing chain with chain tensioner	10 000	daN
Load-bearing capacity per lashing point	10 000	daN
10 x heavy-duty mat	300 x 500	mm
Collision protection: 2 wood beams, adapted to the vehicle	100 x 120	mm
2 x cover tarp	6 x 10	m
2 x lashing strap with ratchet	2 500	daN
8 x lashing strap with ratchet	1 000	daN
Rubber pads for machine protection / placement under chains	5 x 150 x 750	mm
Netting		

Fastening the machine



- (1) Heavy duty mat
(2) Collision protection

! WARNING

Slipping of the transport goods.
Death due to crushing.

- ▶ Use adequately dimensioned and undamaged lashing material.
- ▶ Only fasten lashing material at the lashing points of the transport vehicle.
- ▶ Use sufficient heavy-duty mats so that there is no contact between load and loading surface.
- ▶ Only use clean and undamaged heavy-duty mats.
- ▶ Check the lashing, prior to starting transport, and en route.

1. Lay out the heavy-duty mat on the loading surface of the transport vehicle.
2. Place the transport goods on the transport vehicle.
3. Secure the transport goods with the transport material.
4. Cover the transport goods with a tarp.

6.1 Safety

6.1.1 Personnel qualifications

If improperly executed, personnel can be injured and machine parts can be damaged.

- ▶ Only have Bühler personnel perform installation and commissioning tasks.

6.1.2 Set-up location

In preparing the set-up location the following instructions must be complied with to ensure the greatest possible safety:

- ▶ Keep escape routes free.
- ▶ Maintain the safety distance in accordance with local regulations.
- ▶ Provide sufficient space for peripheral devices, pallets, etc.
- ▶ Ensure accessibility for die change, tie bar pull, service and maintenance.
- ▶ Provide a hoisting device with sufficient load-bearing capacity for die change, service and maintenance.
- ▶ Effectively illuminate and ventilate the workstation of the die-casting machine.
- ▶ Provide a collection trough for hydraulic fluid in accordance with the foundation diagram.
- ▶ Extract gases and fumes that occur when spraying the die. Install a proper extraction system.

6.1.3 Danger of corrosion

Corrosion of machine parts can occur due to the wrong cooling water.

- ▶ Only use cooling water with low sulfate, nitrate and chloride content. See page 19, chapter "Cooling water".
- ▶ If the quality of the cooling water is insufficient, use a water treatment system.

6.2 Preparing the set-up location

- ▶ Prepare for the installation in accordance with the checklist. See page 138, chapter "Checklists: Preparation for installation".

6.2.1 Symbols on the foundation diagram

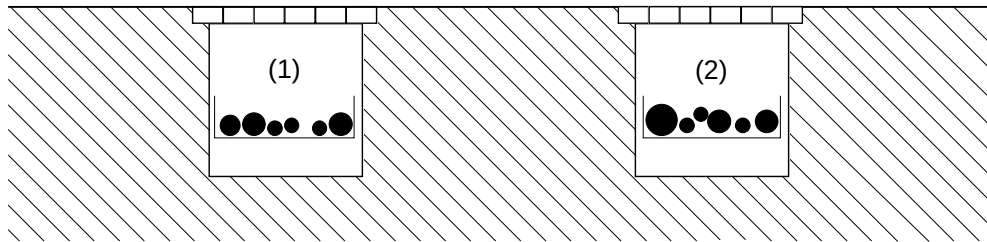
Symbol	Description
	Electrical connection
H ₂ O	Water connection

kN	Floor load
	Zero point

6.2.2 Preparing the foundation

1. Prepare the foundation as specified in the reinforcement drawing.
2. Comply with the curing time specified by the concrete manufacturer.
3. Prepare the energy and water connections in accordance with the foundation diagram.

6.2.3 Preparing the cable ducts



- (1) Control cables of the peripheral devices
 - (2) Connection cable of the energy supply
1. Adequately dimension the cable ducts.
 2. Lay out the control cables of the peripheral devices (1) and connection cable of the energy supply (2) in a separate cable duct.

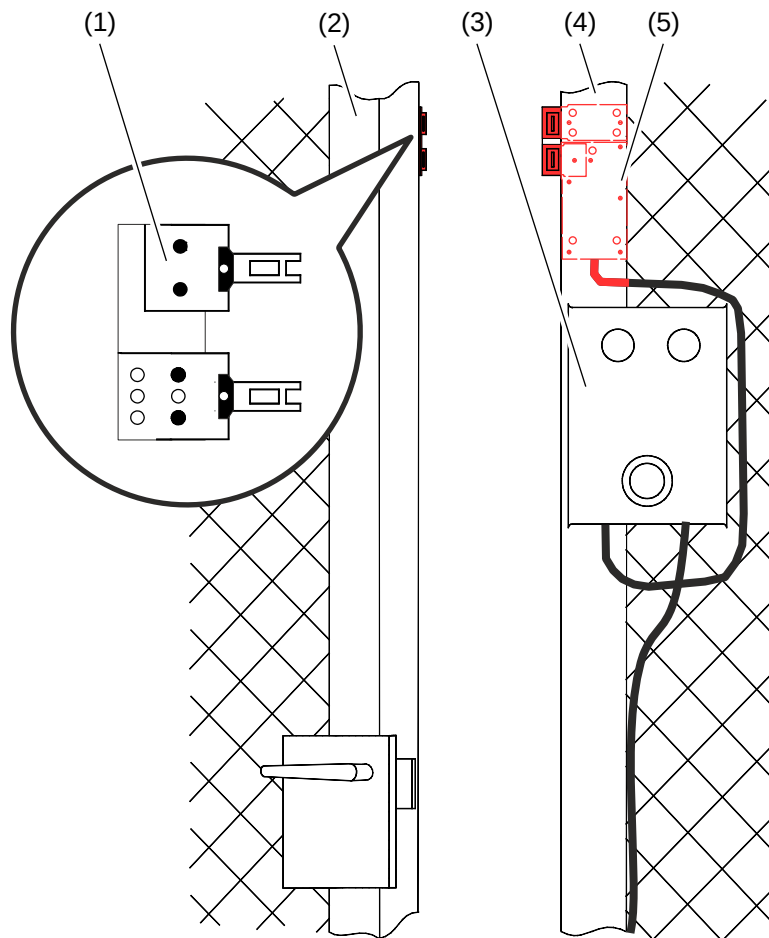
6.3 Installing a protective fence

6.3.1 Selecting a protective fence

1. Select a protective fence that corresponds to the local applicable laws and regulations. Particularly:
 - ▶ Select a protective fence that prevents the danger zones from being reached from the outside.
 - ▶ Select a fence height at which it is not possible to reach over or under the protective fence.
 - ▶ Select a fence grate that makes it impossible to reach through.
2. Provide doors for danger zones that must be entered.

Safeguarding the doors of the protective fence

All doors in the protective fence must be safeguarded with a safety switch that is locked via the controller.



- (1) Activator
- (2) Door
- (3) Operator panel

- (4) Door post
- (5) Safety switch

1. Mount the activator (1) on the door (2).
2. Mount the head of the safety switch (5) in such a manner that the opening for the activator (1) rests on the inside of the protective fence. Use single-use screws for this.
 - When the door (2) is closed the safety switch (5) is activated.
3. Attach the operator panel (3) in the vicinity of the door on the outside of the protective fence.
4. Execute the wiring in accordance with the wiring diagram.
 - ▶ Connect the safety switch (5) on the operator panel (3).
 - ▶ Connect the operator panel (3) on the controller.

7 Commissioning

7.1 Safety

7.1.1 Personnel qualifications

If improperly executed, personnel can be injured and machine parts can be damaged.

- Only have Bühler personnel perform installation and commissioning tasks.

7.1.2 Nitrogen

Explosions can occur if using impure nitrogen.

- Only use nitrogen gas of purity class ≥ 4.5 , $N_2 = 99.99\%$ by volume.

7.2 Preparing for commissioning

- Prepare for commissioning in accordance with the checklist. See page 139, chapter "Checklists: Preparation for commissioning".

7.3 Checking protective devices

No.	Check	✓
1	Each EMERGENCY STOP button on the machine stops all drives relevant to safety.	
2	The machine does not start automatically after unlocking the EMERGENCY STOP button. The machine must be restarted.	
3	Opening the doors of the protective fence is only possible when the signal from the die-casting machine is present. When closing the doors the die-casting machine does not start automatically.	
4	When the doors of the protective fence are open the die-casting machine is in set-up mode.	
5	Core pulls and other movable die parts on which the operator can be injured are safeguarded by the owner.	
6	All safety devices, protective covers, and cladding elements are intact and effective.	
7	All controls and warning systems are functional.	

7.4

Connecting the power supply

Important

The cables for connecting the power supply are not included in the scope of delivery of the die casting machine.

- Provide the cables.

1. Connect the power supply.
 - Comply with locally applicable laws and regulations.
 - Ground the connection.

7.5

Connecting the water supply

NOTICE

Insufficient water quality.

Corrosion.

- Install a water treatment device in the cooling system.
- For cooling water systems from other manufacturers: Install a central filter.

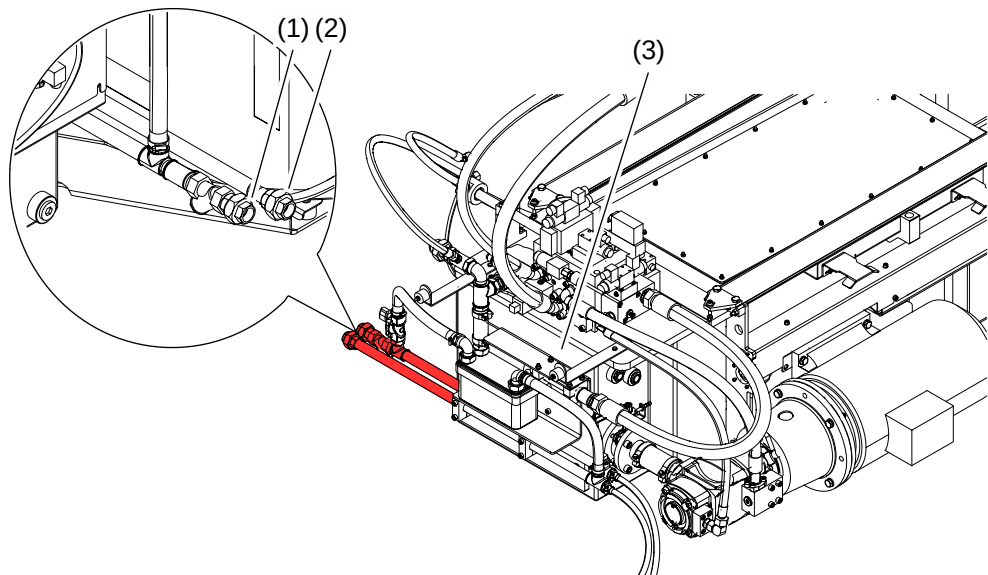
Important

The hoses for connecting the water supply are not included in the scope of delivery of the die casting machine.

- Provide the hoses.

7.5.1

Connecting the cooling circuit of the hydraulic system

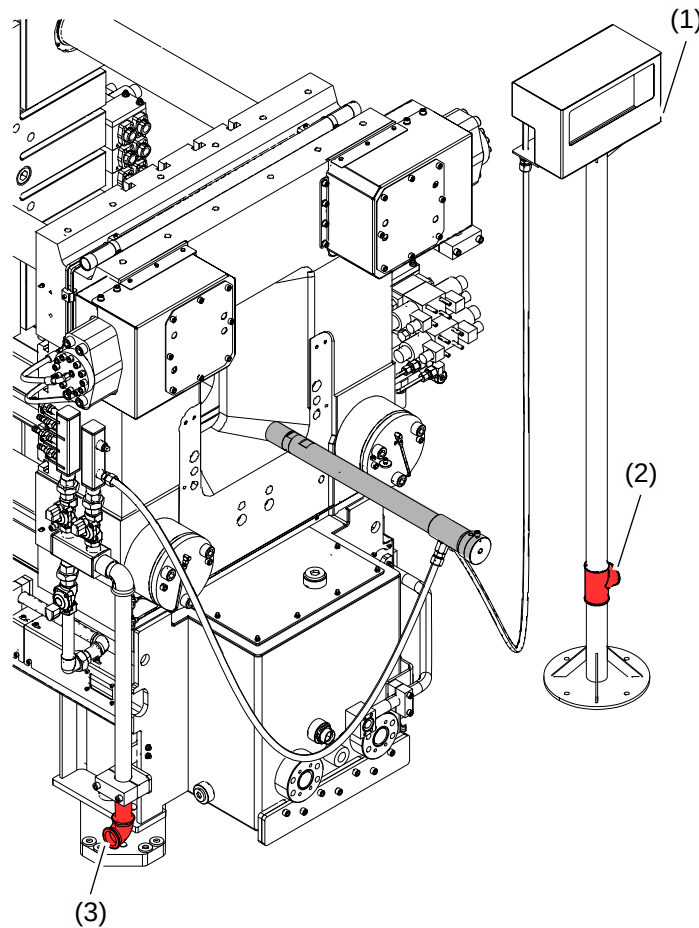


- (1) Water inlet
- (2) Water return
- (3) Drive unit

1. Connect water inlet (1) R 1" and water return (2) R 1" to drive unit (1).
2. Attach a ball valve at the water inlet.

7.5.2

Connecting the cooling circuits of the dies and of the plunger

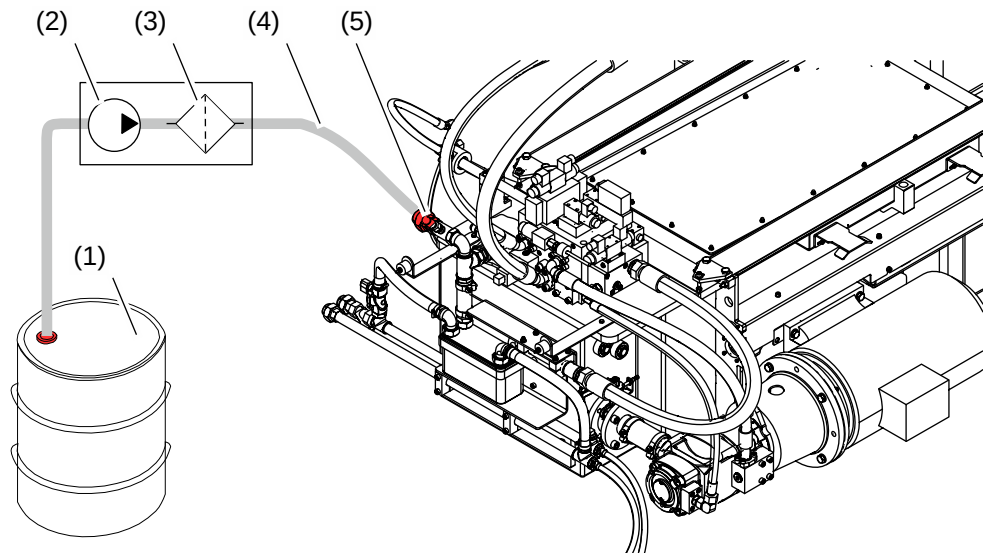


- (1) Water collector
- (2) Water return
- (3) Water inlet

1. Connect water inlet (1) R 1" and water return (2) R 1" at the water collector (1).
2. Attach a ball valve at the water inlet.

7.6

Filling with hydraulic fluid



- (1) Barrel with hydraulic fluid
- (2) External pump
- (3) External filter
- (4) Filling hose
- (5) Filling fitting

NOTICE

Contaminated hydraulic fluid
Damage to hydraulic components. Malfunctions.
► Only use the filling fitting to fill with hydraulic fluid.

1. Connect filling hose (4) to the filling fitting (5) and connect to an external pump (2).
2. Connect barrel with hydraulic fluid (1) and pump hydraulic fluid into the tank until the fill level reaches the middle of the upper oil level sight gauge.
3. Switch on the machine.
4. Move the hydraulic cylinder several times in set-up mode.
 - Close the die and open it.
 - Move the plunger forward and backward.
 - Move the ejector forward and backward.
 - The hydraulic system is being filled with hydraulic fluid.
5. Pump hydraulic fluid into the tank until the fill level reaches the middle of the upper oil level sight gauge.

7.7

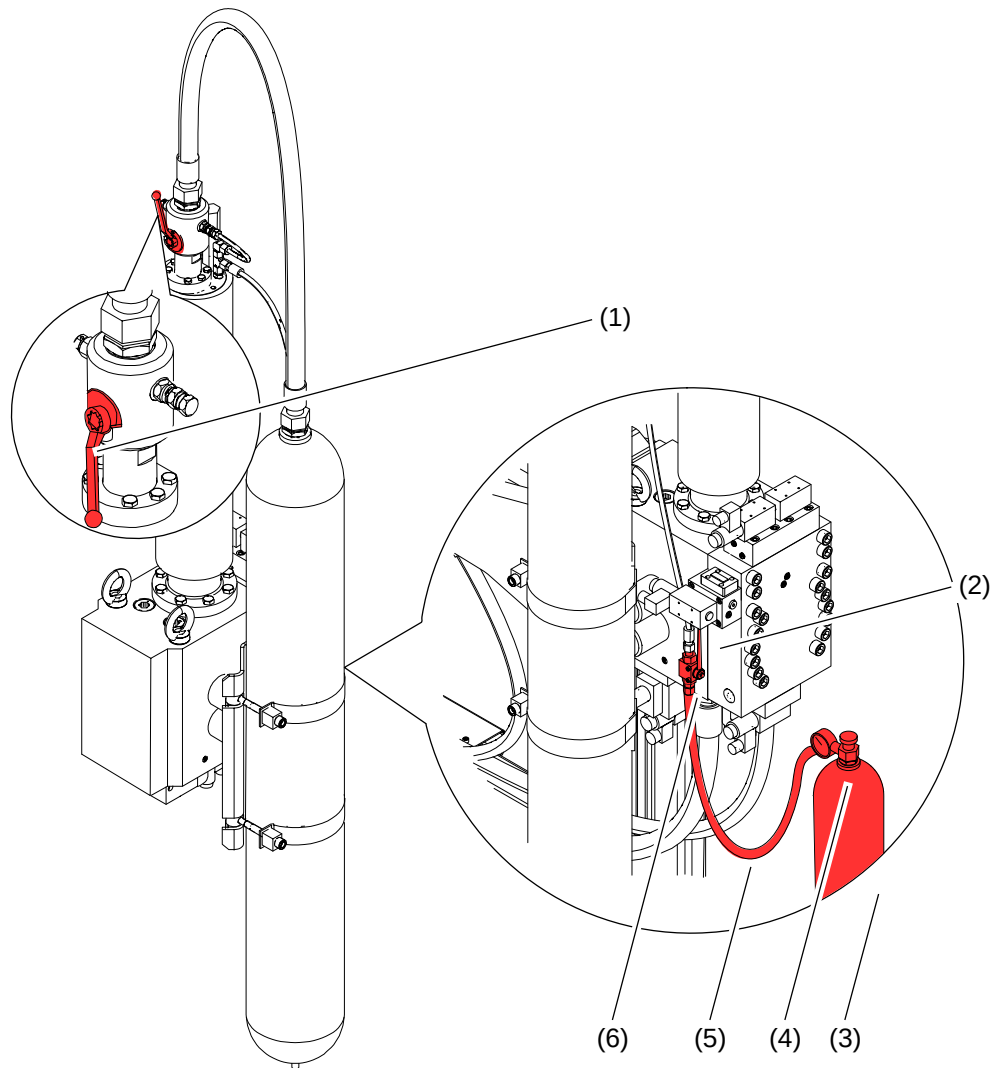
Adjusting the nitrogen pressure

7.7.1

Filling nitrogen into the piston accumulator

Filling the nitrogen into the piston accumulator requires the following work steps:

1. Prepare for filling
2. Connect the nitrogen bottle
3. Fill the piston accumulator
4. Replace the nitrogen bottle
5. Top up nitrogen
6. Conclude the filling process



- (1) Ball valve, 5-FH2AV
- (2) Ball valve 5-FH1AV
- (3) Nitrogen bottle.

- (4) Manometer
- (5) Filling hose
- (6) Filling spout, N₂Acc

Prerequisites:

- Machine must be in home position and in set-up mode.

! DANGER

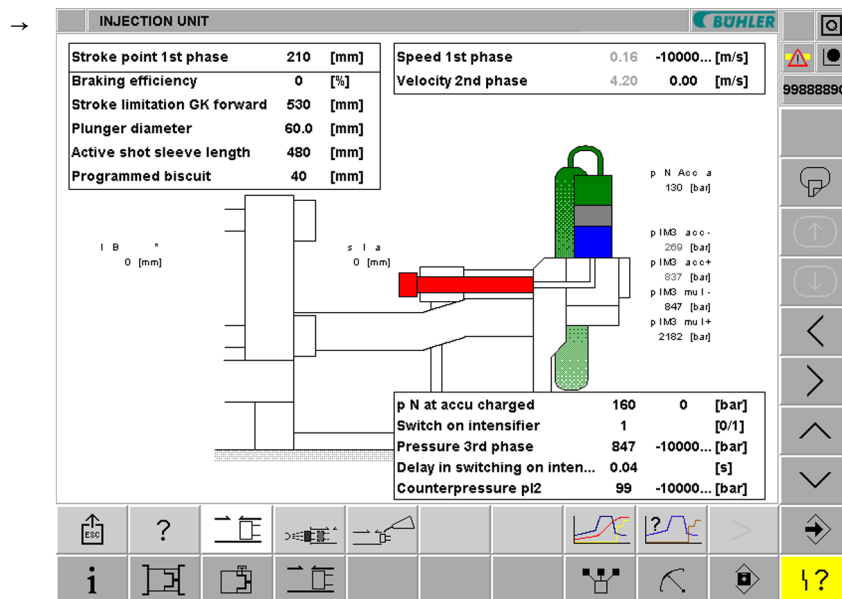
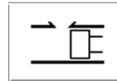
Incorrect gas filling.

Danger of explosion

- Only operate the nitrogen system with nitrogen of purity class ≥ 4.5 , N₂ = 99.99% by volume.

Preparing for filling

1. Open the "Injection unit" screen page.



2. Check the nitrogen pressure.
3. Place the lever on the ball valve (2) in the horizontal position.
→ Ball valve (2) is closed.
4. Remove the plug from the filling spout (6).

Connecting the nitrogen bottle

1. Connect the filling hose (5) to the filling spout (6).
2. Slowly open the nitrogen bottle (3).

Fill the piston accumulator

1. Place the lever on the ball valve (1) in the horizontal position.
→ Ball valve (1) is closed.
2. Slowly bring the lever of the ball valve (2) into a vertical position.
→ Ball valve (2) is open.
Nitrogen flows into the nitrogen system.
3. When the filling pressure of the nitrogen bottle (3) no longer drops, place the lever on the ball valve (2) in the horizontal position.
→ Ball valve (2) is closed.
4. For the parameter "Charge p N with rechargeable battery" enter the desired charge pressure.
5. Select <Manual mode> on the control column.
→ The piston accumulator will be charged.
6. When the piston accumulator is charged, select <Set-up mode> on the control column.
→ The hydraulic pressure in the piston accumulator will be relieved.

7. Fill the piston accumulator repeatedly until the nitrogen bottle (3) is emptied to 20 bar filling pressure.

Replace the nitrogen bottle

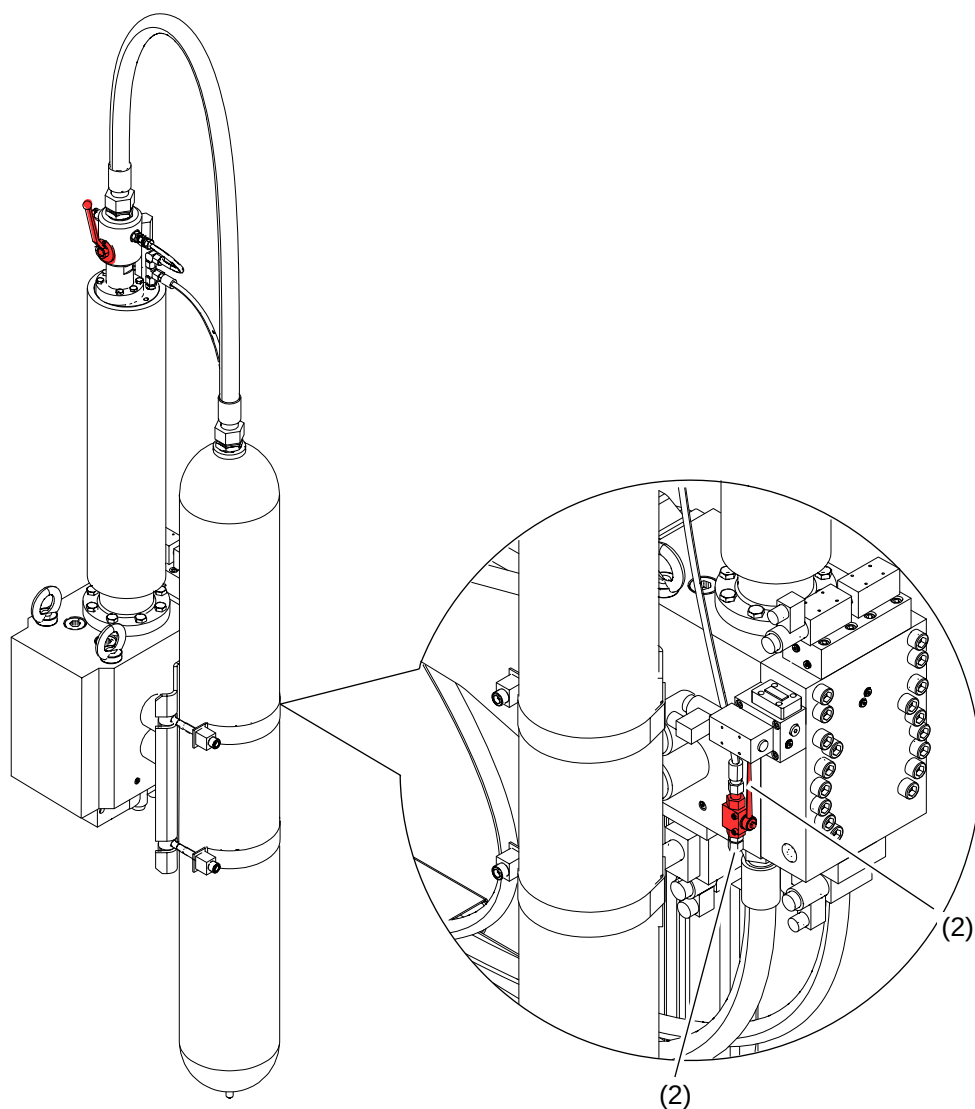
1. Place the lever on the ball valve (2) in the horizontal position.
→ Ball valve (2) is closed.
2. Close the nitrogen bottle (3).
3. Slowly remove the filling hose (5) on the nitrogen bottle (3).
→ The residual pressure in the filling hose (5) will be discharged.
4. Connect the filled nitrogen bottle (3) to the filling hose (5).

Topping up nitrogen

1. Fill the piston accumulator and repeat the changing of the nitrogen bottle until the desired filling pressure is reached.

Concluding the filling process

1. Place the lever on the ball valve (2) in the horizontal position.
→ Ball valve (2) is closed.
2. Close the nitrogen bottle (3).
3. Slowly open the threaded union on the filling hose (5).
→ The residual pressure in the filling hose will be discharged.
4. When the filling hose is depressurized (5) remove the filling hose (5) on the filling spout (6).
5. Attach the plug on the filling spout (6).
6. Place the lever on the ball valve (2) in the vertical position.
→ Ball valve (2) is open.
7. Place the lever on the ball valve (1) in the vertical position.
→ Ball valve (1) is open.

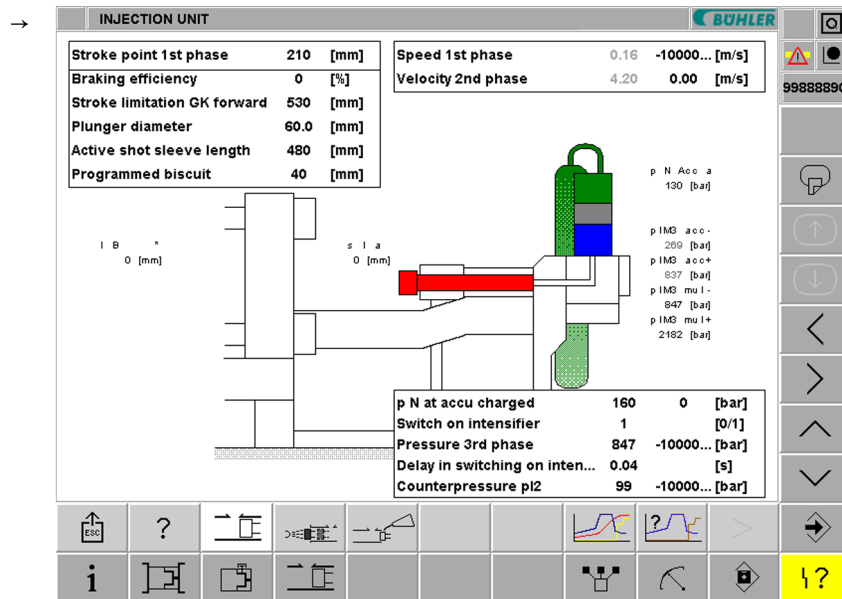
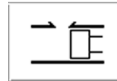


- (1) Ball valve 5-FH1AV
(2) Filling spout, N₂Acc

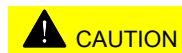
Prerequisites:

- Machine must be stopped.

1. Open the "Injection unit" screen page.



2. Check the nitrogen pressure.
3. Place the lever on the ball valve (1) in the horizontal position.
→ Ball valve (1) is closed.
4. Remove the plug from the filling spout (2).



CAUTION Spraying leak oil. Eye injuries. Cover the filling spout with a cloth.

5. Slowly bring the lever of the ball valve (1) into a vertical position.
→ Ball valve (1) is open.
Accumulator pressure is reduced.
6. Read off the dropping accumulator pressure on the parameter "p N Acc" on the "Injection unit" screen page.
7. When the accumulator pressure "p N Acc" has reached the desired pressure, place the lever on the ball valve (1) in the horizontal position.
→ Ball valve (1) is closed.
8. Attach the plug on the filling spout (2).
9. Place the lever on the ball valve (1) in the vertical position.
→ Ball valve (1) is open.

7.8

Filling with lubricant

1. Ensure that the plunger is lubricated.
 - ▶ Fill with lubricant. See page 107, chapter "Topping up the lubricant tank".
 - ▶ Manually trigger plunger lubrication on the operating panel until oil escapes on the lubricating bar.
2. Fill the central lubrication unit via a filling sieve to the upper mark of the oil level indicator. See page 17, chapter "Central lubricating oil CGLP".

No.	Check	✓
1	Nitrogen bottles are filled with nitrogen, purity class ≥ 4.5 , $N_2 = 99.99\%$ by volume.	
2	Hydraulic fluid is filled and the hydraulic system is vented.	
3	Cooling water is connected.	
4	The direction of rotation of the motors agrees with the arrow on the pumps.	
5	Electrical installation is executed as prescribed.	
6	All transport locking devices are removed.	
7	Machine is cleaned.	
8	There are no foreign objects in the direct vicinity of the machine.	
9	No one is in the danger zone of the machine.	

8 Operation

8.1 Safety

8.1.1 Personnel qualifications

Improper operation can injure personnel and damage machine parts.

- ▶ The machine must only be operated by trained operators.
- ▶ Only machine setters who have been technically trained or who have participated in training offered by the manufacturer can perform set-up tasks.

8.1.2 Protective devices

During operation all protective devices must be installed and function correctly. This particularly applies for:

- Protective enclosure
- Protective doors
- EMERGENCY STOP buttons and EMERGENCY STOP photoelectric sensors
- Protective fence
- Splash guard that protects against metal splashes that occur when opening the die
- Protective hood, plunger with extension
- ▶ Never render protective devices inoperable or remove them.
- ▶ Perform work only with the protective hood of the plunger closed.

8.1.3 Personal protective equipment

During operation, machine surfaces are hot and touching them can result in burns due to liquid metal and hot surfaces.

- ▶ Wear protective clothing.
- ▶ Wear safety shoes.
- ▶ Wear hand protection.
- ▶ Wear head and eye protection.

8.1.4 One-person operation

If several people are working on the machine at the same time, life-threatening injuries can occur.

- ▶ Only have one person operate the machine.
- ▶ In set-up mode, ensure that a second person is not in the danger zone of the machine.
- ▶ If a second person is necessary in set-up mode, maintain eye contact.

8.1.5 Mobile climbing aids

If mobile climbing aids, such as steps or ladders fall over, personnel can be injured.

- ▶ Only use free-standing climbing aids.
- ▶ Never lean climbing aids against the machine.
- ▶ Do not climb onto the machine.

8.2 Preparing for production

8.2.1 Pre-heating the die

The die must be pre-heated before the start of production. Temperature-control devices or gas heaters can be used for pre-heating.

Prerequisites:

- Machine must be stopped.
 - Machine must be in set-up mode.
1. Turn the switch <Close die in set-up mode> to the right.
→ Closing of the die in set-up mode is enabled.
 2. Move the die by continuously pressing the two-hand start button on the control column.
 - ▶ When using a temperature control device, move the die closed until the parting surfaces of the die touch.
 - ▶ When using a gas heater, close the die far enough so that the parting surfaces of the die are as close as possible to the gas heater.
 3. Pre-heat the die.

8.3 Switching the machine on and off

8.3.1 Switching on the machine

1. Remove the padlock from the control cabinet.
2. Set master switch to “1”.
→ After initialization the “Start Mask” screen page will be displayed.



8.3.2 Switching off the machine

1. Stop the machine. See page 77, chapter "Stopping the machine".
2. Set the master switch at the control cabinet to "0".
3. Attach a padlock.
4. Remove the key from the padlock and carry it with you.
→ Machine is switched off.

8.4 Starting and stopping the machine

8.4.1 Starting the machine

1. Switch on the machine.
2. Switch on the drive motor on the control column with the <Drive motor on> key.

8.4.2 Starting after emergency stop

1. Eliminate fault.
2. Unlock the emergency stop button
3. Switch on the drive motor on the control column with the <Drive motor On/ Off> button.

8.4.3 Stopping the machine

Stopping in semi-automatic mode

- ▶ Wait until the end of the machine cycle, until the machine and all peripheral devices are in home position.
- ▶ Switch off the drive motor on the control column with the <Drive motor On/ Off> button.
→ Machine is stopped.

Stopping in automatic mode

1. Press semi-automatic mode on the control column.
→ At the end of the machine cycle the machine stops in home position.
2. Wait until all peripheral devices are in home position.
3. Switch off the drive motor on the control column with the <Drive motor On/ Off> button.
→ Machine must be stopped.

Stopping in set-up mode and manual mode

1. Switch off the drive motor on the control column with the <Drive motor On/ Off> button.
→ Machine is stopped.

8.5 Starting a machine cycle

A machine cycle is started in "automatic" mode and "semi-automatic" mode.

Prerequisites:

- The machine is switched on.
- The machine is started.

 DANGER

Machine movements in the area of the die.

Life-threatening crushing injuries.

- ▶ Ensure that no one is in the area of the die during the casting process.
- ▶ Ensure that the protective devices are correctly functioning.

 WARNING

Incorrectly positioned metal delivery chute.

Severe burns due to liquid metal.

- ▶ Wear protective gloves, protective clothing and safety shoes.
- ▶ Wear head and eye protection.
- ▶ When using dosing furnaces, always monitor the position of the metal delivery chute.

1. Switch on the peripheral devices: Press the buttons for switching on the connected peripheral devices on the control column.
2. Move the machine into home position: Press the <Machine home position> key.
3. Switch on the peripheral devices: Press the buttons for home position of the connected peripheral devices.
4. Select the mode: <Automatic> key or <Semi-automatic> key.
5. Press the two-hand start buttons on the control column simultaneously.
 - Machine cycle starts.

8.6

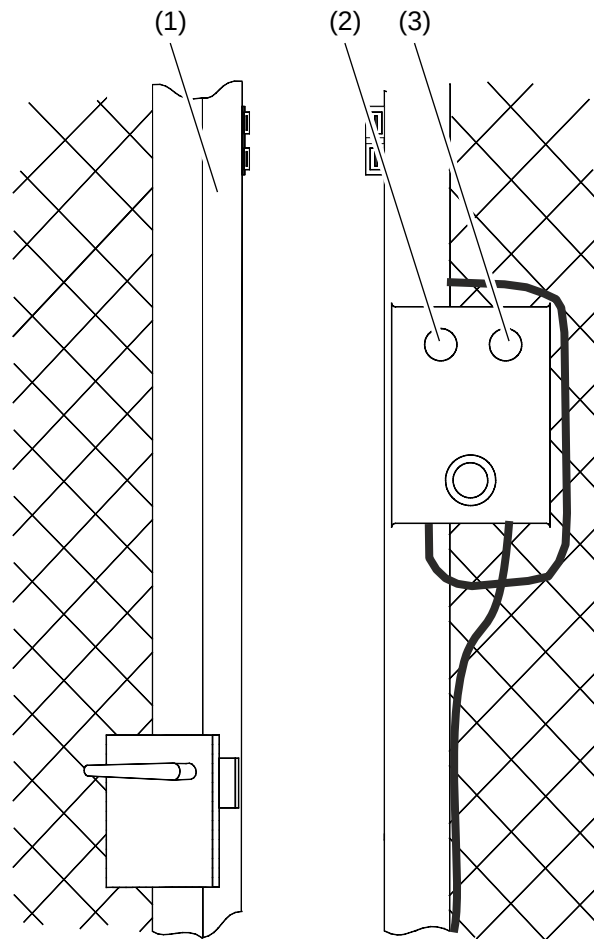
Checks during operation

8.6.1

Safety checks

During operation, protective devices, as well as dangerous machine parts and peripheral devices must be constantly monitored.

- ▶ Check that protective devices are correctly functioning.
- ▶ Ensure that protective devices are not rendered inoperable.
- ▶ When using a dosing furnace, always monitor the position of the metal delivery chute.
- ▶ Ensure that no individuals climb over protective devices.



- (1) Door
- (2) <Unlock door> key
- (3) <Lock door> key (blue)

WARNING

Misuse of the emergency unlock.

Severe injuries due to unexpected start-up of the machine.

- ▶ Only use the emergency unlock if the machine is de-energized.
- ▶ Keep the key for resetting the emergency unlock under lock and key and only make it accessible in an exceptional case.

1. Press the <Semi-automatic mode> button.
→ The machine cycle is concluded. Machine changes to set-up mode.
2. Press the <Unlock door> button.
→ The door will be unlocked.
3. Open door.
→ <Lock door> button is illuminated.
4. Secure the machine to prevent unexpected start-up, e.g. by attaching a personal padlock in the stop of the door.
5. Execute tasks within the protective fence, exit the fenced-enclosed area and close the door.
6. Press the <Lock door> button.
→ Door is locked.

Machine is ready to operate and can be started.

8.7

Set-up

8.7.1

Automatic tie bar pulling

Automatic tie bar pulling requires the following work steps:

1. Clean the tie bars.
2. Extend the tie bars.
3. Retract the tie bars.
4. Grease the tie bars.

Cleaning the tie bars

Prerequisites:

- Machine must be stopped.
 - Machine must be secured against unexpected startup.
1. Remove metal splashes.
 2. Clean the tie bars.

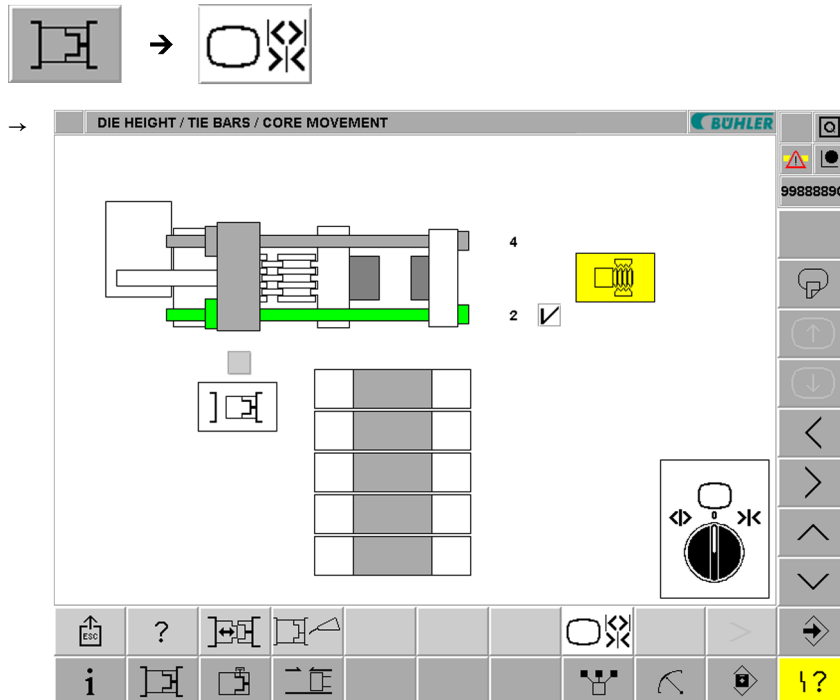
Pulling tie bars

Prerequisites:

- Machine must be in home position and in set-up mode.
- Tie bars must be cleaned.

Important The activated process will be canceled. Leave the screen page on the operator panel open until the process is concluded.

1. Open the screen page “Die height, tie bars, core movement” on the operating panel.



2. Preselect tie bar.

3. Turn the <Run preselected movement> switch to the left.
 - The selected tie bars will be pulled.
 When the tie bars are completely pulled out of the die area, the movement stops.

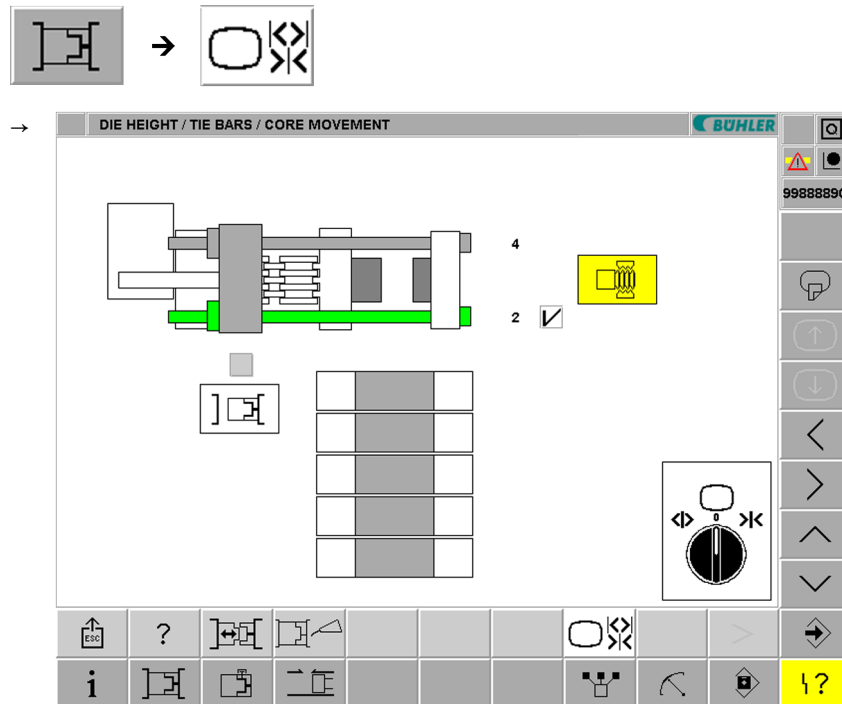
Inserting the tie bars

Prerequisites:

- Tie bars must be pulled.

Important The activated process will be canceled. Leave the screen page on the operator panel open until the process is concluded.

1. Open the screen page “Die height, tie bars, core movement” on the operating panel.



2. Preselect tie bar.
3. Turn the <Run preselected movement> switch to the right.
 - The selected tie bars will be inserted.
 When the tie bars are completely inserted out of the die area, the movement stops.
4. Check the position of the scale ring. See page 115, chapter “Checking the load of the tie bars”.

Greasing the tie bars

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.
- Tie bars must be retracted.

1. Grease the tie bars.

Pulling the tie bars manually

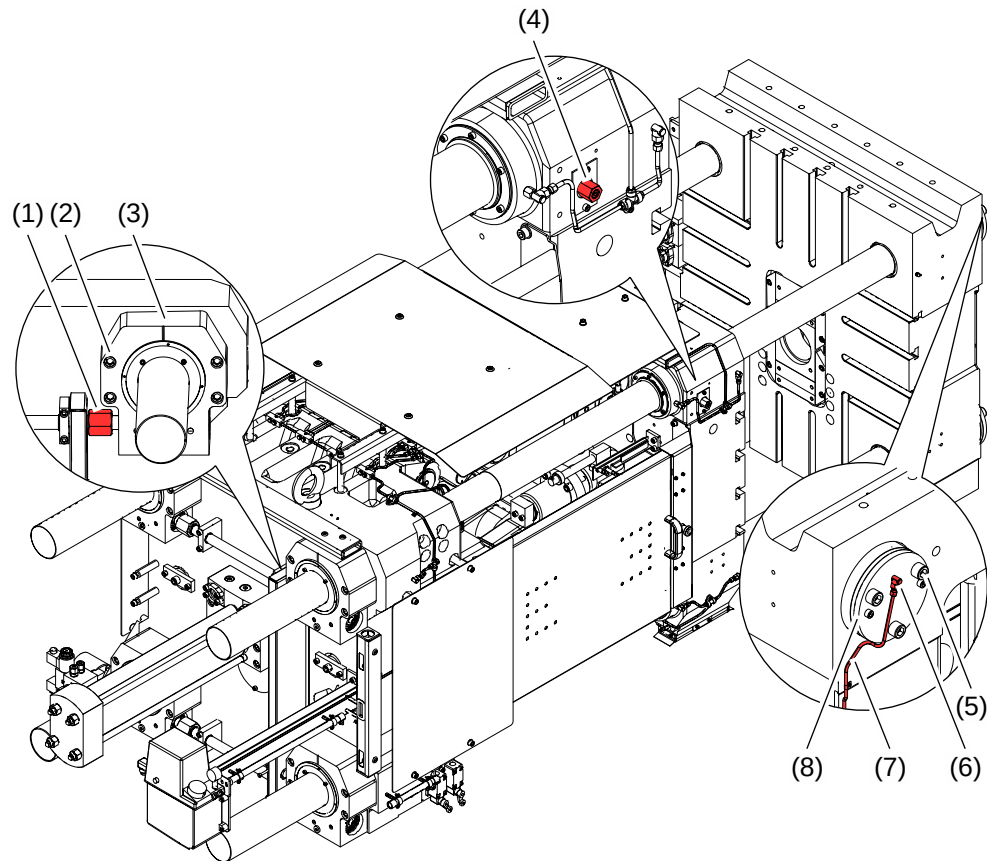
Cleaning the tie bars

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Remove metal splashes.
2. Clean the tie bars.

Pulling tie bars



- | | |
|-----------------------|----------------------------|
| (1) Coupling | (5) Positioning screw |
| (2) Screws | (6) Screws |
| (3) Worm gear housing | (7) Lubrication connection |
| (4) Clamp bolt | (8) Tie bar nut |

Prerequisites:

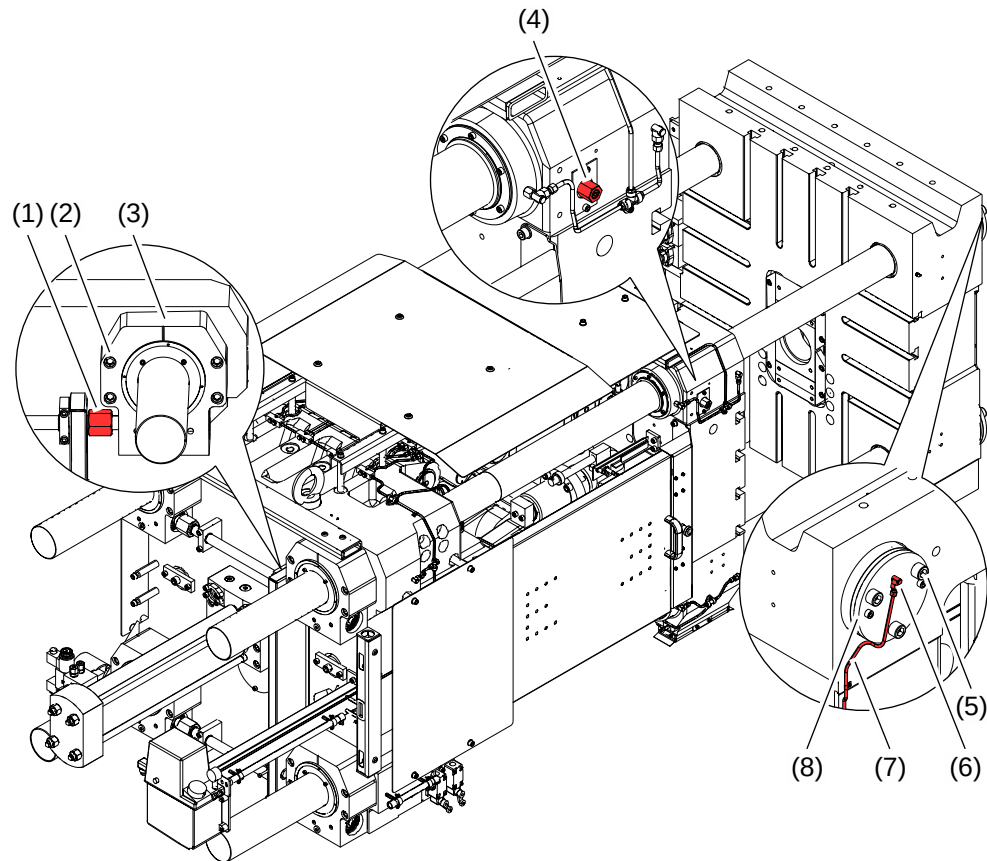
- Machine must be in home position and in set-up mode.
- Tie bars must be cleaned.

1. Turn the switch <Close die in set-up mode> on the control column to the right. This activates closing of the die in set-up mode.
2. Close the die by continuously pressing the two-hand start button on the control column until the parting surfaces of the die touch.
3. Turn the switch <Close die in set-up mode> on the control column to the left. This deactivates closing of the die in set-up mode.
4. Switch off the machine.

NOTICE Activate the die height adjustment element with tie bars pulled. The position of the tie bar nuts varies. The tie bar load must be compensated again. If the tie bar is pulled, do not activate the die height adjustment unit.

5. Detach the worm gear housing (3)
 - ▶ Uncouple the coupling(1)
 - ▶ Remove screws (2).
6. Remove tie bar nut (8)
 - ▶ Remove lubrication connection (7).
 - ▶ Remove positioning screw (6).
 - ▶ Remove screws (5).
 - ▶ Remove tie bar nut (8).
7. Switch on the machine.
8. Select <Set-up mode> on the control column.
9. Tighten clamp bolt(4).
10. Turn the switch <Die, move core pull units> to the left.
 - The die will open.
11. Unscrew clamp bolt(4).
12. Turn the switch <Close die in set-up mode> on the control column to the right.
13. Close the die by continuously pressing the two-hand start button on the control column.
14. Repeat steps 9 to 13 until the tie bar is flush with the moving plate.
15. Tighten clamp bolt(4).
16. Select <Set-up mode> on the control column.
17. Turn the switch <Die, move core pull units> to the left.
 - The die will open.

Inserting the tie bars



- | | |
|-----------------------|----------------------------|
| (1) Coupling | (5) Positioning screw |
| (2) Screws | (6) Screws |
| (3) Worm gear housing | (7) Lubrication connection |
| (4) Clamp bolt | (8) Tie bar nut |

Prerequisites:

- Tie bars must be pulled.
- Machine must be in set-up mode.

1. Turn the switch <Close die in set-up mode> on the control column to the right. This activates closing of the die in set-up mode.
2. Close the die by continuously pressing the two-hand start button on the control column until the parting surfaces of the die touch.
3. Turn the switch <Close die in set-up mode> on the control column to the left. This deactivates closing of the die in set-up mode.
4. Unscrew clamp bolt(4).
5. Turn the switch <Die, move core pull units> to the left.
→ The die will open.
6. Tighten clamp bolt(4).
7. Repeat steps 1 to 6 until the worm gear housing (3) is flush with the cylinder place.
8. Unscrew clamp bolt(4).
9. Switch off the machine.
10. Mount the worm gear housing(3).
 - ▶ Install the screws (2).
 - ▶ Couple the coupling(1).
 - ▶ Check the position of the scale rings.

11. Mount the tie bar nut (8).

- ▶ Clean tie bar threads.
- ▶ Grease the tie bar threads and the facing side of the tie bar.
- ▶ Turn the tie bar nut (8) until it is flush with the stationary plate.
- ▶ Attach the positioning screw (6).
- ▶ Install the screws (5).
- ▶ Mount the lubrication connection (7).

Greasing the tie bars

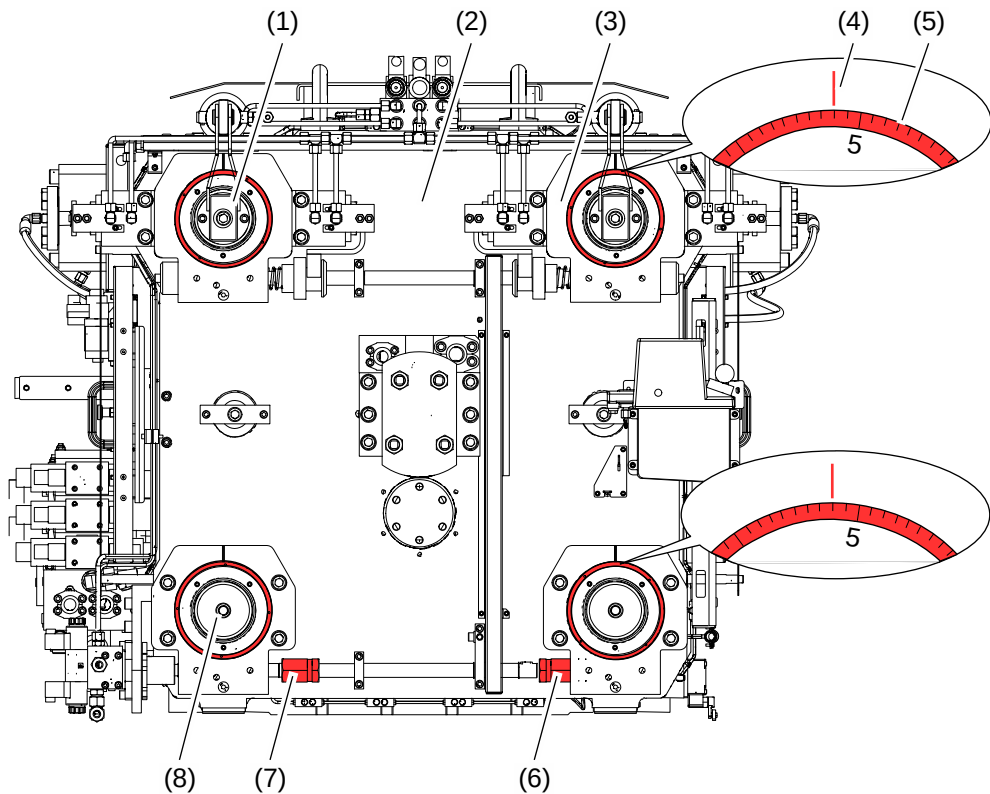
Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.
- Tie bars must be retracted.

1. Grease the tie bars.

8.7.3

Correcting the load of the tie bars



- | | |
|--|---|
| (1) Tie bars with automatic tie bar pull | (5) Scale ring |
| (2) Cylinder plate | (6) Tie bar coupling (uncoupled) |
| (3) Tie bar nut | (7) Tie bar coupling (coupled in) |
| (4) Marking | (8) Tie bars without automatic tie bar pull |

NOTICE

Different loads on the tie bars.

Breakage of the tie bars.

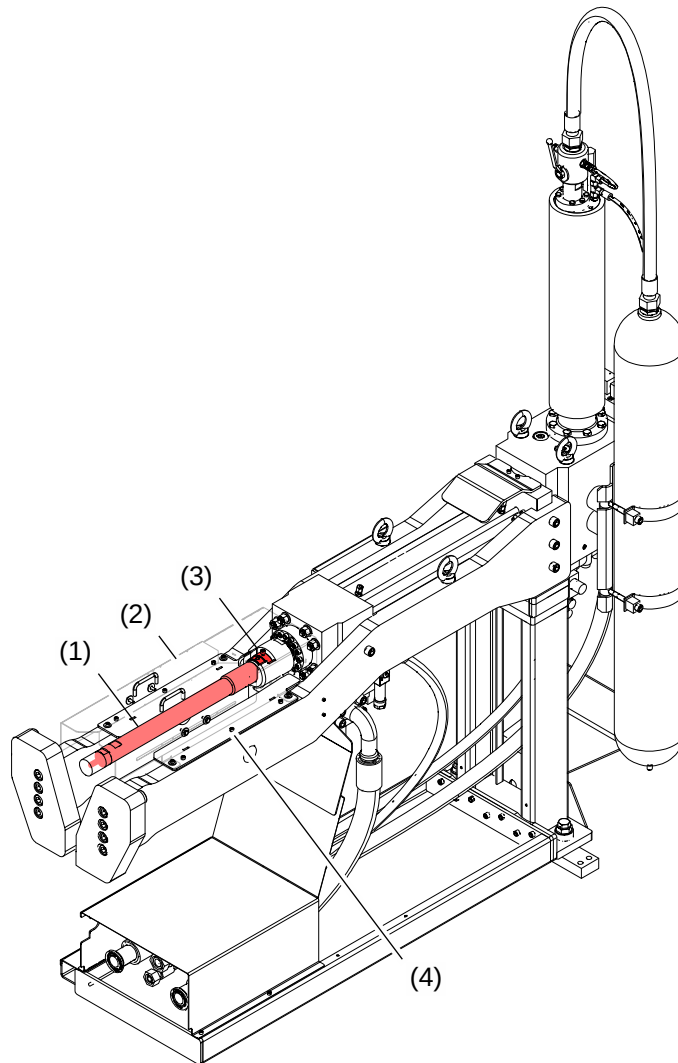
High wear of the locking unit.

- ▶ Use parallel dies.
- ▶ Manually correct the load on the tie bars.

1. Uncouple the tie bar coupling.
 - ▶ For tie bars with automatic tie bar pull (1) slightly extend the tie bars. See page 80, chapter “Automatic tie bar pulling”.
 - ▶ For tie bars without automatic tie bar pull (8) slide the tie bar coupling (6) to the outside.
2. Adjust the load uniformly on all four tie bars of the tie bar coupling (6). In this process ensure that the deviation on the scale ring (5) is maximum 3 scale lines.
3. Couple in the tie bar coupling and retract the tie bars with the automatic tie bar pull unit (1).

8.7.4

Changing the plunger rod



(1) Plunger rod

(2) Guard

(3) Spacer disk

(4) Screws

Changing the plunger rod requires the following work steps:

1. Remove the plunger rod
2. Insert the plunger rod

Removing the plunger rod

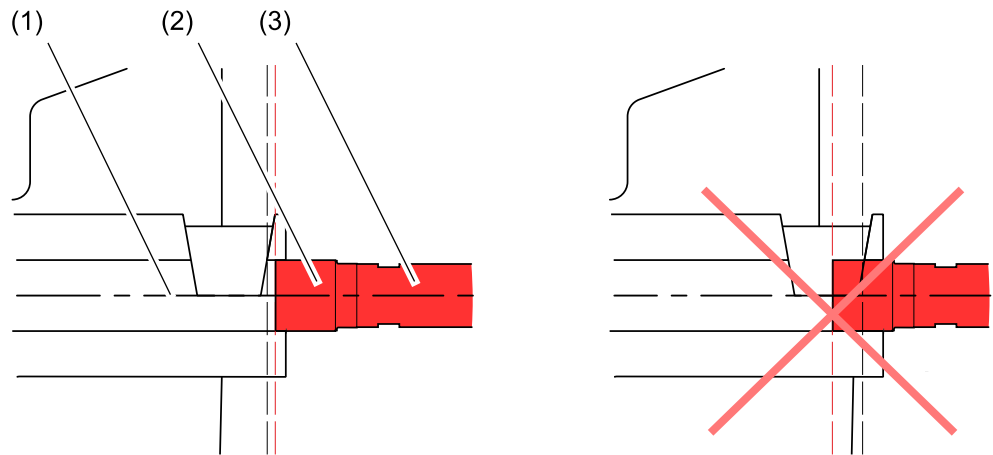
Prerequisites:

- Machine must be stopped.
 - The water cooling system must be switched off on the control column with the <Cool injection set> switch.
 - Machine must be secured against unexpected startup.
1. Unscrew the screws.
 2. Take off the protective cover of the plunger rod.
 3. Unscrew the connections for cooling the casting unit. In this process catch any water that escapes.
 4. Remove the spacer disk.
 5. Remove the plunger rod.

Inserting the plunger rod

Prerequisites:

- Machine must be stopped.
 - The water cooling system must be switched off on the control column with the <Cool injection set> switch.
 - Machine must be secured against unexpected startup.
1. Insert the plunger rod
 2. Place the spacer disk in the coupling flange.
 3. Check the length of the plunger rod (3). In this regard, ensure that the plunger (2) does not protrude into the filling opening of the shot sleeve (1).

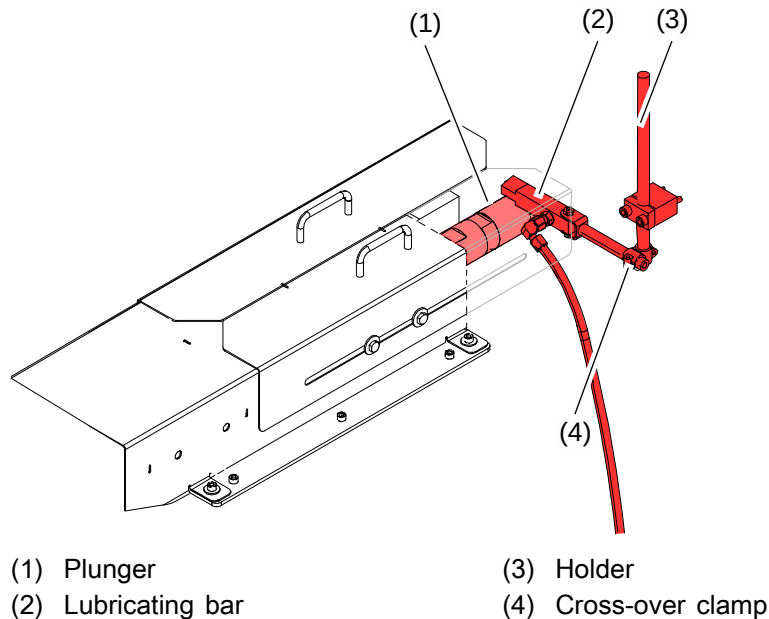


4. If the plunger rod is too short, extend the plunger rod with the plunger follow-through.
 - ▶ Remove the spacer disk.
 - ▶ Remove the plunger rod.
 - ▶ Screw the plunger follow-through onto the plunger rod.
 - ▶ Insert the plunger rod.
 - ▶ Place the spacer disk in the coupling flange.
5. Connect the hoses for the cooling to the plunger rod.

6. Attach the protective cover of the plunger rod with screws.
7. If necessary adjust the length of the protective cover.

8.7.5

Adjusting the lubricating bar



Prerequisites:

- Machine must be stopped.
 - Machine must be secured against unexpected startup.
1. Position the holder (3) with the lubricating bar (2) at the correct height.
 2. Fix the holder (3) with the lubricating bar (2) in place on the cross-over clamp (4).
 3. Position the lubricating bar (2) centered over the plunger (1).
 - ▶ Place the lubricating bar (2) on the shot sleeve.
 - ▶ Ensure that both sides of the plunger (1) are lubricated.
 4. Tighten the screws on the cross-over clamp (4).

8.7.6

Mounting the die

Mounting the die requires the following work steps:

1. Clean and grease the fixing surfaces.
2. Clamp the die.
3. Support the die.
4. Connect the die cooling system.

NOTICE

Excessive surface pressure.
Damage to the plate.

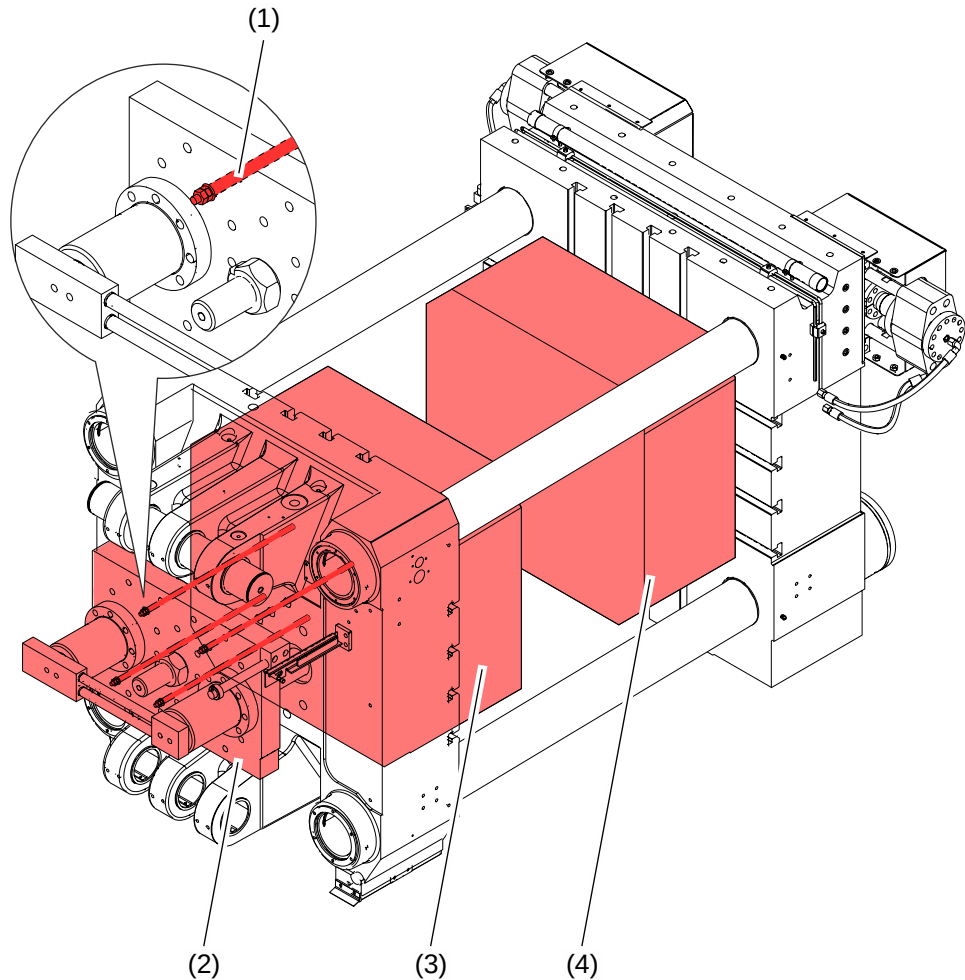
- ▶ Maintain the smallest permissible clamping dimensions.

Cleaning and greasing of the plate surfaces

Prerequisite

- Die is unclamped.
- 1. Clean the plate surfaces of the fixed and moving die halves.
- 2. Lubricate the plate surfaces.

Clamping the die



- | | |
|--------------------|----------------------|
| (1) Ejection rod | (3) Movable die half |
| (2) Ejection plate | (4) Fixed die half |

Prerequisites:

- Machine must be in home position and must be stopped.
 - Machine must be secured against unexpected startup.
 - Plate surfaces must be cleaned and greased.
 - The shot position must be adjusted and the shot sleeve must be positioned.
1. If there is danger of collision with the upper tie bars when retracting the die, pull the upper tie bars.
 2. Move the die into the machine with a crane.
 3. Guide the die onto the fixed plate and position it.
 4. Position the fixed die half (4) on the fixed plate using tension jaws. Tighten the nuts firmly in this process.

5. Insert the ejection rods (1) into the bores of the moving plate and the ejector plate (2).
6. Use a crane to move the movable die half (3) to the moving plate, so that the ejection rods (1) can be screwed in.
7. Screw ejection rods (1) into the threaded bores of the movable die half (3).
8. Position the movable die half on the moving plate so that it is flush.
9. Position and secure the movable die half (3) on the moving plate using die clamps. Tighten the nuts firmly in this process.
10. Using the nuts and the lock nuts, tighten the ejection rods (1) on the ejector plate (2) of the machine. Tighten the nuts and lock nuts so that the gap between the washer and the ejector plate is 1 mm.

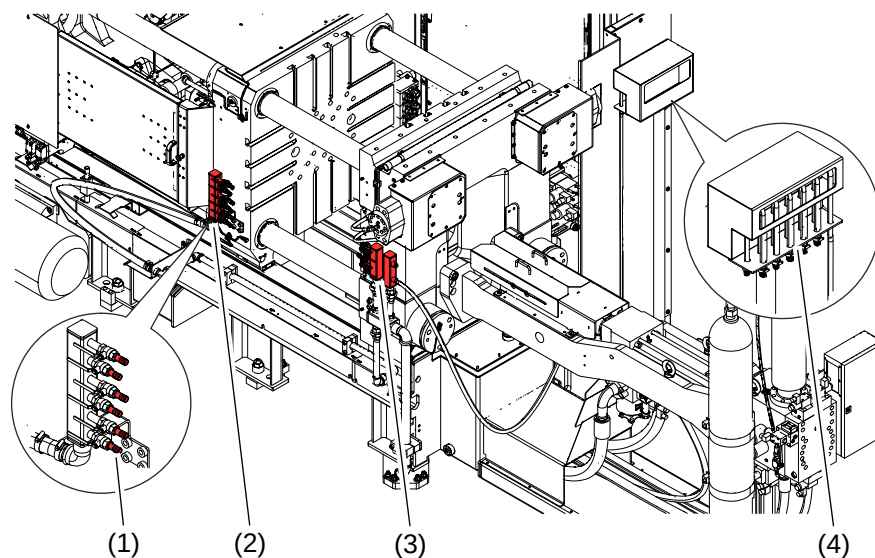
Supporting the die

For heavy dies a die carrier is fastened on the die.

- Die is opened.
- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Adjust die carrier.

Connecting the die cooling system

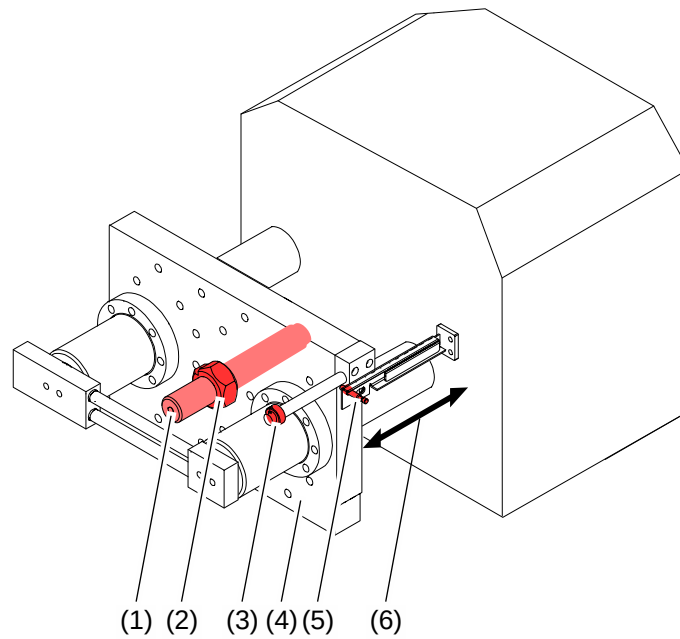


- | | |
|--|---|
| (1) Quick-release coupling | (3) Cooling water connection, fixed plate |
| (2) Cooling water connection, moving plate | (4) Water collector |

Prerequisites:

- Machine must be stopped.
- The water cooling system is switched off on the control column with the <Cool die> button.
- Machine must be secured against unexpected startup.

1. Connect hoses to the cooling connections of the die.
2. Connect hoses to the quick-release couplings (1).
3. Adjust the cooling effect at the ball valves of the cooling water connections.

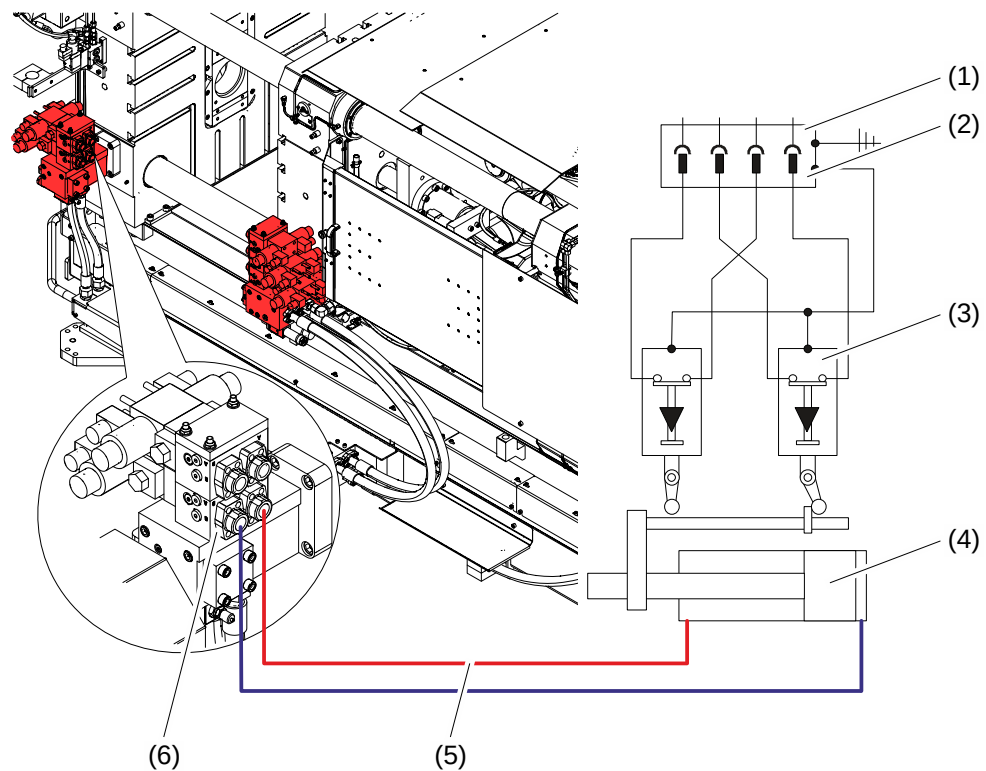


- | | |
|-------------|----------------------|
| (1) Spindle | (4) Ejector plate |
| (2) Nut | (5) Proximity switch |
| (3) Cam | (6) Ejection stroke |

Prerequisites:

- Machine must be stopped.
 - Machine must be secured against unexpected startup.
1. Adjust the ejection stroke (6).
 - ▶ Loosen nut (2).
 - ▶ Adjust spindle (1) until the desired stroke is reached.
 - ▶ Tighten nut (2).
 2. Start up the machine.
 - Machine is in set-up mode.
 3. Move the <Move ejector> switch on the control column to the right.
 - The ejectors have moved forward.
 4. Stop the machine.
 5. Secure the machine against unexpected startup.
 6. Adjust cam (3) of proximity switch (5).
 7. Check ejection stroke (6).

Connecting the core pulls



- | | |
|-----------------------|------------------------|
| (1) Electrical outlet | (4) Hydraulic cylinder |
| (2) Plug | (5) Hydraulic lines |
| (3) Limit switch | (6) Connection block |

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

! WARNING

Moving core pull units.
Severe crushing injuries.

- ▶ Safeguard core pull units, so personnel cannot reach into them.

NOTICE

Collision of core pulls with die.
Damage to the die.

- ▶ Connect core pulls individually, electrically and hydraulically.
- ▶ Do not use limit switches with engagement in the end positions.
- ▶ Do not expose limit switches to extreme heat and moisture.
- ▶ Ground limit switches.

1. Specify the numbering for the core pulls.
2. For each core pull, use two mechanical limit switches with IP 65 degree of protection.
3. Equip limit switches (3) with close contacts for the “Core pull retracted” and “Core pull extended” end positions.

Important Insufficient inner diameters of the hydraulic lines reduce the retraction and extension speed and increase the cycle time. Select an inner diameter for the hydraulic lines (5) that is suitable for the hydraulic cylinder (4).

4. Connect the hydraulic lines. In this regard, comply with the numbering of the core pulls and connect core pull M1 to core pull unit M1, for example.
 - ▶ Install the hydraulic line between the cylinder side “Extend” and the connection block “OUT”.
 - ▶ Install the hydraulic line between the cylinder side “Retract” and the connection block “IN”.

NOTICE Moisture in electrical connections. Damage to the electrical installation. Attach guard panels to electrical outlets that are not in use. Install plugs so that no moisture can penetrate.

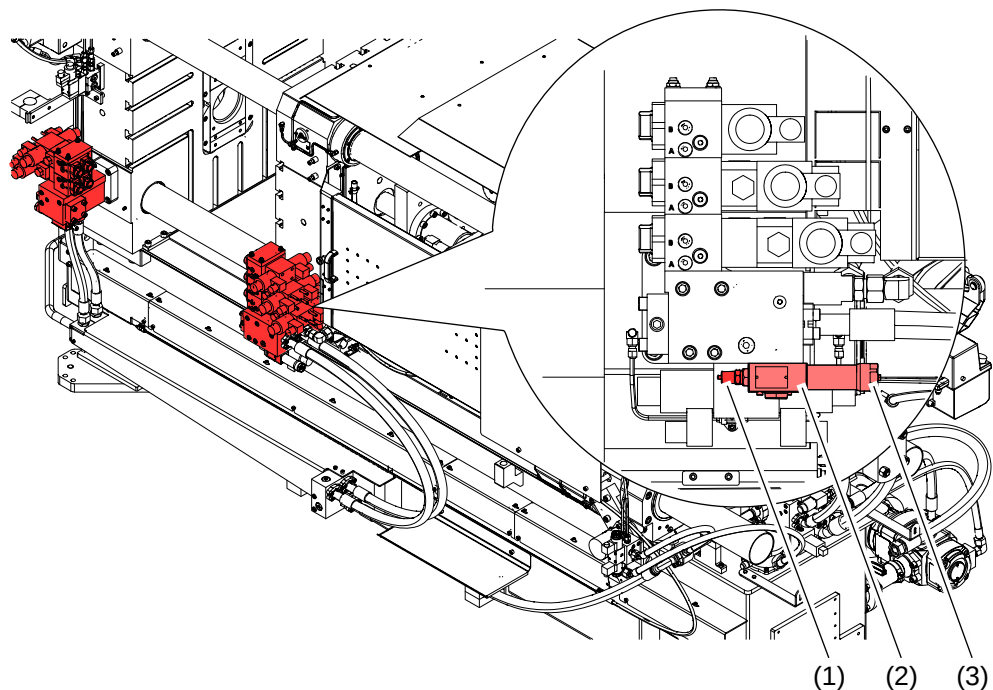
5. Plug the electrical plug (2) into the electrical outlet (1). In this regard, comply with the numbering of the core pull units and connect the plug for core pull unit M1 into the electrical outlet M1, for example.
6. Bleed the cylinders via the measurement connections on the connection block.

8.7.9

Adjusting the central pressure control system

Adjustment of the central pressure control system requires the following work steps:

1. Connecting the manometer
2. Adjusting the pump pressure
3. Adjusting the pressure regulator
4. Removing the manometer



- (1) Measuring point
- (2) Pressure regulator
- (3) Adjusting screw

■ Machine must be in the home position and in set-up mode.

⚠ WARNING

Unexpected start-up of the machine.

Risk of severe injuries.

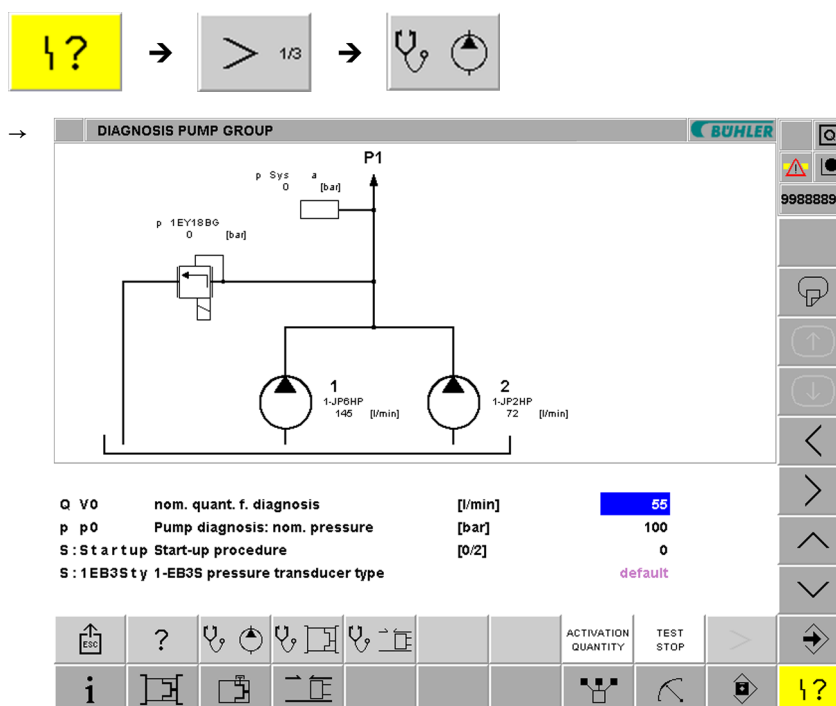
- ▶ Ensure that no one else can start the machine.
- ▶ Attach a warning sign on the machine.

Connecting the manometer

1. Remove the metal cap from the measuring point (1) of the pressure regulator (2).
2. Connect the manometer to the measuring point (1).

Adjusting the pump pressure

1. Select the “Drive unit diagnostics” page with authorization level “4” on the operating panel.



NOTICE Pressure too high. Pump damage Do not set pressure above 210 bar.

2. Set the following values on the “Drive unit diagnosis” page.
 - ▶ Set the “Q V0” parameter to the displayed value of pump 2 “1-JP2HP”.
 - ▶ Set the “p p0” parameter to value “210”.
3. Press the [Activation quantity] button on the operating panel.
 - The [Activation quantity] button is illuminated.
 - The set pressure is built up in the hydraulic system.

Adjusting the pressure regulator

1. Remove the protective cap from the pressure regulator (2).
2. Loosen the jam nut on the pressure regulator (2).
3. Turn the adjusting screw (3). While doing so, observe the pressure on the manometer.

4. When the desired pressure is displayed on the manometer, tighten the jam nut.

Removing the manometer

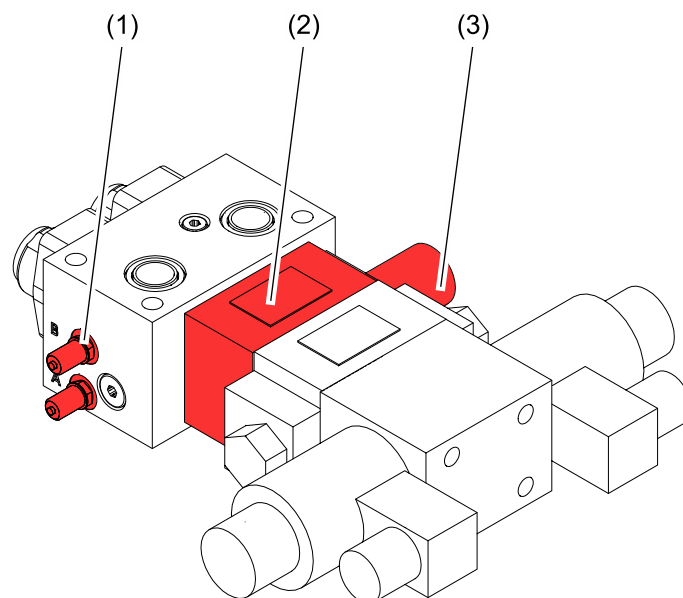
1. Press the [TEST STOP] button on the "Drive unit diagnostics" page on the operating panel.
 - The [Activation quantity] button is deactivated.
 - Pressure in the hydraulic system is dissipated.
2. Remove the manometer.
3. Attach the metal cap to the measuring point (1).

8.7.10

Adjusting the pressure control system on the core pull unit

The adjustment of the pressure control system on the core pull unit requires the following work steps:

1. Connecting the manometer
2. Adjusting the pump pressure
3. Adjusting the pressure regulator
4. Removing the manometer



- (1) Measuring point
- (2) Pressure regulator
- (3) Adjusting screw

- Machine must be in the home position and in set-up mode.

WARNING

Unexpected start-up of the machine.

Risk of severe injuries.

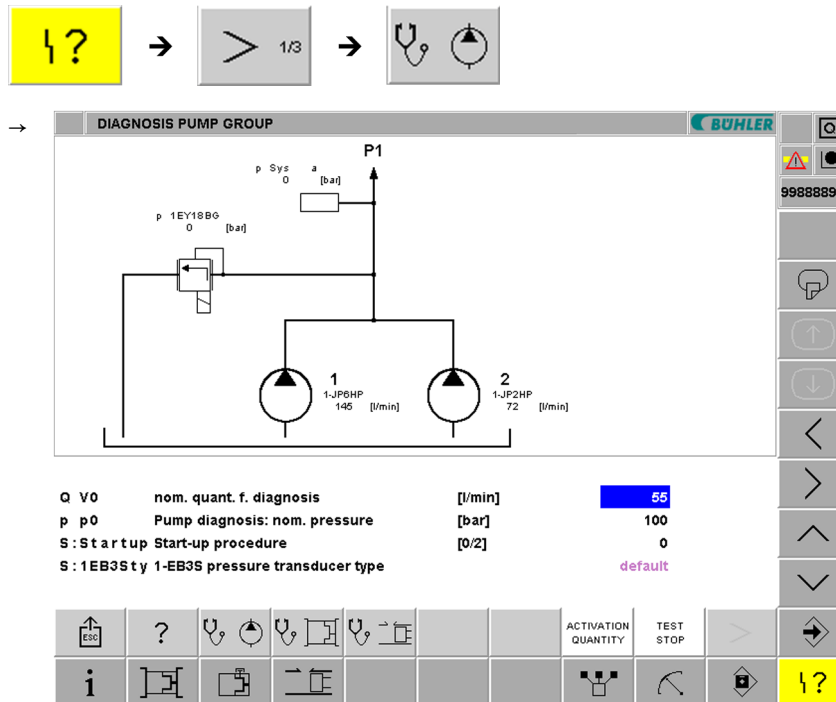
- Ensure that no one else can start the machine.
- Attach a warning sign on the machine.

Connecting the manometer

1. Remove the metal cap from the measuring point (1) of the pressure regulator (2).
2. Connect the manometer to the measuring point (1).

Adjusting the pump pressure

1. Select the "Drive unit diagnostics" page with authorization level "4" on the operating panel.



NOTICE Pressure too high. Pump damage Do not set pressure above 210 bar.

2. Set the following values on the "Drive unit diagnosis" page.
 - ▶ Set the "Q V0" parameter to the displayed value of pump 2 "1-JP2HP".
 - ▶ Set the "p p0" parameter to value "210".
3. Press the [Activation quantity] button on the operating panel.
 - The [Activation quantity] button is illuminated.
 - The set pressure is built up in the hydraulic system.

Adjusting the pressure regulator

1. Remove the protective cap from the pressure regulator (2).
2. Loosen the jam nut on the pressure regulator (2).
3. Turn the adjusting screw (3). While doing so, observe the pressure on the manometer.
4. When the desired pressure is displayed on the manometer, tighten the jam nut.

Removing the manometer

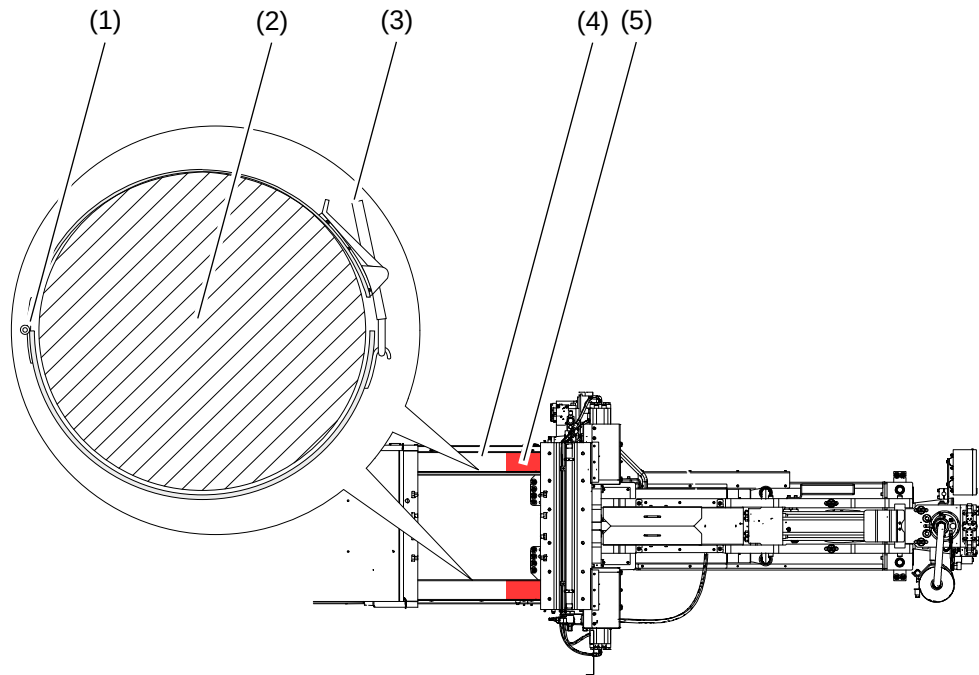
1. Press the [TEST STOP] button on the "Drive unit diagnostics" page on the operating panel.
 - The [Activation quantity] button is deactivated.
 - Pressure in the hydraulic system is dissipated.

2. Remove the manometer.
3. Attach the metal cap to the measuring point (1).

8.7.11

Attaching the tie bar protection

To protect the tie bars from liquid metal splashes, a tie bar protection element can be attached.



- | | |
|-------------|--------------------|
| (1) Hinge | (4) Tie bar |
| (2) Tie bar | (5) Protected area |
| (3) Closure | |

Prerequisites:

- Machine must be stopped and must be in home position.
- Machine must be secured against unexpected startup.

1. Clean and lubricate tie bars (4).
2. Position the tie bar protection element so that it is centered on the parting area of the die. In this regard, ensure that the tie bar sections that can be damaged by metal splashes are covered.
3. Close the lock (3) on the tie bar protection.

8.7.12

Adjusting the nitrogen pressure

1. Adjust the nitrogen pressure as needed. See page 68, chapter "Adjusting the nitrogen pressure".

9 Troubleshooting

9.1 Safety

9.1.1 Personnel qualifications

If improperly executed, personnel can be injured and machine parts can be damaged.

- The plant operator must assign responsibilities and make these known.

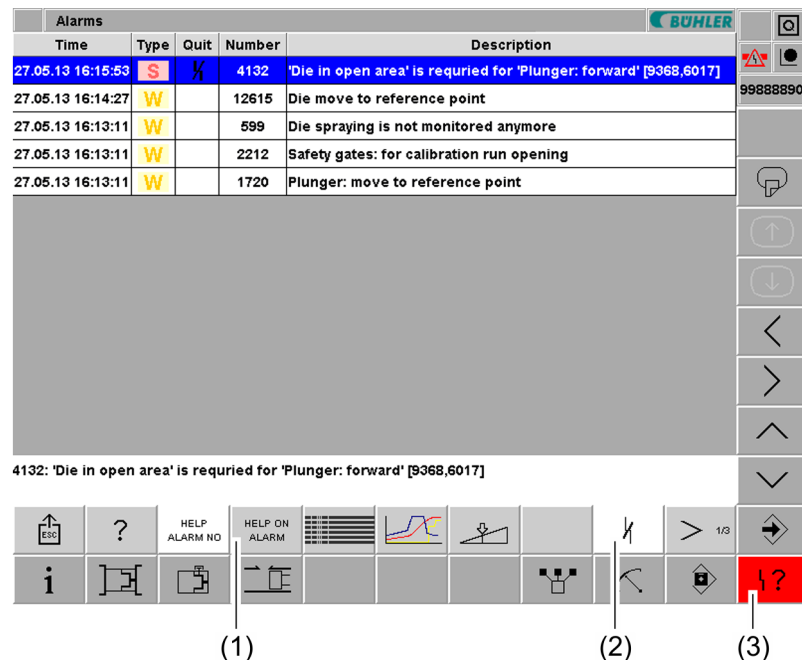
9.1.2 Mobile climbing aids

If mobile climbing aids, such as steps or ladders fall over, personnel can be injured.

- Only use free-standing climbing aids.
- Never lean climbing aids against the machine.
- Do not climb onto the machine.

9.2 Alarms

If there are faults, required maintenance messages, operation errors, and faulty settings, an alarm message will be generated. Alarm messages are displayed on the operator panel and can be displayed via the keys on the control cabinet.



- (1) Alarm help buttons
- (2) [Acknowledge] button
- (3) [Alarms] button

Two different types of alarm messages are possible: Warnings and faults.

9.2.1 Warnings

Warnings indicate required maintenance or faulty settings.

A warning causes the following:

- [Alarms] (3) button is illuminated yellow.
- Status symbol "Alarm" is yellow.



9.2.2 Faults

Faults indicate malfunctions and critical / faulty machine settings.

A fault causes the following:

- [Alarms] (3) is illuminated red.
- Status symbol "Alarm" is red.



- The machine stops automatically.

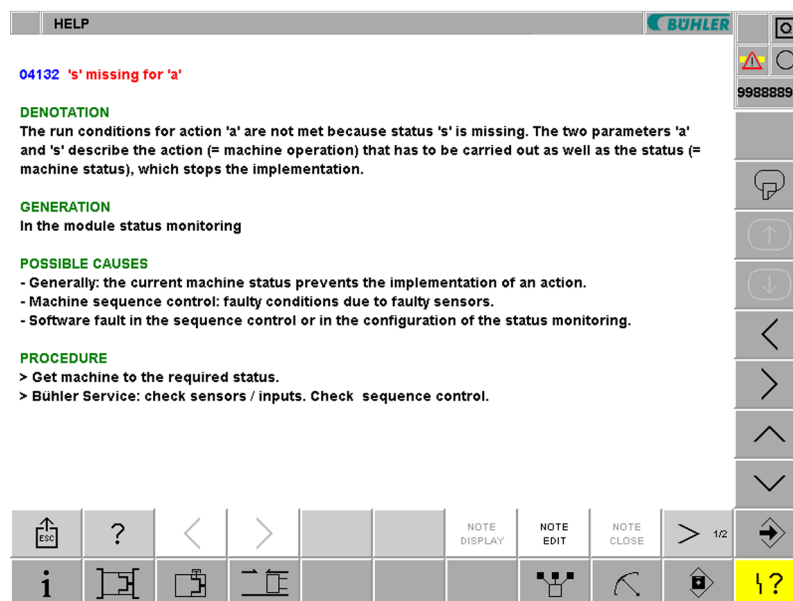
9.3 Displaying current alarm messages

All current alarm messages can be displayed.

- ▶ Press the [Alarms] button on the operator panel.
 - Warnings are marked in yellow and faults are marked in red.

9.4 Displaying help for alarm messages

Help pages for alarm messages contain information on the type of alarm, causes, and remedy.



Help pages for alarm messages can be displayed in two ways:

- Display through highlighting
- Display via alarm number

9.4.1 Highlighting and displaying an alarm message

1. Highlighting alarm messages.
2. Press the button [Help for alarm].

9.4.2 Displaying alarm message via alarm number

1. Press the button [Help alarm no.].
2. Use the number pad to enter the number of the alarm.
3. Confirm the entry with the [Enter] button on the number pad.
4. Display the alarm number with the [display] button.

9.5 Procedure for alarm messages

1. Follow the instructions on the help pages.
2. If acknowledgement is necessary press the [Acknowledge] button.
 - Alarm message is deleted.

9.6 Procedure for maintenance alarm messages

On the operator panel an alarm message can appear that indicates necessary maintenance tasks.

1. Execute the required maintenance task.
2. Press the [Maintenance] button on the operator panel.
3. Press the button for executed maintenance interval.
 - The maintenance message will disappear from the list of alarm messages.

10 Maintenance

10.1 Safety

10.1.1 Personnel qualifications

If improperly executed, personnel can be injured and machine parts can be damaged.

- ▶ The plant operator must assign responsibilities and make these known.

10.1.2 Mobile climbing aids

If mobile climbing aids, such as steps or ladders fall over, personnel can be injured.

- ▶ Only use free-standing climbing aids.
- ▶ Never lean climbing aids against the machine.
- ▶ Do not climb onto the machine.

10.2 Symbols used



(1)



(2)



(3)

- (1) Time required
- (2) Tools required
- (3) Material required

- ▶ Comply with the information on the service and maintenance tasks.

10.3 Cleaning

NOTICE

Water accumulation and foreign objects between moving components.
Premature wear and rust formation.

- ▶ When cleaning with compressed air do not blow any foreign objects between moving machine parts.
- ▶ Do not clean moving machine parts with a steam cleaner.
- ▶ Immediately clean up water accumulation and regrease surfaces.

10.3.1

Cleaning schedule

Interval	Machine part	Duration	Measure
24 h / once a day	Wear bands and slide surfaces of the tie bars	00:05	► Clean the wear bands and slide surfaces of the tie bars.
120 h / once a week	Collection trough	00:30	► Remove oil, grease, and spray agent from the collection trough.
	Machine	01:00	► Clean the machine.
500 h / once a month	Drive unit	00:10	► Clean the vent grille of the drive motor.
	Water cooling system, hydraulic fluid	00:15	► Clean the water inlet line strainer.
6,000 h / once a year	Plate heat exchanger	01:00	► Clean the plate heat exchanger.
	Central lubrication unit	01:00	► Clean intake screen. ► Clean the lubricant tank.

10.3.2

Cleaning wear bands and slide surfaces of the tie bars

 5 minutes

 Tear-proof, lint-free cleaning cloth

 Gas-oil mixture

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Soak cleaning cloth in the detergent.
2. Clean wear bands.
3. Clean slide surfaces of the tie bars.

10.3.3

Cleaning the machine

 1 hour

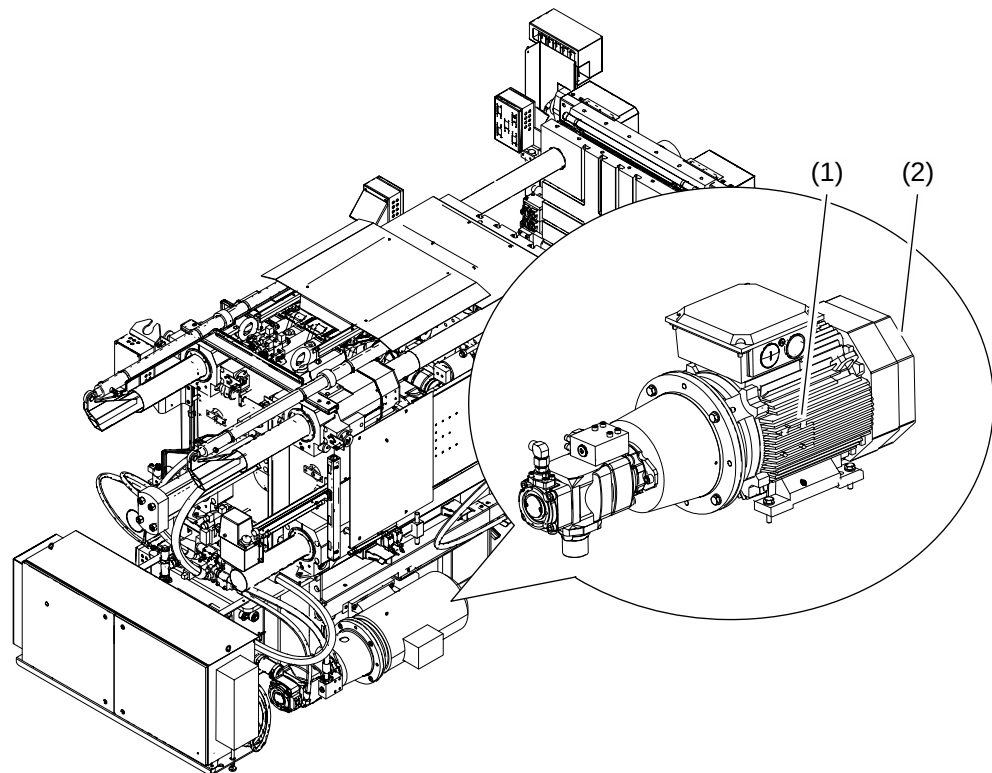
 Tear-proof, lint-free cleaning cloth

 Detergent

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Soak cleaning cloth in the detergent.
2. Wipe fouling off of the machine.



- (1) Drive motor
(2) Ventilation grating with fastening

🕒 10 minutes

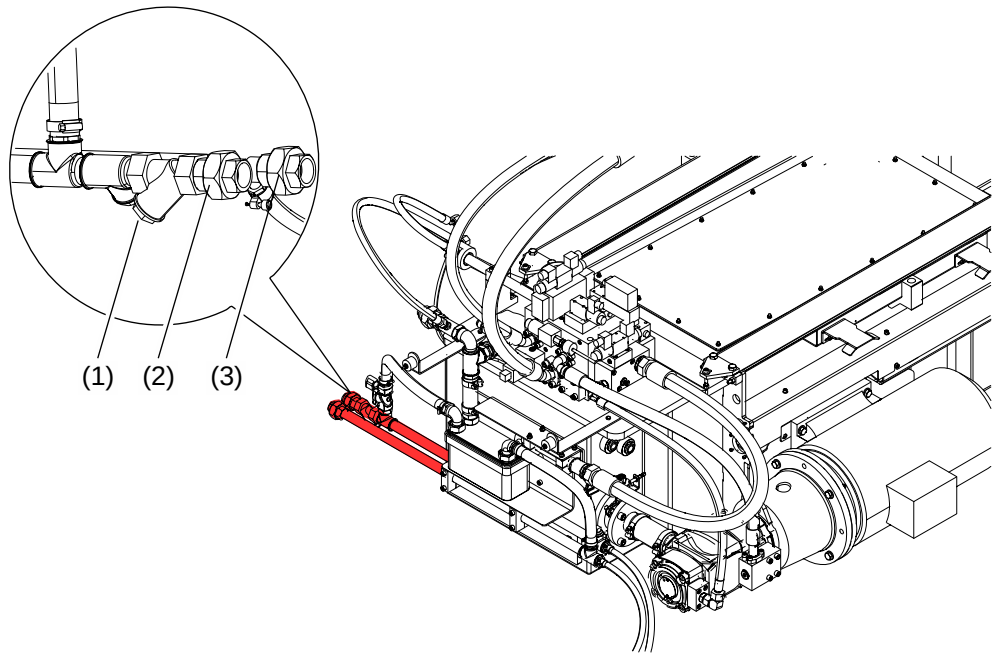
🔧 Tear-proof, lint-free cleaning cloth; compressed air

🛒 Detergent

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Remove the vent grille (2).
2. Clean the vent grille (2).
3. Clean the fan.
4. Install the vent grille (2).



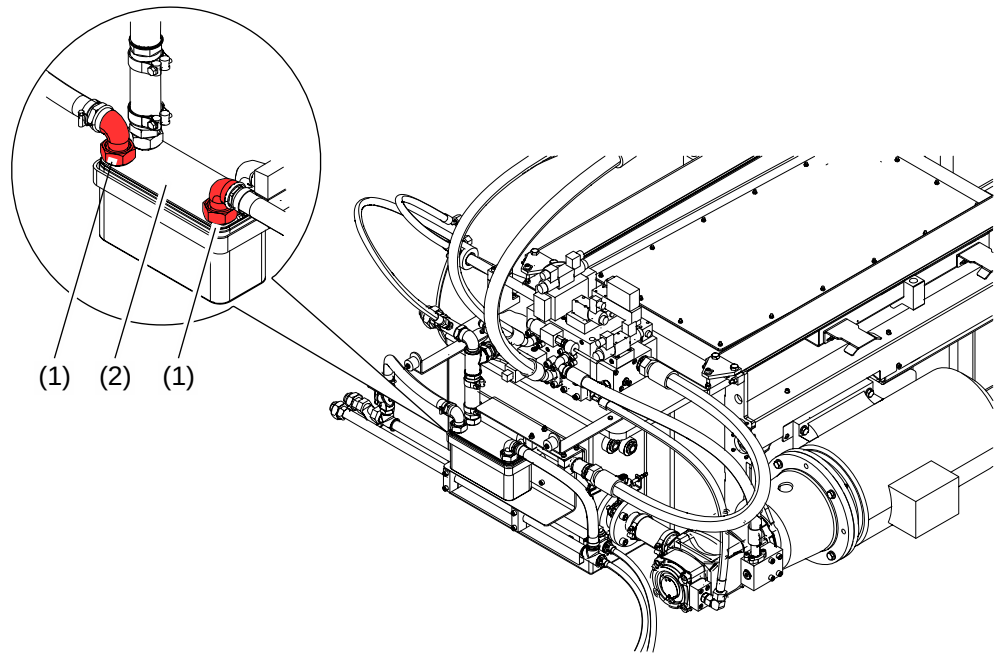
- (1) Y-type strainer
- (2) Water inlet
- (3) Water return

⌚ 15 minutes

🔧 Collection container, compressed air

Prerequisites:

- Machine must be stopped.
 - Machine must be secured against unexpected startup.
 - Water inlet must be shut off.
1. Place the collection container under the line strainer (1).
 2. Screw off the plug.
 3. Pull out intake screen.
 4. Clean or replace intake screen.
 5. Install intake screen.
 6. Firmly screw on plug.
 7. Open water inlet (2).
 8. Check line strainer (1) for leaks.
 9. Remove collection container.



- (1) Water flow and water return on the plate heat exchanger
 (2) Plate heat exchanger

🕒 1 hour

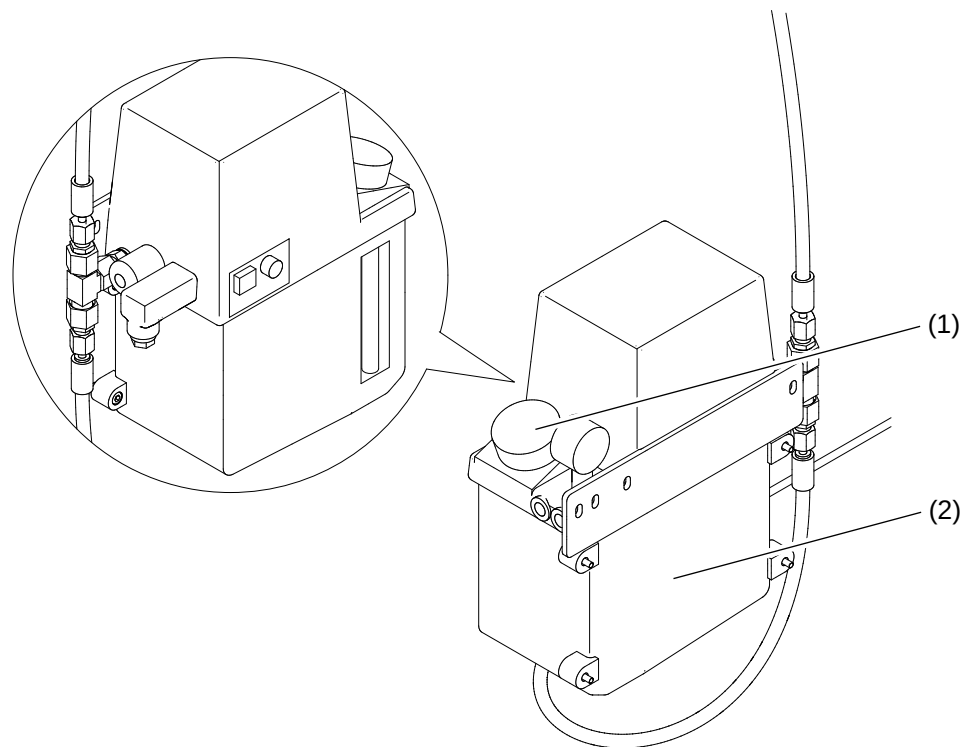
🔧 Collection container

🛒 5% phosphoric acid or 5% oxalic acid

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.
- Water flow must be shut off.

1. Place collection container under the plate heat exchanger (2).
2. Unscrew the cooling water flow and cooling water return (1) from the plate heat exchanger (1).
3. Allow cleaning solution to flow through the heat exchanger (2) several times counter to the water flow direction.
4. Thoroughly rinse out the plate heat exchanger (2) with water.
5. Screw the cooling water flow and cooling water return (1) onto the plate heat exchanger (1).
6. Remove collection container.



- (1) Filling opening with intake screen
- (2) Container for central lubricant

🕒 1 hour

🔧 Tear-proof, lint-free cleaning cloth; compressed air

🛒 Detergent

1. Pull the intake screen out of the filling opening (1).
2. Blow out the intake screen with compressed air.
3. Reinsert the intake screen.
4. Wipe off the container (2) exterior with a cleaning cloth.

10.4

Lubrication

10.4.1

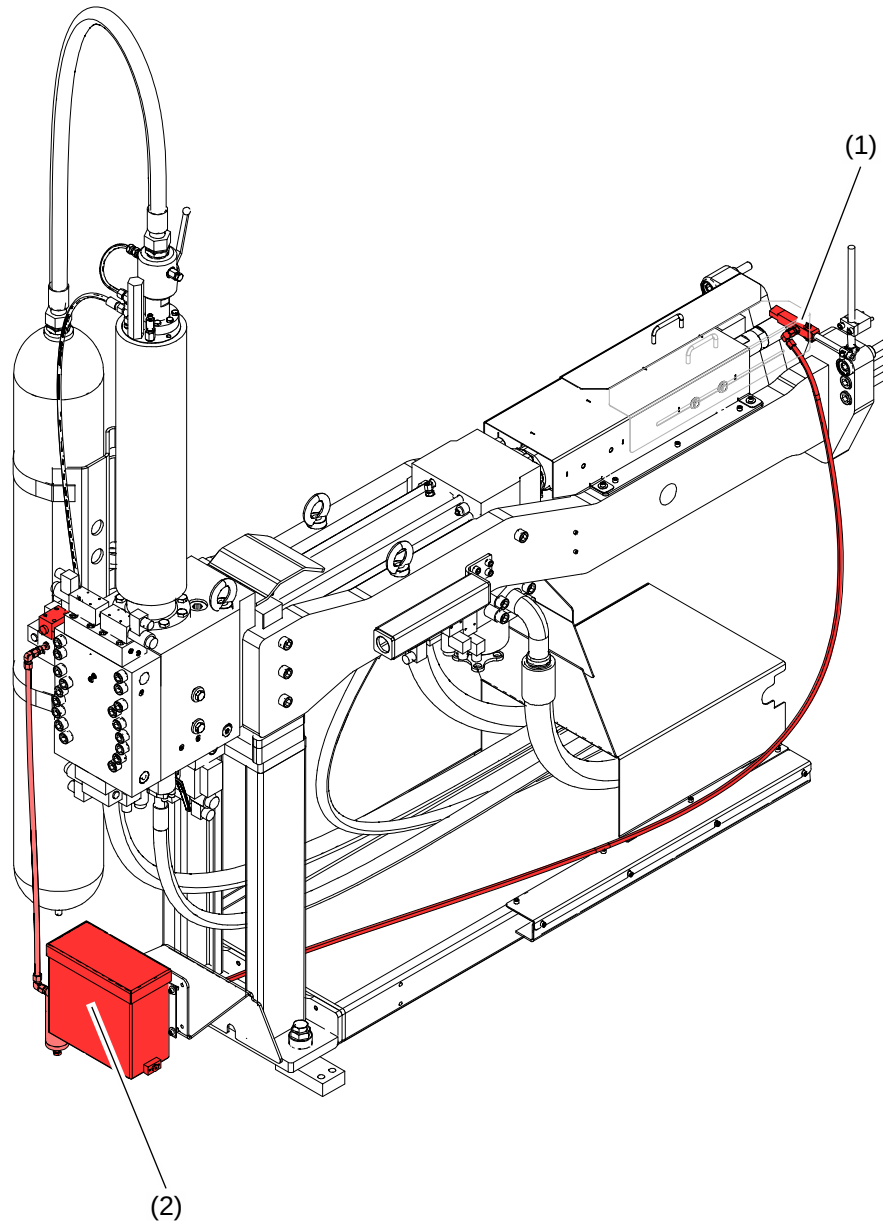
Lubrication schedule

Interval	Machine part	Dura- tion	Measure
120 h / once a week	Plunger lubrication system	00:05	▶ Top up lubricant tank.
500 h / once a month	Fixed plate	00:10	▶ Grease tie bar bearings.

Interval	Machine part	Duration	Measure
6,000 h / once a year	Fixed plate	00:10	► Grease split nuts.
	Tie bars	01:00	► Grease tie bar threads.

10.4.2

Topping up the lubricant tank



- (1) Lubricating bar
- (2) Lubricant tank

🕒 5 minutes

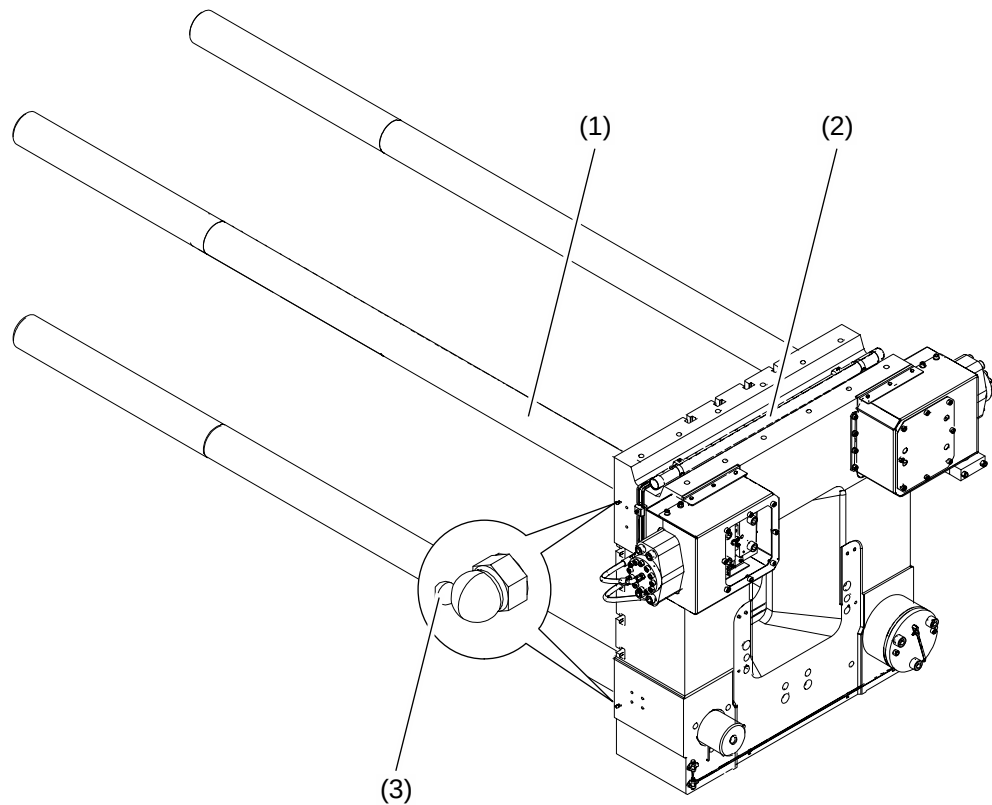
🛒 Lubricant. See page 19, chapter “Lubricant”.

1. Open the lid of the lubricant tank (2) at the rear of the injection unit.
2. Check the lubricant inside the tank (2) for contamination. If necessary, dispose of the lubricant.
3. Use a sieve to top up the lubricant tank (2).

4. Close the lid of the lubricant tank (2).

10.4.3

Greasing tie bar bearings of the fixed plate



- (1) Tie bars
- (2) Fixed plate
- (3) Grease nipple of the tie bar bearings

🕒 10 minutes

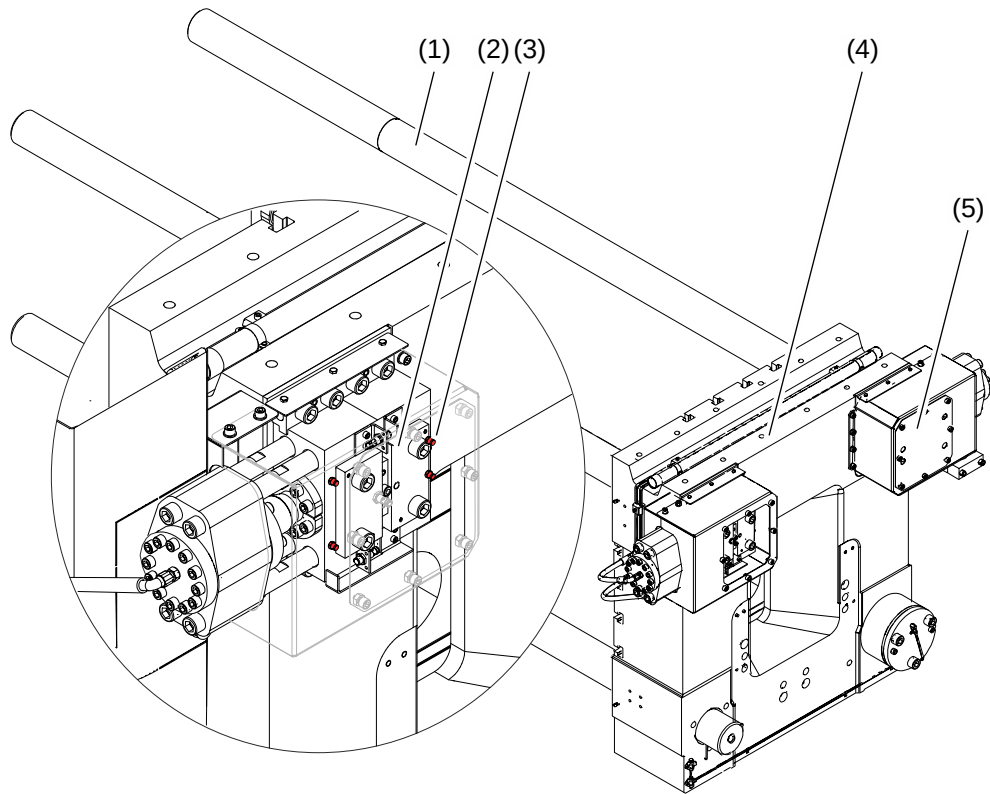
🔧 Grease gun

🛒 Grease See page 19, chapter "Grease for higher requirements".

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Press grease into the grease nipple (3) until grease escapes between the tie bar (1) and the bearing bush.



- | | |
|-------------------------------------|----------------------------------|
| (1) Tie bars | (4) Fixed plate |
| (2) Split nut (without cover sheet) | (5) Split nut (with cover sheet) |
| (3) Grease nipple | |

🕒 20 minutes

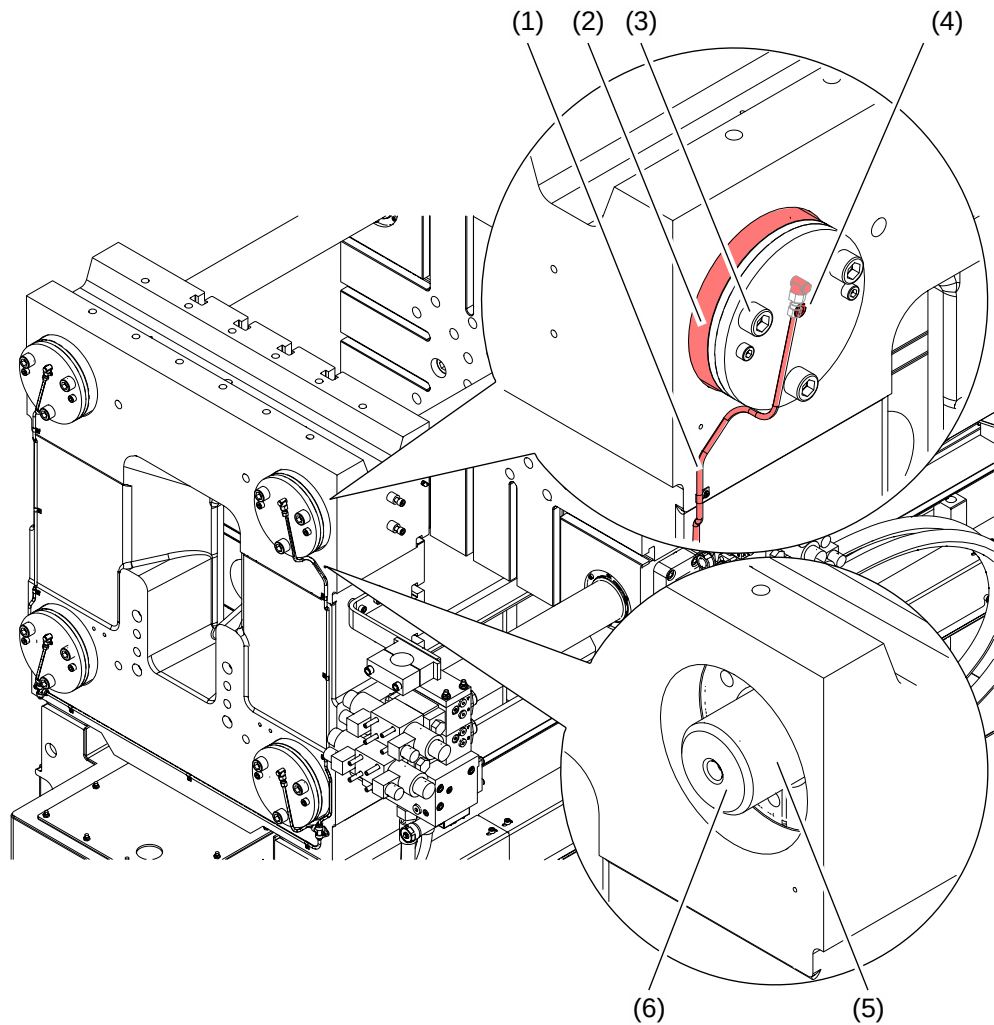
🔧 Grease gun

🛒 Grease. See page 19, chapter “Grease for higher requirements”.

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Unscrew the guard panel of the split nut (5).
2. Use a grease gun to get grease into the 4 grease nipples (3).
3. Pull the tie bars. See page 80, chapter “Automatic tie bar pulling”.
 - Split nut (2) opens.
4. Check whether the slide surface between the split nuts (2) and the fixed plate (4) is greased.
5. Repeat the lubrication process if lubrication is inadequate.
6. Insert the tie bars.
7. Screw on the guard panel.



- (1) Lubrication connection
- (2) Tie bar nut
- (3) Screw

- (4) Positioning screw
- (5) Tie bar thread
- (6) Facing side of the tie bar

🕒 1 hour

🛒 Grease. See page 19, chapter “Grease for higher requirements”.

Prerequisites:

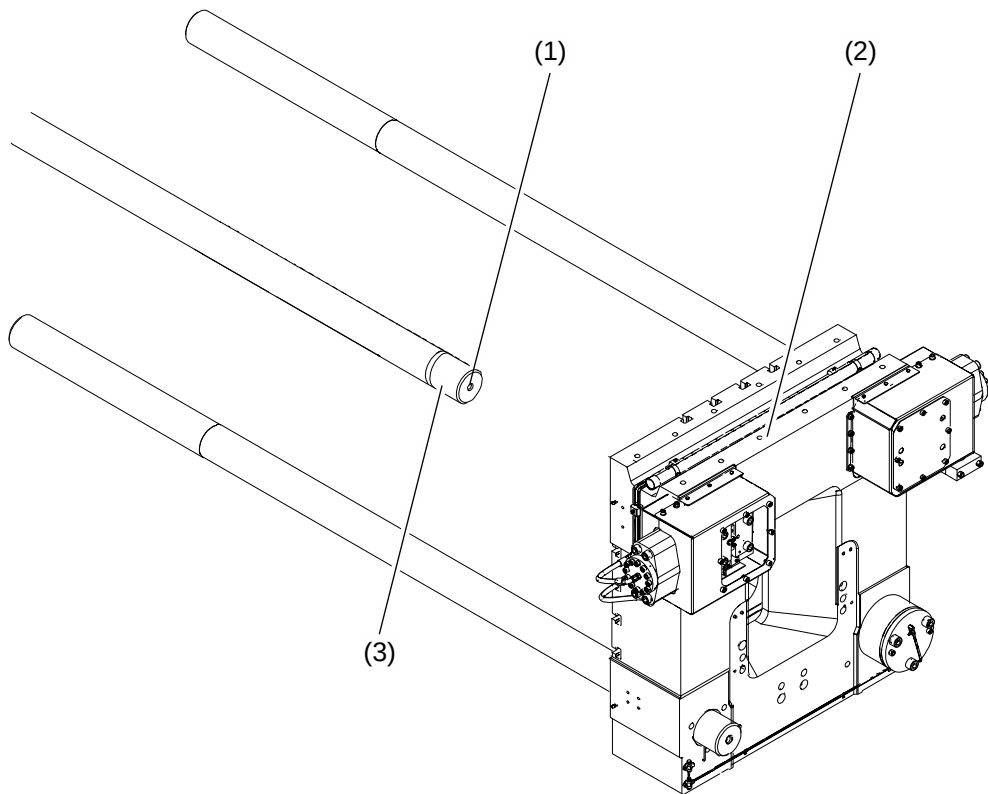
- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Remove lubrication connection (1).
2. Remove positioning screw (4).
3. Remove screws (3).
4. Remove tie bar nut (2)
5. Clean the thread of the tie bar nut (2).
6. Clean tie bar thread (5).
7. Grease tie bar thread (5) and front side of the tie bar (6).
8. Grease the thread of the tie bar nut (2).
9. Fit the tie bar nut (2) and align it in such a manner that the positioning screw (4) can be correctly introduced into the bore hole.
10. Fit positioning screw (4).
11. Fit screws (3) and firmly tighten.

12. Fit the lubrication connection (1).

10.4.6

Greasing the tie bar threads for automatic tie bar pull



- (1) Front side
- (2) Fixed plate
- (3) Tie bar thread

🕒 15 minutes

🔧 Spatula

🛒 Grease. See page 19, chapter “Grease for higher requirements”.

1. Pull tie bars and stop in the middle between the moving and the fixed plate (2). See page 80, chapter “Pulling tie bars”.
2. Stop the machine.
3. Secure the machine against unexpected startup.
4. Grease thread (3) and front side (1) of the tie bars.
5. Start up the machine.
6. Pull the tie bars. See page 81, chapter “Inserting the tie bars”.

10.5

Maintenance

⚠️ WARNING

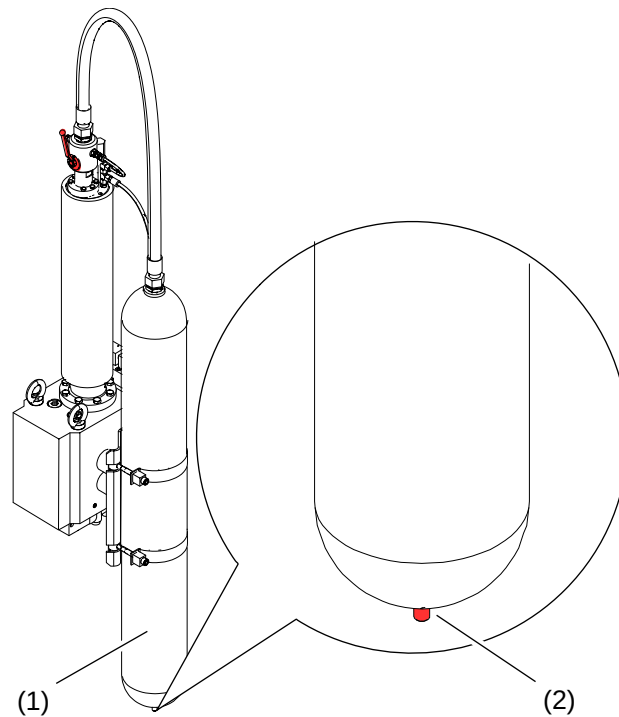
Parts flying about.

Danger of injury.

- Prior to task on the nitrogen system, discharge the nitrogen.

Interval	Machine part	Duration	Measure
24 h / once a day	Switch cabinet	00:05	▶ Check doors (visual inspection). ▶ Lock unlocked doors.
	Terminal boxes on the machine	00:05	▶ Check doors (visual inspection). ▶ Lock unlocked doors.
120 h / once a week	Protective doors	00:05	▶ Check the locking function of the protective doors.
	Hydraulic system	00:05	▶ Check the components for leaks. (Visual inspection)
	Injection unit	00:10	▶ Drain leak oil from the nitrogen bottles. See page 114, chapter "Draining leak oil from the nitrogen bottles".
500 h / once a month	Operating column	00:05	▶ Check switches and signal lights. Replace defective parts.
	Tie bar nuts	__:__	▶ Check the load on the tie bars. See page 115, chapter "Checking the load of the tie bars".
1,500 h / 3 months	Hydraulic fluid	00:10	▶ Take a sample and have it analyzed. ▶ Change contaminated or used hydraulic fluid.
	Hydraulic tank	00:10	▶ Replace the filter elements of the tank air filters. See page 116, chapter "Replacing the filter elements of the tank air filters".
3,000 h / 6 months	Die height shifting	00:30	▶ Check the chain drive. See page 117, chapter "Checking the chain drive".
	Operating panel	00:05	▶ Replace the protective foil on the touch screen.
6,000 h / once a year	Machine and control system	01:00	▶ Check the status of the machine and control system. Draw up a list of defects and rectify the defects. ▶ Check the switch cabinet and the control boxes for leaks. Rectify defects.
	Protective doors	00:10	▶ Check the belt tension of the protective doors.

Interval	Machine part	Duration	Measure
	Injection unit	00:45	▶ Change the sleeve on the displacement sensor of the plunger. ▶ Check holder, holder flange, and bottle retention of pressure vessels for the formation of cracks (visual inspection). Check all bolts/screws for tightness.
	Hydraulic hoses	00:30	▶ Check hydraulic hoses for abrasion and cracks. ▶ Replace defective hoses.
	Central lubrication system	01:00	▶ Check lines and manifold for leaks. ▶ Replace defective components.
	Worm gear housing	00:10	▶ Check the oil level in the worm gear housing. See page 120, chapter "Check the oil level in the worm gear housing".
	Slide plates and slide belts	04:00	▶ Replace the slide plates and slide belts. See page 123, chapter "Replacing the slide plates and slide belts". ▶ Replace scraper plates.
18,000 h / 3 years	Control PC in switch cabinet	00:15	▶ Replace the buffer battery in the control computer. See page 131, chapter "Replacing the battery in the control computer".
30,000 h / 5 years	Hydraulic hoses	02:00	▶ Replace hydraulic hoses.



- (1) Nitrogen bottles
 (2) Measuring points with metal cap

🕒 10 minutes

🔧 Collecting container, vent hose (machine accessories)

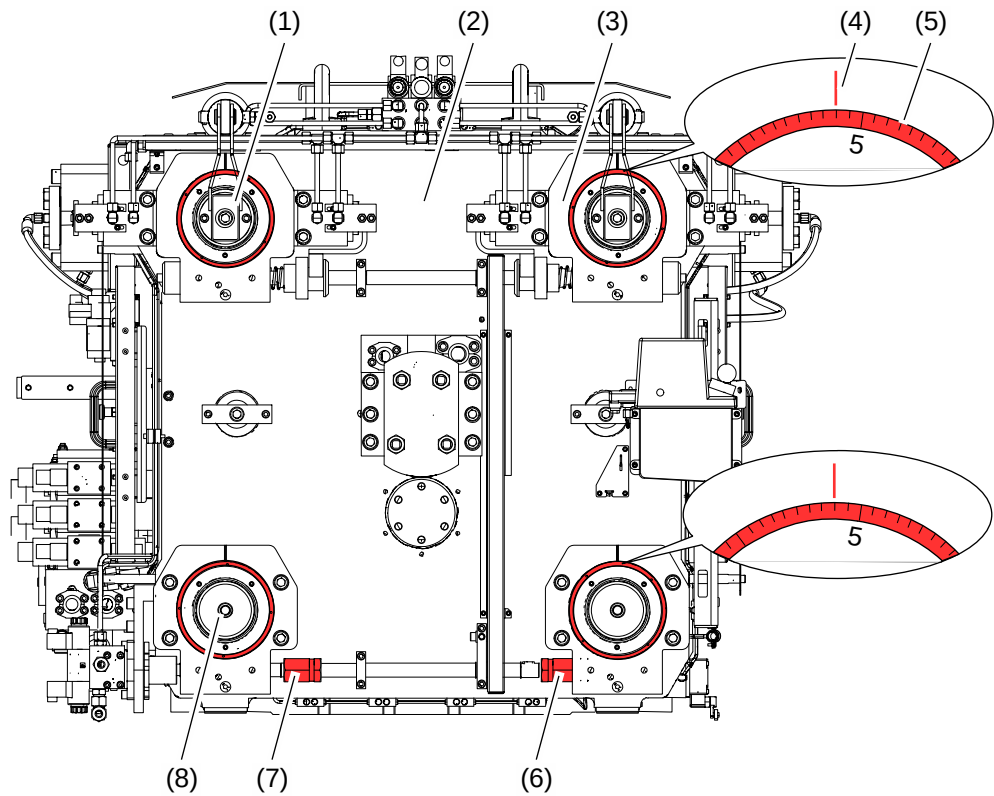
Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

Important

- ▶ If a lot of leak oil flows out, check piston seals of the piston accumulator.

1. Place collection container under the measuring point (2) of the nitrogen bottle (1).
2. Remove the metal cap of the measuring point (2).
3. Firmly screw the ventilation hose (2) onto the measuring point.
 → Leak oil will flow off.
4. When leak oil no longer escapes, unscrew the ventilation hose.
5. Mount the metal cap of the measuring point (2).



- | | |
|--|---|
| (1) Tie bars with automatic tie bar pull | (5) Scale ring |
| (2) Cylinder plate | (6) Tie bar coupling (uncoupled) |
| (3) Tie bar nut | (7) Tie bar coupling (coupled in) |
| (4) Marking | (8) Tie bars without automatic tie bar pull |

⌚ 5 minutes

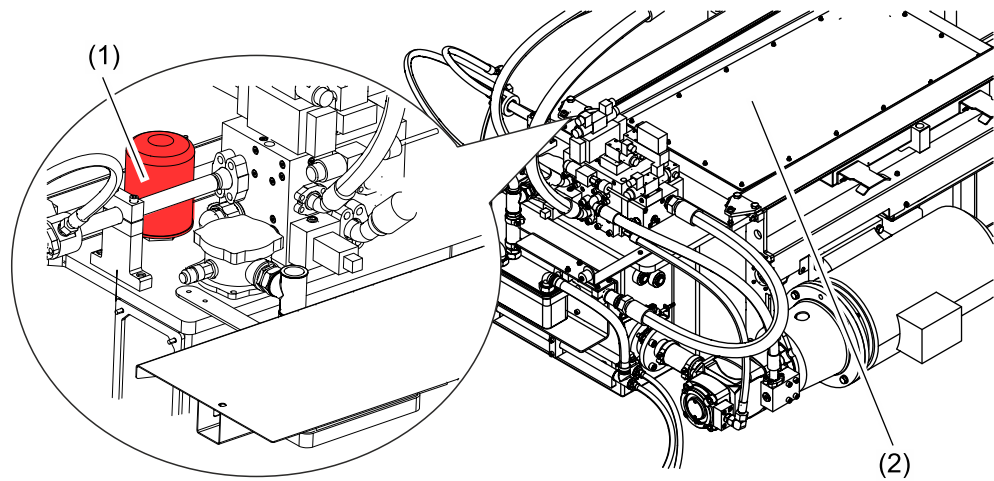
Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

NOTICE

- Different loads on the tie bars.
Breakage of the tie bars.
High wear of the locking unit.
- ▶ Use parallel dies.
 - ▶ Manually correct the load on the tie bars.

1. Check the load of the tie bars on all four tie bars. In this process ensure that the deviation on the scale ring (5) is 3 scale lines maximum. Adjust the load of the tie bars, if necessary. See page 85, chapter "Correcting the load of the tie bars".
2. Check the position of the tie bar coupling.



- (1) Tank air filter
(2) Hydraulic tank

🕒 10 minutes

🛒 Filter element spare part

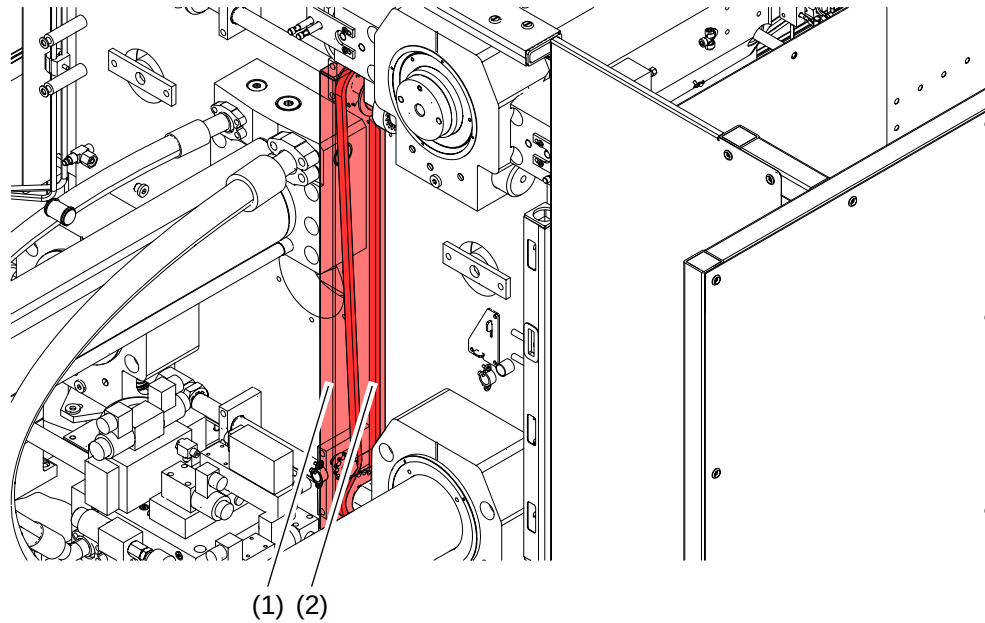
Prerequisites:

- Machine must be stopped.
 - Machine must be secured against unexpected startup.
1. Dismount the protective hood of the tank air filter (1).
 2. Remove the filter element from the tank air filter (1).
 3. Clean protective hood and support surface of the filter element.
 4. Insert new filter element.
 5. Mount the protective hood.

10.5.5

Checking the chain drive

If the chain of the chain drive is excessively tight, problems can occur with the die height adjustment.



(1) Chain box

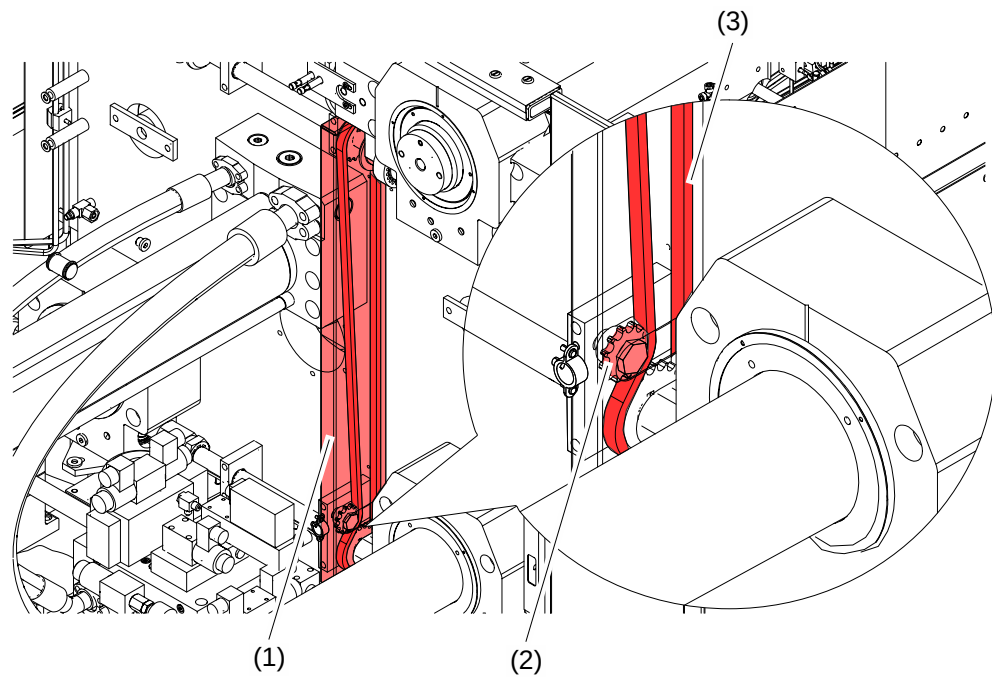
(2) Chain

⌚ 30 minutes

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Dismount the chain box (1).
2. Clean chain (2).
3. Check chain(2).
 - ▶ Check chain (2) for mechanical damage. If the chain (2) is damaged, replace chain (2).
 - ▶ Check chain tension. If the chain (2) is excessively tightened or insufficiently tightened adjust the chain tension. See page 118, chapter “Adjust the chain tension on the chain drive.”.
4. Lubricate the chain (2).
5. Mount the chain box (1).



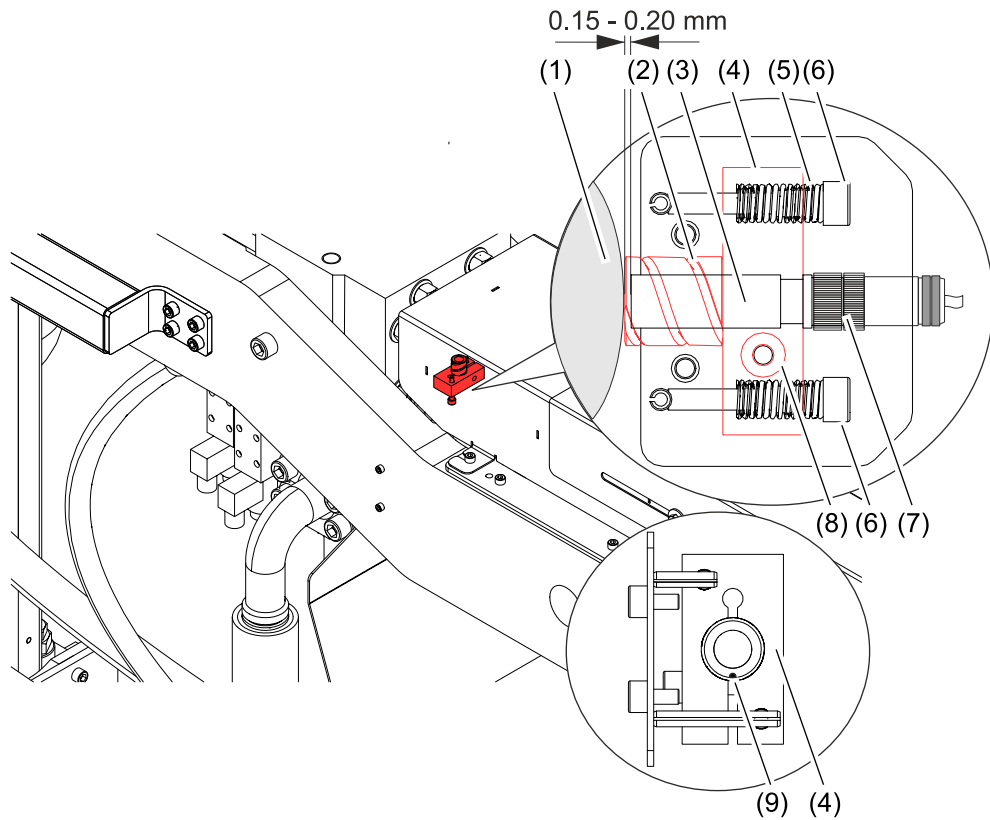
- (1) Chain box
- (2) Chain tensioner
- (3) Chain

🕒 10 minutes

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Dismount the chain box (1).
2. Adjust the tension of the chain (3) on the chain tensioner (2).
3. Mount the chain box (1).



- | | |
|-------------------------|---------------|
| (1) Piston rod | (6) Screw |
| (2) Sleeve | (7) Plug |
| (3) Displacement sensor | (8) Lock bolt |
| (4) Clamping plate | (9) Mark |
| (5) Spring | |

🕒 30 minutes

🛒 Sleeve (spare part)

Prerequisites:

- Machine is switched off.

NOTICE

Metal chips at the magnetic displacement sensor.
 Damage to the sensor and the piston rod.
 ► Carefully remove metal chips.

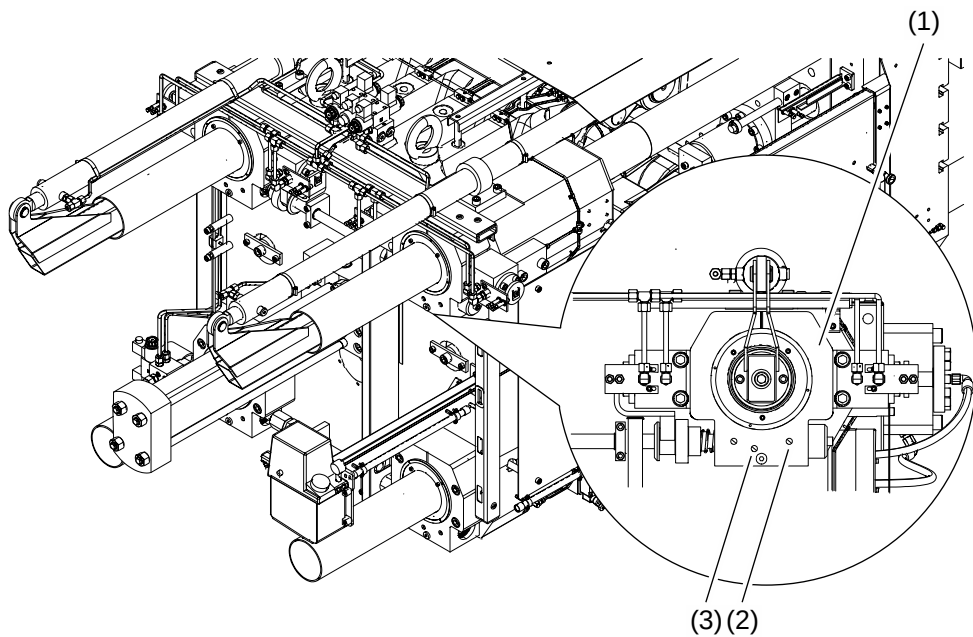
1. Loosen screws (6) and remove clamping plate (4).
2. Loosen lock bolt (8).
3. Remove displacement sensor (3) and sleeve (2).
4. Clean entire device.

NOTICE Forgotten spacer film. Damage to the displacement sensor. Remove spacer film before fitting the clamping plate.

5. Position displacement sensor (4) and new sleeve (3) in clamping plate.
 - ▶ Attach sticky side of spacer film to the displacement sensor (4).
 - ▶ The mark (9) should point downward.
 - ▶ Align displacement sensor (4) with sleeve
 - ▶ Tighten lock bolt (8).
 - ▶ Peel off non-sticky backing of spacer film.
 - There should be a 0.15 - 0.20 mm gap between the displacement sensor (4) and the end of the sleeve (2).
6. Attach clamping plate (4).
 - ▶ Position clamping plate (4).
 - ▶ Screw in screws (6) with springs (5) and tighten.

10.5.8

Check the oil level in the worm gear housing



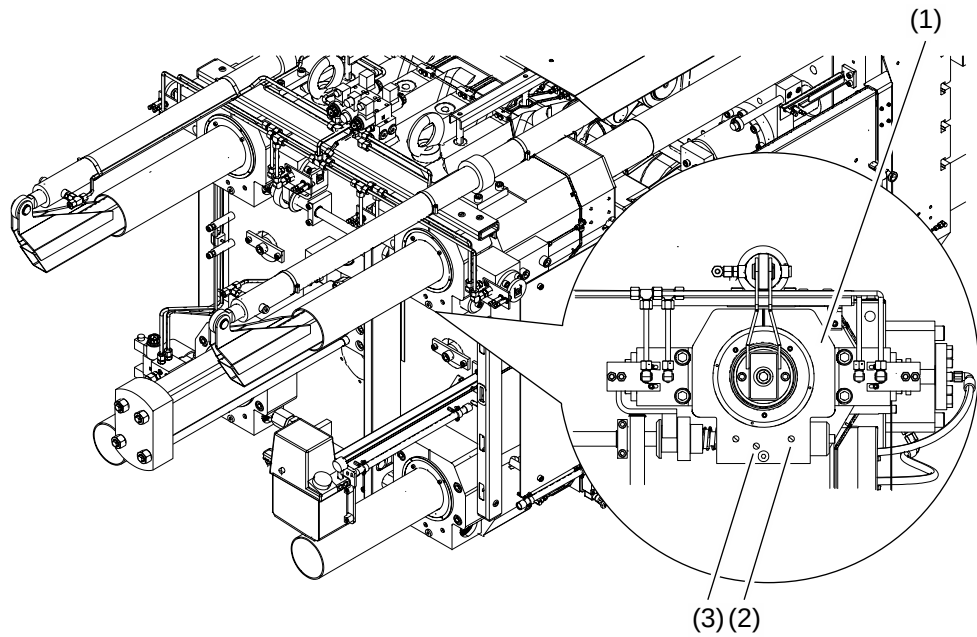
- (1) Worm gear housing
- (2) Plug
- (3) Plug

⌚ 10 minutes

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Remove plug (3).
 - ▶ If gear oil escapes immediately push the plug back into the opening.
 - ▶ If no gear oil escapes, top up gear oil.



- (1) Worm gear housing
- (2) Plug
- (3) Plug

🕒 10 minutes

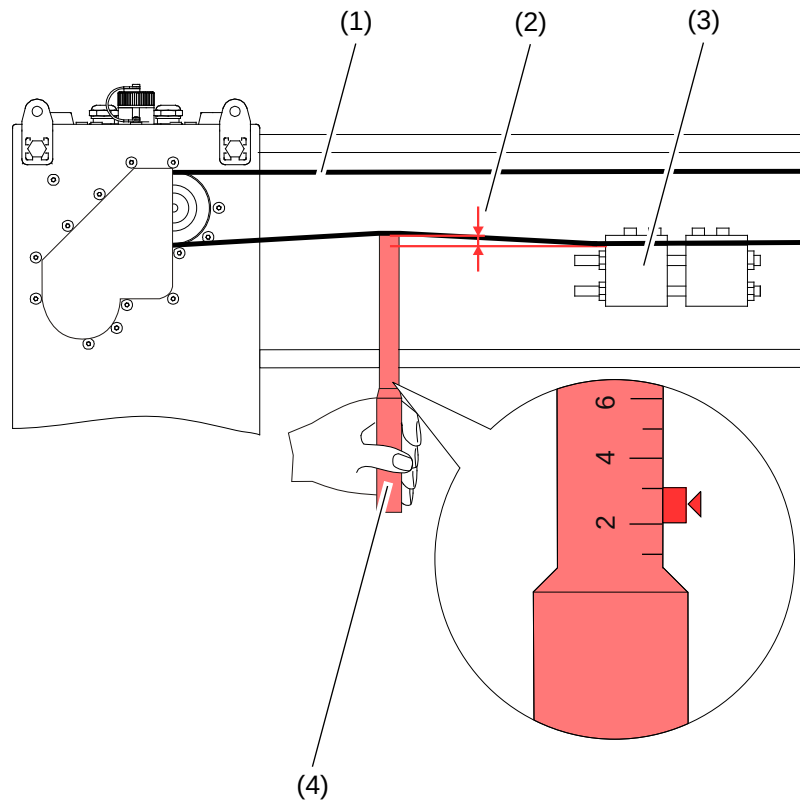
🔧 Oil can.

🛒 Gear oil. See page 18, chapter “CLP gear oil”.

Prerequisites:

- Machine is stopped.
- Machine must be secured against unexpected startup.

1. Remove plug (2) and plug (3).
2. Fill gear oil via the upper borehole, until the gear oil comes out of the lower borehole.
3. Re-attach plug (2) and plug (3).



(1) Timing belt

(2) Guide dimension

(3) Tension device

(4) Tension meter

⌚ 10 minutes

🔧 Tension meter

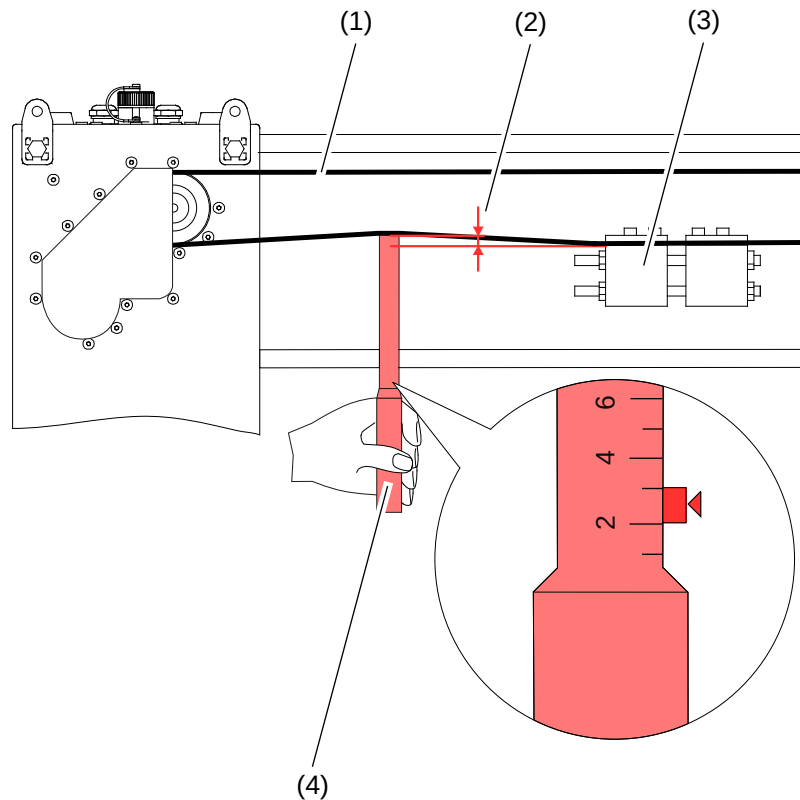
Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

1. Close the protective door by hand.
2. Mark the position of the timing belt (1).
3. Measure belt tension.
 - ▶ Press the tension meter (4) in the middle of the timing belt (1) upward from below.
 - ▶ Measure the distance to the mark: Guide dimension (2) ~ 10 mm.
4. If there is deviation from the guide dimension (2), adjust belt tension.

10.5.11

Adjusting the belt tension of the protective door



- | | |
|---------------------|--------------------|
| (1) Timing belt | (3) Tension device |
| (2) Guide dimension | (4) Tension meter |

⌚ 10 minutes

🔧 Tension meter

Prerequisites:

- Machine must be stopped.
 - Machine must be secured against unexpected startup.
1. Uncouple the tension device (3) from the protective door.
 2. Correctly adjust the belt tension (3) with the tension device.
 - ▶ Press the tension meter (4) in the middle of the timing belt (1) upward from below.
 - ▶ Measure the distance to the mark: Guide dimension (2) ~ 10 mm.
 3. Connect the tension device (3) to the protective door.
 4. Check belt tension.

10.5.12

Replacing the slide plates and slide belts

There are two ways of lifting the cylinder platen in order to change the slide plates and the slide belts:

- With a hoisting device. See page 124, chapter “Replacing the slide plates and slide belts with a hoisting device”.
- If a crane is unavailable: With the aid of a transport safeguard. See page 127, chapter “Replacing the slide plates and slide belts with the aid of the transport safeguard”.



NOTICE

Applying too much force to lift the cylinder platen.

Damage to the machine base.

- ▶ Raise the cylinder platen by 10 mm at the most.
- ▶ If a crane is available, lift the cylinder platen with a hoisting device.

10.5.13

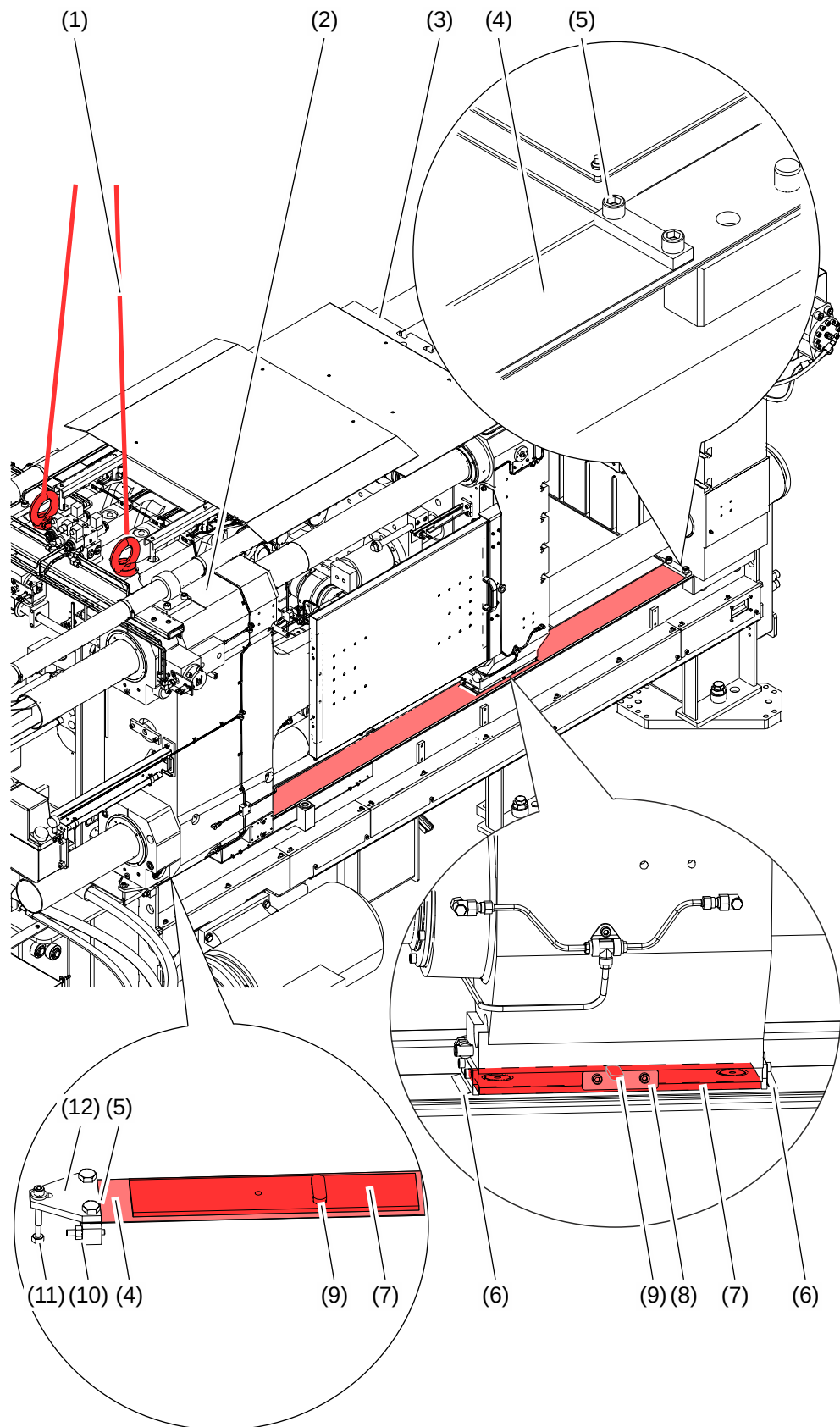
Replacing the slide plates and slide belts with a hoisting device

Replacing the slide plates and slide belts requires the following work steps:

1. Remove slide plates and slide belts.
2. Attach slide belts.
3. Attach slide plates.
4. Lubricate slide plates.

🕒 4 hours

🛒 Slide plates, slide belts, scrapers



- (1) Hoisting device
- (2) Cylinder platen
- (3) Moving plate
- (4) Slide belt

- (7) Slide plate
- (8) Wedge securing plate
- (9) Wedge
- (10) Tension bolt

- (5) Fastening screw
- (6) Scraper

- (11) Adjusting screw
- (12) Clamp plate



NOTICE

Applying too much force to lift the cylinder platen.

Damage to the machine base.

- Raise the cylinder platen by 10 mm at the most.
- If a crane is available, lift the cylinder platen with a hoisting device.

Removing slide plates and slide belts

- Machine must be in home position and in set-up mode.
 - Die is removed.
1. Turn the <Close die in set-up mode> key-operated switch to the right.
 2. Open the "Die height, tie bars, core movement" screen page and pre-select "die height".



3. Open the locking unit all the way using the <Run preselected movement> switch.
4. Switch off the machine and secure it against unexpected startup.
5. Remove scrapers (6) and wedge securing plates (8).
6. Slowly lift cylinder platen (2) with a crane by 10 mm at the most.
7. Remove slide plates (4) and wedge (9).
 - Push the slide plate (4) on the cylinder platen (2) toward the inside of the machine.
 - Push the slide plates (4) on the moving plate (3) toward the outside of the machine.
8. Remove the fastening screws (5) of the slide belts and remove the slide belt.

Attaching the slide belts

1. Clean the support surface of the slide belt (4) on the machine base and apply a corrosion preventative.
2. Put on a new slide belt (4).
3. Screw in the fastening bolts (5) until they rest on the slide belt. Do not tighten.
4. Unscrew the clamp screws (10).
5. Align the clamp plate (12) horizontally with the adjustment screw (11).
6. Clamp the slide belts (4) with the clamp screws (10).
7. Fix the clamp screws (10) in place with the counter nuts.
8. Tighten the fastening screws (5).
9. Clean and oil the slide belt (4). See page 17, chapter "Central lubricating oil CGLP".

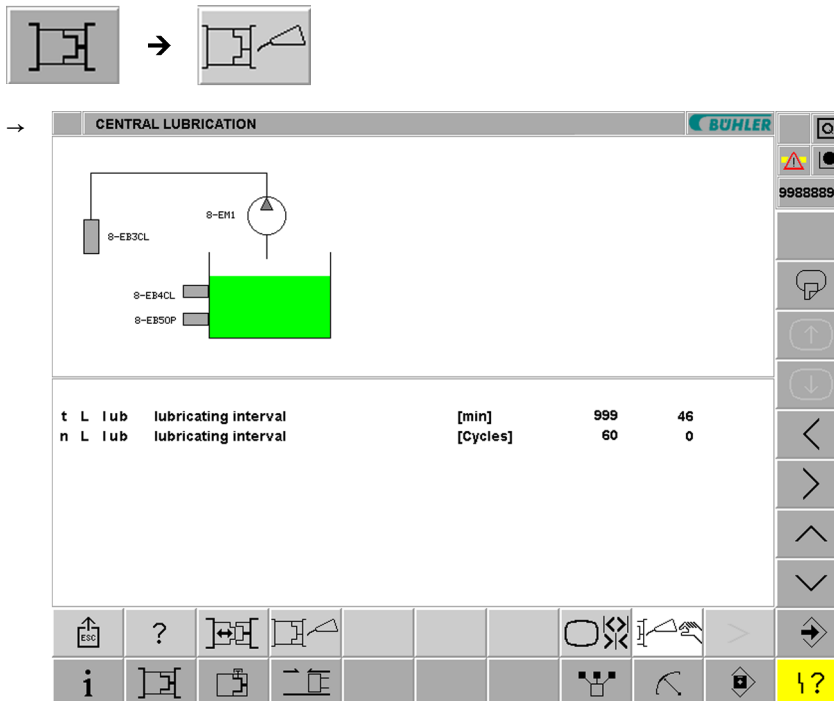
Attach slide plates.

1. Push the slide plate (7) on the moving plate (3) with an inserted wedge (9) on the slide belt (4) toward the inside of the machine.

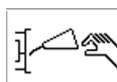
2. Push the slide plate (7) on the cylinder platen (2) with an inserted wedge (9) on the slide belt (4) toward the outside of the machine. In this process, lower the cylinder platen until the wedge (9) is fixed in place in the groove.
3. Slowly lower the cylinder plate (2).
4. Remove hoisting device (1).
5. Install scrapers (6) and wedge securing plates (8).

Lubricating the slide plates

1. Switch on the machine.
2. Open the "Central lubrication" screen page.



3. Press the button.



→ Lubricating cycle will be executed.

4. Wait 10 seconds.
→ Dosing pressure is completely dissipated.
5. Trigger a total of 20 lubrication cycles with pauses.
→ Sliding plates are lubricated.

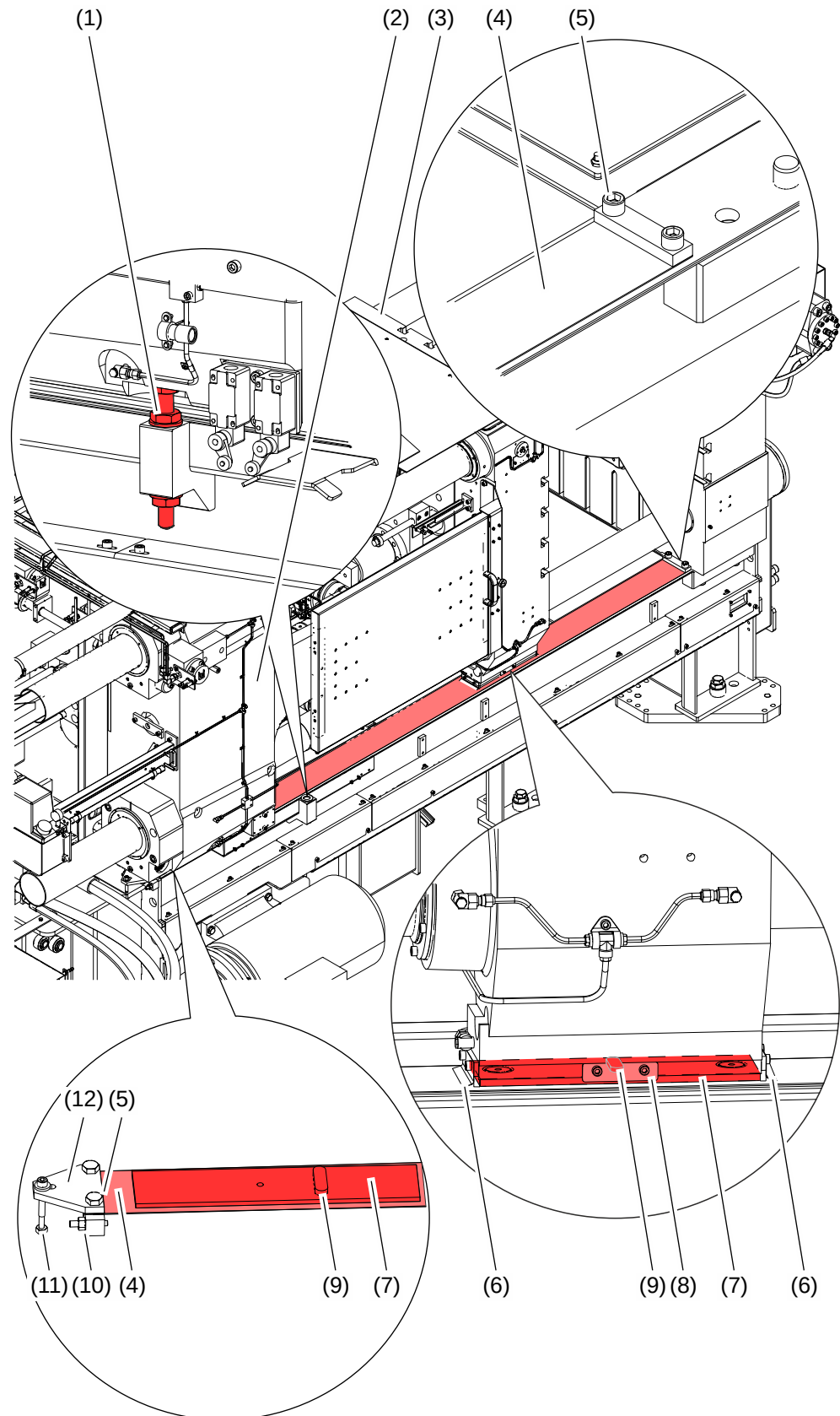
10.5.14 Replacing the slide plates and slide belts with the aid of the transport safeguard

Replacing the slide plates and slide belts requires the following work steps:

1. Remove slide plates and slide belts.
2. Attach slide belts.
3. Attach slide plates.
4. Lubricate slide plates.

⌚ 4 hours

 Slide plates, slide belts, scrapers



(1) Screw of the transport safe-guard

(2) Cylinder platen

(3) Moving plate

(7) Slide plate

(8) Wedge securing plate

(9) Wedge

- | | |
|---------------------|----------------------|
| (4) Slide belt | (10) Tension bolt |
| (5) Fastening screw | (11) Adjusting screw |
| (6) Scraper | (12) Clamp plate |



NOTICE

Applying too much force to lift the cylinder platen.

Damage to the machine base.

- ▶ Raise the cylinder platen by 10 mm at the most.
- ▶ If a crane is available, lift the cylinder platen with a hoisting device.

Removing slide plates and slide belts

- Machine must be in home position and in set-up mode.
 - Die is removed.
1. Move the cylinder platen (2) into transport position.
 2. Switch off the machine and secure it against unexpected startup.
 3. Remove scrapers (6) and wedge securing plates (8).
 4. Fit the screws of the transport safeguard (1).
 5. Evenly lift the cylinder platen (2) with the screws of the transport safeguard on both sides of the machine by 10 mm.
 6. Remove slide plates (4) and wedge (9).
 - ▶ Push the slide plate (4) on the cylinder platen (2) toward the inside of the machine.
 - ▶ Push the slide plates (4) on the moving plate (3) toward the outside of the machine.
 7. Remove the fastening screws (5) of the slide belts and remove the slide belt.

Attaching the slide belts

1. Clean the support surface of the slide belt (4) on the machine base and apply a corrosion preventative.
2. Put on a new slide belt (4).
3. Screw in the fastening bolts (5) until they rest on the slide belt. Do not tighten.
4. Unscrew the clamp screws (10).
5. Align the clamp plate (12) horizontally with the adjustment screw (11).
6. Clamp the slide belts (4) with the clamp screws (10).
7. Fix the clamp screws (10) in place with the counter nuts.
8. Tighten the fastening screws (5).
9. Clean and oil the slide belt (4). See page 17, chapter "Central lubricating oil CGLP".

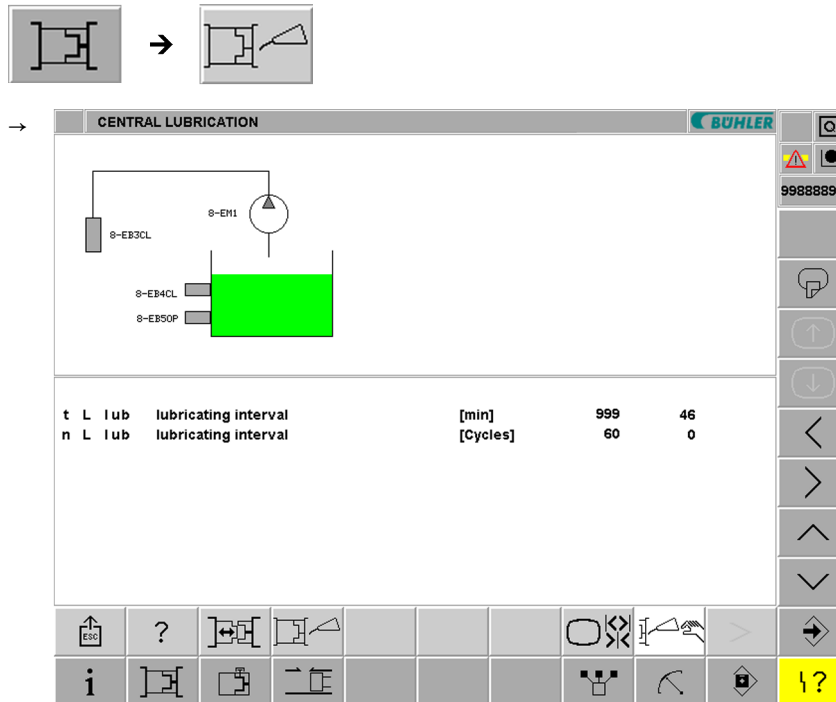
Attach slide plates.

1. Push the slide plate (7) on the moving plate (3) with an inserted wedge (9) on the slide belt (4) toward the inside of the machine.
2. Push the slide plate (7) on the cylinder platen (2) with an inserted wedge (9) on the slide belt (4) toward the outside of the machine. In this process, lower the cylinder platen until the wedge (9) is fixed in place in the groove.
3. Lower cylinder plate (2).
4. Remove screws of the transport safeguard (1).
5. Fit new scrapers (6).

6. Fit wedge securing plates (8).

Lubricating the slide plates

1. Switch on the machine.
2. Open the "Central lubrication" screen page.

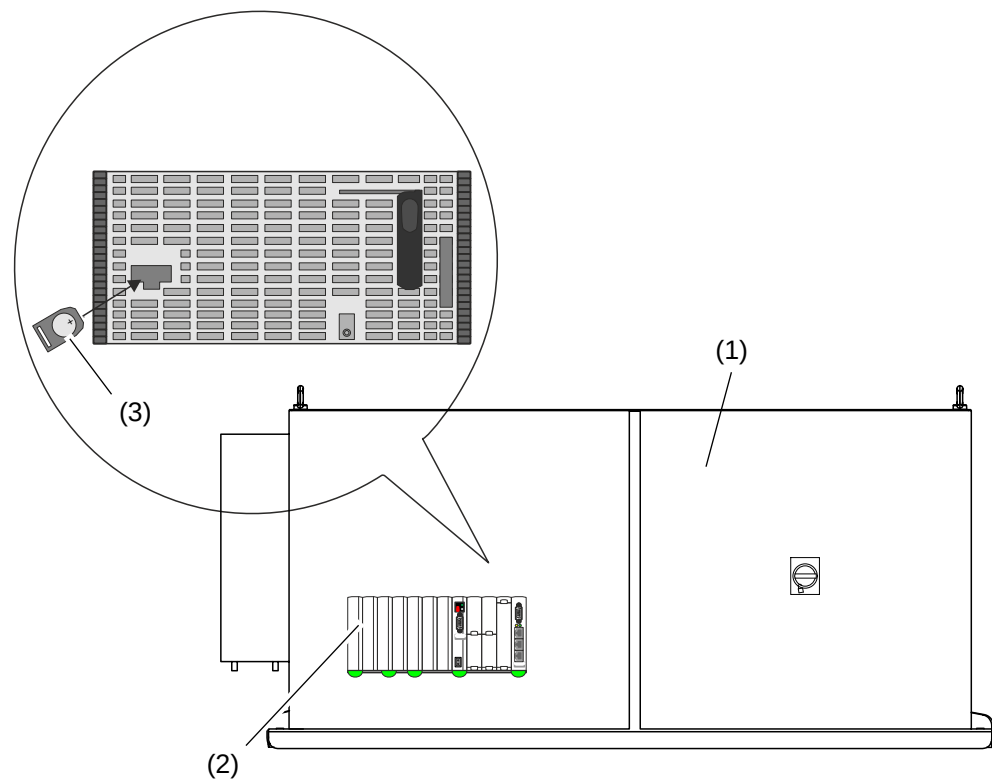


3. Press the button.



→ Lubricating cycle will be executed.

4. Wait 10 seconds.
→ Dosing pressure is completely dissipated.
5. Trigger a total of 20 lubrication cycles with pauses.
→ Sliding plates are lubricated.



- (1) Control cabinet
- (2) Control computer (in control cabinet behind the door)
- (3) Battery

🕒 15 minutes

🛒 Battery type CR2032, 3 V/220 mAh

Prerequisite:

- The machine must be switched on.

1. Open control cabinet (1).
2. Pull the battery compartment out of the top of the control computer (2).
3. Replace the battery (3), in this process ensure correct polarity.
4. Insert battery compartment into the control computer (2).
5. Close door of control cabinet (1).
6. Properly dispose of the removed battery.

11 Repair

11.1 Safety

11.1.1 Personnel qualifications

If improperly executed, personnel can be injured and machine parts can be damaged.

- ▶ The plant operator must assign responsibilities and make these known.

11.1.2 Mobile climbing aids

If mobile climbing aids, such as steps or ladders fall over, personnel can be injured.

- ▶ Only use free-standing climbing aids.
- ▶ Never lean climbing aids against the machine.
- ▶ Do not climb onto the machine.

11.2 Symbols used



(1)



(2)



(3)

- (1) Time required
- (2) Tools required
- (3) Material required

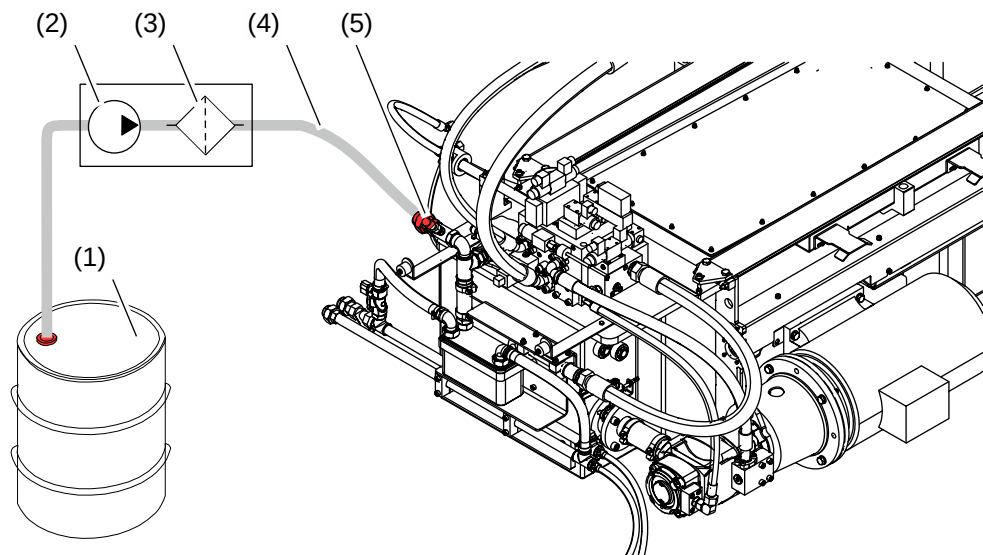
- ▶ Comply with the information on the service and maintenance tasks.

11.3 Filling with hydraulic fluid

The filling level of the hydraulic tank and the filling quantity depend on the machine mode.

The piston accumulator is charged in manual mode, semi-automatic mode or automatic mode. For this reason, the filling level of the hydraulic tank will drop.

The piston accumulator is relieved when the machine has stopped or the set-up mode has been selected. This will raise the filling level of the hydraulic tank.



- (1) Barrel with hydraulic fluid
- (2) External pump
- (3) External filter
- (4) Filling hose
- (5) Filling fitting

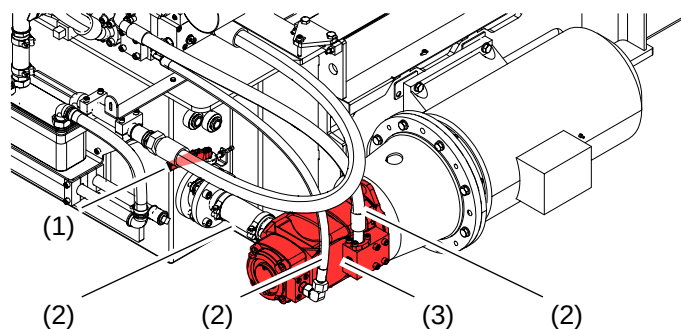
NOTICE

Contaminated hydraulic fluid
 Damage to hydraulic components. Malfunctions.
 ► Only use the filling fitting to fill with hydraulic fluid.

1. Connect filling hose (4) to the filling fitting (5) and connect to an external pump (2).
2. Connect barrel with hydraulic fluid (1) and pump hydraulic fluid into the tank.
 - When the machine has stopped: Monitor the filling level via the oil sight gauges of the hydraulic tank and on the operating panel.
 - During manual mode, semi-automatic mode or automatic mode: Pay attention to the filling quantity. See page 25, chapter "Machine base".

11.4

Replacing the fixed displacement pump



- (1) Ball valve
- (2) Hydraulic lines
- (3) Fixed displacement pumps

⌚ 2 hours

🔧 Collection container

🛒 New fixed displacement pump (spare part)

Prerequisites:

- Machine must be stopped.
- Machine must be secured against unexpected startup.

NOTICE

Commissioning procedure with insufficient hydraulic fluid.

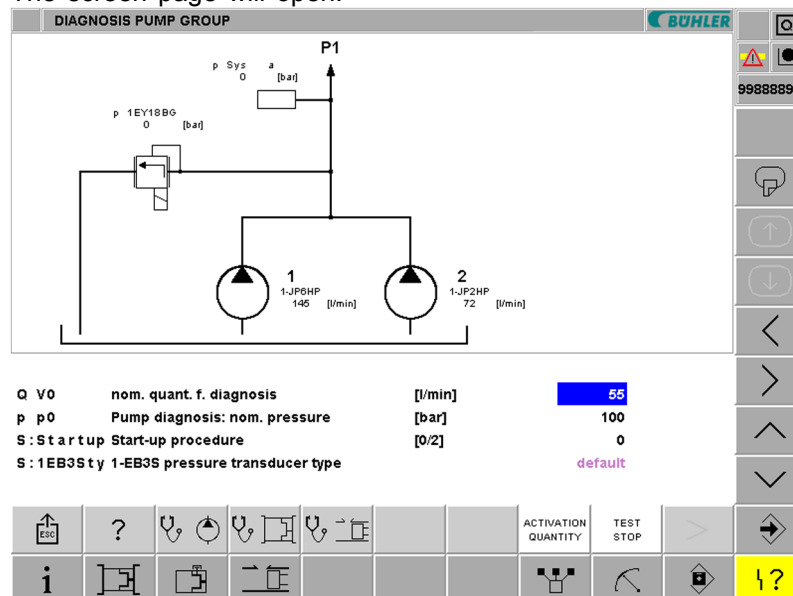
Damage to the fixed-displacement pumps.

- Prior to commissioning, check the hydraulic fluid level via the oil sight gauge. Top up if necessary.

1. Close the ball valve (1).
2. Dismount the hydraulic lines (2) on the fixed displacement pumps (3). Use the collecting container to collect any escaping hydraulic oil.
3. Dismount the fixed displacement pumps (3).
4. Check the new fixed displacement pumps (3) for transportation damage and contamination. If required, remove contamination and oil residue.
5. Install fixed displacement pumps (3).
6. Attach the hydraulic lines (2) to the fixed displacement pumps (3).
7. Open ball valve (1).
8. Briefly switch drive motor on and off 5 times.
→ Bleed the hydraulic lines.
9. Let the fixed displacement pumps (3) run in a depressurized state for at least 15 minutes.
10. Select set-up mode on the control column.
11. Open the "Pump group diagnosis" screen page.



→ The screen page will open.

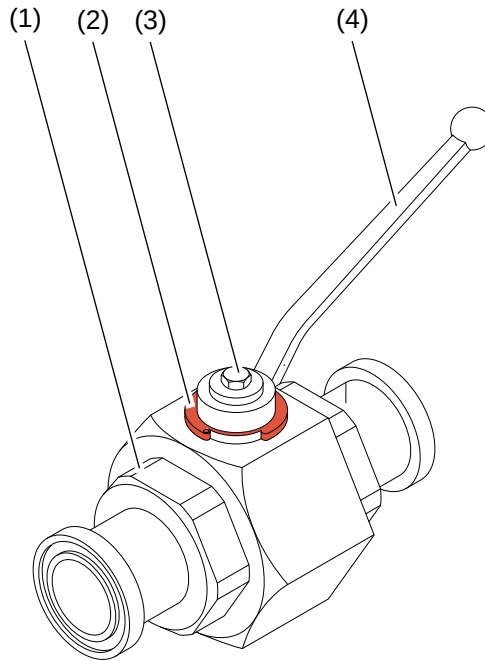


12. Set parameter "S:Startup" to "1".
13. Press the [Activation Quantity] button.
→ The start-up procedure starts and lasts for approx. 42 minutes.
14. Set parameter "S:Startup" to "0".
→ Machine is ready to operate.

11.5

Fixing the ball valve in place

The ball valve can close unintentionally in operation. To prevent this the ball valve can be fixed in place.



- | | |
|----------------|----------------------|
| (1) Ball valve | (3) Screw |
| (2) Limit disk | (4) Activation lever |

1. Unscrew the screw on the ball valve (1) and take off the activation lever (4).
2. Change the limit disk (2).
→ The semi-circular aperture is on the limitation pin.
3. Mount the activation lever (4) and screw it down with the screw.
4. To close the ball valve (1), for example, when filling nitrogen, remove the fixation element.

12 Decommissioning

12.1 Safety

12.1.1 Personnel qualifications

If improperly executed, personnel can be injured or machine parts can be damaged.

- ▶ Decommissioning must only be executed by personnel who have been trained for this task.

12.1.2 Mobile climbing aids

If mobile climbing aids, such as steps or ladders fall over, personnel can be injured.

- ▶ Only use free-standing climbing aids.
- ▶ Never lean climbing aids against the machine.
- ▶ Do not climb onto the machine.

12.2 Shutting down

Prerequisites:

- Production is ended.
- Machine is switched off.
- Water inlet is shut off.

NOTICE

Frozen cooling water.

Frost damage.

- ▶ Completely drain the cooling water.

1. Disconnect the supply connections from the machine.
 - ▶ Disconnect the machine from the power supply.
 - ▶ Remove water inlet and water return.
2. Drain and collect operating supplies
 - ▶ Cooling water: Empty heat exchanger and lines.
 - ▶ Central lubrication unit: Empty tank.
 - ▶ Plunger lubrication system: Drain tank and lines.
 - ▶ Hydraulic lines: Drain tank and lines.
 - ▶ Nitrogen system: Discharge nitrogen.
3. Clean the machine and protect it against corrosion.

12.3

Dismantling



WARNING

Suspended load.

Serious injuries or death.

- ▶ Use only undamaged hoisting devices with a sufficient rating.
- ▶ Do not step or stand under suspended loads.
- ▶ Ensure that there is no one in the danger zone.

- ▶ Dismantle the machine is assembly groups.
- ▶ Secure assembly groups with a crane when dismantling.

12.4

Storage

Prerequisites:

- Machine has been properly shutdown and dismounted.
- Assembly groups no longer contain any fluids.
- ▶ Do not store the machine and assembly groups outdoors.
- ▶ Protect the machine and assembly groups from the effects of weather.
- ▶ Prevent temperature fluctuations because they can cause corrosion due to condensation.
- ▶ Safeguard assembly groups from falling
- ▶ Safeguard the storage location against unauthorized access.

12.5

Disposal

12.5.1

Disposal of operating materials

- ▶ Remove operating materials completely from the machine and dispose of them properly in accordance with the local legal requirements.

12.5.2

Disposing of machine

1. Dismantle machine into its components.
2. Sort components according to basic materials and dispose of them in compliance with local applicable laws and regulations.

13 Appendix

13.1 Checklists: Preparation for installation

These checklists must be filled out and submitted to the project manager at least 2 weeks prior to the planned beginning of installation for signing. Open points must be discussed with the project manager. This is a precondition for delegating experts.

13.1.1 Checklist - installation location

Check the following items based on the checklist:

No.	Check	✓
1	Escape routes are accessible.	
2	Hoisting device with appropriate load-bearing capacity is present and available for the time needed.	
3	Recesses for the horizontal fixation of the die-casting machine are provided according to the foundation plan (add current photos).	
4	The foundation is designed according to the Bühler drawing and is ready to be put in (add current photos of the foundation).	
5	Foundation for the peripheral devices is ready.	
6	There is sufficient covered storage space available for the complete delivery.	
7	The introduction of the die-casting machine up to the final point is clarified.	
8	Safeguards are present for the danger zones.	
9	For die-casting machines with more than 2000 tons of die locking force, there is a movable working platform available.	
10	Cleaning cloths / cleaning agents for removing oil / Tectyl are provided.	
11	Lockable office room with desk and telephone is provided for the service engineer.	
12	Broadband Internet connection (DSL) with free Internet access is available near the installation site.	

13.1.2

Checklist - energy connections

Check the following items based on the checklist:

No.	Check	✓
1	The required amount of energy is available. (See technical data of the devices)	
2	Electrical supply available for the controls.	
3	Cable between the controls and drive motors are available in the required cross section and correct length.	
4	Safeguard of the external feeder unit is present.	
5	Cable ducts and feed-throughs are available.	
6	Water quality meets the requirements.	
7	Cleaning / disposal of the service water is assured.	
8	Required amount of cooling water is available.	
9	Required amount of compressed air is available.	
10	Preparations have been made for the peripheral devices.	
11	The peripheral devices provided by the customer are completely available.	

13.2

Checklists: Preparation for commissioning

13.2.1

Checklist of technical data / operating supplies

Check the following items based on the checklist:

No.	Check	✓
1	Required hydraulic fluid available.	
2	Required central lubricating oil available.	
3	Required plunger lubricating oil or granulate is available.	
4	Required nitrogen is available.	
5	Lubricating grease for die-casting machine is available.	

13.3

Requirements for delegation

We hereby confirm that all of the above-marked points have either been already completed or have been discussed with the project manager.

Based on his wishes, the commissioning technician will arrive on the following date:

Desired date of arrival

Name: Signature

Sender: Date

